12 CHAPTER

Bioinformatics

Introduction

Bioinformatics is the application of computer technology to the management of biological information. Computers are used to gather, store, analyze and integrate biological and genetic information which can then be applied to gene-based drug discovery and development. The science of Bioinformatics, which is the melding of molecular biology with computer science, is essential to the use of genomic information in understanding human diseases and in the identification of new molecular targets for drug discovery and to study fundamental biomedical problems.

The terms bioinformatics and computational biology are often used interchangeably. However put more simply, bioinformatics is concerned with the information while computational biology is concerned with the hypotheses.

Biological Databases

Data generated by molecular biology is deposited in publically accessible databases. The data include nucleotide sequences of genes or amino acid sequences of proteins. Furthermore information about function, structure, localization (both cellular and chromosomal), clinical effects of mutations as well as similarities of biological sequences and structures can be found.

Types of Databases

1. Nucleotide sequence databases
2. Protein sequence databases
3. Molecular structure databases

Examples of Databases

Primary Nucleotide Sequence Databases	
EMBL	EMBL (European Molecular Biology Laboratory) is maintained by European Bioinformatics Institute, UK.
GenBank	This nucleotide database is maintained by NCBI (National center for Biotechnology Information), which itself is a metadatabase and is a part of National Institute of Health, USA.
DDBJ	The DNA Data Bank of Japan is a collaboration between EMBL and GenBank
Other Nucleotide Sequence Databases	
UniGene	Non-redundant database of gene-oriented clusters, derived from GenBank
SGD Genome	Saccharomyces Genome database stores information on molecular biology and genetics of Saccharomyces cerevisiae
EBI Genomes	Provides statistics for the complete genomes.
Ensembl	Joint project between EMBL & EBI and Sanger Institute to develop software to produce and maintain automatic annotation of eukaryotic genomes.
	Protein sequence Databases
SWISS-PROT,	Swiss-Prot is a manually curated biological database of protein sequences. TrEMBL TrEMBL, Translated EMBL is a very large protein database in Swiss-Prot format generated by computer translation of the genetic information
PIR	Protein Information Resource is an integrated public bioinformatics resource to support genomic and proteomic research, and scientific studies.

UniProt	UniProt is the universal protein database, a central repository of protein data created by combining Swiss-Prot, TrEMBL and PIR.
	PROSITEIt is a database of Protein Families and Domains
	PfamProtein families database of alignments and Hidden Markov Models
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Protein Structure Databases	
PDB	The Protein Data Bank (PDB) is a repository for 3-D structural data of proteins and nucleic acids. These data, typically obtained by X-ray crystallography or NMR spectroscopy
SCOP	Structural Classification of Proteins classifies protein 3D structures in a hierarchical scheme.
CATH	CATH is a hierarchical classification of protein domain structures, which clusters proteins at four major levels, Class(C), Architecture(A), Topology(T) and Homologous super family (H).
Other Databases	
KEGG	KEGG (Kyoto Encyclopedia of Genes and Genomes) is a collection of online databases dealing with genomes, enzymatic pathways, and biological chemicals.
	COGsCluster Of Orthologous Genes/Groups

Sequence Alignment

The similarities and differences-at the level of individual bases or amino acids-are analyzed, with the aim of inferring structural, functional, and evolutionary relationships among the sequences under study. The most common comparative method is sequence alignment, which provides an explicit mapping between the residues of two or more sequences. Some important terms related to sequence alignment are:

(*a*) **Gaps and insertions :** In an alignment, one may achieve much better correspondence between two sequences if one allows a gap to be introduced in one sequence. Equivalently, one could allow an insertion in the other sequence. Biologically, this corresponds to a mutation event that eliminates a part of a gene, or introduces new DNA into a gene.

(*b*) **Global alignment :** An alignment that assumes that the two proteins are basically similar over the entire length of one another. The alignment attempts to match them to each other from end to end, even though parts of the alignment are not very convincing. Needleman-Wunsch algorithm is used for global alignment. e.g.

```
LGPSTKDFGKISESREFDN
 |   ||||  |
LNQLERSFGKINMRLEDA
```

(*c*) **Local alignment:** An alignment that searches for segments of the two sequences that match well. There is no attempt to force entire sequences into an alignment, just those parts that appear to have good similarity. Smith -Waterman algorithm is used for local alignment. e.g.

```
------FGKI------
      ||||
------FGKI------
```

ypes of sequence alignments

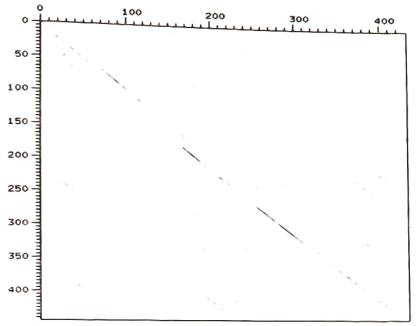
) **Pair wise sequence alignment**

This method is used to find the best-matching piecewise (local) or global alignments of two query sequences. Pair wise alignments can only be used between two sequences at a time, but they are efficient to calculate and

are often used for methods that do not require extreme precision. Primary methods of producing pair wise alignments are dot-matrix methods, dynamic programming. Tools include BLAST, FASTA.

(*a*) **Dynamic Programming :** It is a general algorithm for solving certain optimization problems. The word "programming" does not mean that it has to be a computer program; this is just mathematical jargon for using a fixed set of rules to arrive at a solution. The technique of dynamic programming can be applied to produce global alignments via the Needleman-Wunsch algorithm, and local alignments via the Smith-Waterman algorithm. In typical usage, protein alignments use a substitution matrix to assign scores to amino-acid matches or mismatches, and a gap penalty for matching an amino acid in one sequence to a gap in the other

(*b*) **Dot Matrix/Dot Plots :** A dot plot is a visual representation of the similarities between two sequences. Each axis of a rectangular array represents one of the two sequences to be compared. A window length is fixed, together with a criterion when two sequence windows are deemed to be similar. Thus, when two sequences share similarity over their entire length a diagonal line will extend from one corner of the dot plot to the diagonally opposite corner. If two sequences only share patches of similarity then that will be revealed by diagonal stretches.



(*ii*) Multiple Sequence Alignment

This is an extension of pair wise alignment to incorporate more than two sequences at a time. Multiple alignment methods try to align all of the sequences in a given query set. Multiple alignments are often used in identifying conserved sequence regions across a group of sequences hypothesized to be evolutionarily related. Alignments are also used to aid in establishing evolutionary relationships by constructing phylogenetic trees. MSA can be done by dynamic programming, progressive methods and by motif finding.

(*a*) **Progressive Method:** Progressive, hierarchical, or tree methods generate a multiple sequence alignment by first aligning the most similar sequences and then adding successively less related sequences or groups to the alignment until the entire query set has been incorporated into the solution.

(*b*) **Motif Finding:** Motif finding, also known as profile analysis, constructs global multiple sequence alignments that attempt to align short conserved sequence motifs among the sequences in the query set.

Tools for Multiple Sequence Alignment:

- ClustalW
- T-Coffee
- MAP
- PIMA (Pattern induced Multiple alignment)
- DIALIGN 2

Search Tools

The usefulness of the databases can be increased by using certain software systems to provide searching, indexing and cross referencing in the databases.

(i) Text-based Search Tools

(a) **ENTREZ:** for nucleotide, protein, GenBank, EMBL, DDBJ, NCBI searching

(b) **SRS (Sequence Search and Retrieval Tool):** PIR and other Protein sequence database searching

(c) **ExPASy (Expert Protein Analysis System):** SWISS-PROT & others searching.

(ii) Sequence Similarity Based Search Tools

(a) **FASTA :** It is a DNA and Protein sequence alignment software package first described (as FASTP) by David J. Lipmann and William R. Pearson. The FASTA program follows a largely heuristic method which contributes to the high speed of its execution. It initially observes the pattern of word hits, word-to-word matches of a given length, and marks potential matches before performing a more time-consuming optimized search using a Smith-Waterman type of algorithm. The size taken for a word, given by the parameter ktup, controls the sensitivity and speed of the program. Increasing the ktup value decreases number of background hits that are found. From the word hits that are returned the program looks for segments that contain a cluster of nearby hits. It then investigates these segments for a possible match.

(b) **BLAST :** It stands for Basic Local Alignment Search Tool.

Different types of BLAST are:

Program	Details
BLASTN	Matches nucleotide query against nucleotide database
BLASTP	Matches protein query against protein database
BLASTX	Matches 6 frame translated nucleotide query against protein database
TBLASTN	Matches protein query against translated (6 frame) nucleotide database
TBLASTX	Matches translated nucleotide query against translated nucleotide database
PSI-BLAST	(Position Specific Iterative BLAST) A list of all closely related proteins is created. These proteins are combined into a general "profile" sequence, which summarizes significant features present in these sequences. A query against the protein database is then run using this profile, and a larger group of proteins is found. This larger group is used to construct another profile, and the process is repeated.
PHI-BLAST	Pattern Hit Initiated BLAST searches for particular patterns in protein queries

Structure Alignments

Structural alignments, which are usually specific to protein and sometimes RNA sequences, use information about the secondary and tertiary structure of the protein or RNA molecule to aid in aligning the sequences.

(i) DALI : The DALI method, or distance matrix alignment, is a fragment-based method for constructing structural alignments based on contact similarity patterns between successive hexapeptides in the query sequences. It can generate pair wise or multiple alignments and identify a query sequence's structural neighbors in the Protein Data Bank

(ii) SSAP : Sequential structure alignment program is a dynamic programming-based method of structural alignment that uses atom-to-atom vectors in structure space as comparison points. It has been extended since its original description to include multiple as well as pair wise alignments.

Phylogenetic Analysis

Phylogeny refers to study of evolutionary relationships. The final goal of phylogenetic analysis is to prepare an evolutionary tree describing the relationship of the various taxa with respect to each other. Phylogenetic trees illustrate the evolutionary relationships among groups of organisms, or among a family of related nucleic acid or protein sequences.

Type of philosophies of building trees :

(i) Cladistic : Trees are drawn based on conserved characters. These methods construct a tree called cladogram where branch order is shown and branch lengths are meaningless.

(ii) Phenetic : Trees are drawn based on morphological characters. These methods build a tree called phylogram which show both branch order and length.

While making a tree one must know about the different types of sequences:

Building of a Tree

Homolgs : Sequences having a common origin

Orthologs : Homologs derived from speciation

Paralogs : Homologs derived from a common ancestral gene that underwent duplication and divergence

Xenologs : Homologs resulting from horizontal transfer of genes

Steps of Building a Tree

(i) Alignment

This step involves using multiple sequence alignment to align the sequences.

(ii) Determining Substitution Model

DNA substitution models :

Jukes Cantor Model-gives independent probability of base substitutions at all sites

Kimura- gives different rates of transitions and transversions.

Amino Acid Substitution Models :

PAM : PAM matrices are a common family of score matrices. PAM stands for Percent Accepted Mutations, where "accepted" means that the mutation has been adopted by the sequence in question. Thus, using the PAM 250 scoring matrix means that about 250 mutations per 100 amino acids may have happened, while with PAM 10 only 10 mutations per 100 amino acids are assumed, so that only very similar sequences will reach useful alignment scores. Other PAM matrices are PAM 50,PAM 500.

BLOSUM : BLOSUM (BLOcks of Amino Acid SUbstitution Matrix) is a substitution matrix used for sequence alignment of proteins. For example, BLOSUM80 is used for less divergent alignments, and BLOSUM45 is used for more divergent alignments. BLOSUM62 is the matrix calculated by using the observed substitutions between proteins which have at most 62% sequence identity. BLOSUM has proved better at scoring distantly related sequences than the once-widely-used Point Accepted Mutation (PAM) matrices.

(iii) Tree Construction

There are several methods available for tree construction:

(a) Distance methods (used for recognizable similarity):

- UPGMA(Unweighted Group Method With Arithmetic Mean)
- Transformed Distance Methods,
- Least Squares Methods. Fitch and Kitsch of the Phylip package (Jo Felsentein,Univ. Washington) are least squares method (Fitch and Margoliash)
- Neighbor-joining methods
- Neighbor of the Phylip package (Jo Felsentein, Univ. Washington),
- ClustalW (D. Higgins, EMBL (check),
- Distnj in Protml package (Adachi and Hasegawa, Univ. Tokyo)
- Darwin (Gaston Gonner, ETH, Zurich, via mail server or WWW)

(b) Character-based methods:

- Protein Maximum likelihood (used for very less similarity)
- Protml (Adachi and Hasegawa, Univ. Tokyo) (very cpu intensive)
- Puzzle (Korbinian Strimmer and Arndt von Haeseler, Univ. Munich) (A heuristic method much faster than Protml)

(c) Protein maximal parsimony (used for strong similarity):

- Protpars (Jo Felsentein, Univ. Washington)
- Paup (David Swofford, latest version will be commercial)

(iv) Tree Evaluation

To determine the reliability and robustness of a tree generated, there are two strategies:

(a) **Bootstrap :** It is a statistical technique that can use random re-sampling of data to determine sampling error for tree topologies.

(b) **Jackknife :** Replicates are created by dropping one or more sites within each replicate

Computer Softwares for Phylogenetics.

1. PHYLIP(Phylogenetic Inference Package) **2.**CLUSTALX **3.**BioEdit **4.** MEGA 4 **5.**TreeView and PAUP

Protein Identification

Techniques such as circular dichroism spectroscopy, optical rotatory dispersion, X-ray crystallography, and nuclear magnetic resonance are extremely powerful in determining structural features, but their execution requires many hours of highly skilled, technically demanding work. There are computational techniques that allow for biological discovery based on the protein sequence itself or on their comparison to protein families.

(i) Protein Identity Based on Composition

(a) AA Compident: uses the amino acid composition of an unknown protein to identify known proteins of the same composition.

(b) AA Compsim: rather than using an experimentally derived amino acid composition as the basis for searches, the sequence of a SWISS-PROT protein is used instead. A theoretical pI and molecular weight are computed before computation of the difference scores using Compute pI/MW

(c) Propsearch: this technique is more robust than AACompSim in that 144 different physical properties are used in performing the analysis

(ii) Physical Properties Based on Sequence.

(a) PeptideMass : Designed for use in peptide mapping experiments, PeptideMass determines the cleavage products of a protein after exposure to a given protease or chemical reagent.

(b) TGREASE : TGREASE calculates the hydrophobicity of a protein along its length

(c) SAPS : The Statistical Analysis of Protein Sequences (SAPS) algorithm provides extensive statistical information for any given query sequence

(d) Compute pI/MW and ProtParam : Compute pI/MW is a tool that calculates the isoelectric point and molecular weight of an input sequence. ProtParam takes this process one step further. Based on the input sequence, Prot- Param calculates the molecular weight, isoelectric point, overall amino acid composition, a theoretical extinction coefficient the protein's grand average of hydrophobicity (GRAVY) value etc.

(iii)Protein Structure prediction: Several online servers like : JPRED, PSIPRED etc, are available for secondary structure prediction of protein sequences. Homology Modeling for protein sequences showing similarity to other sequences using servers like Modeler, prime, Swiss Prot etc.

Ab-initio structure prediction : For protein sequences that hardly share any similarity to known protein structures. Several servers can be used for the same : I-Tasser, Rossetta etc.

Gene Prediction

Even before sequencing data reach this highly polished state, investigators can begin to ask whether or not given stretches of sequence represent coding or noncoding regions.These predictions can be made with help of certain tools of bioinformatics. Some examples of these tools are:

(i) **GRAIL :** Gene Recognition and Analysis Internet Link is the elder statesman of the gene prediction techniques. GRAIL 1 makes use of a neural network method to recognize coding potential in fixed-length. GRAIL 1a expands on this method by considering regions immediately adjacent to regions deemed to have coding potential. Most recent version of GRAIL is GRAIL-EXP.

(ii) **FGENEH/FGENES :** FGENEH, developed by Victor Solovyev and colleagues, is a method that predictsinternal exons by looking for structural features such as donor and acceptor splice etc. The method makes use of linear discriminant analysis. FGENES, can be used in cases when multiple genes are expected in a given stretch of DNA

(iii) **GENSCAN :** GENSCAN can identify introns, exons, promoter sites, and polyA signals, as do a number of the other gene identification algorithms and relies on what the author terms a "probabilistic model" of genomic sequence composition and gene structure

(iv) **GeneID :** The original version of this program was among the fastest in that it used a rule-based system to examine the putative exons and assemble them into the "most likely gene" for that sequence.

(v) **HMMgene:** HMMgene predicts whole genes in any given DNAsequence using a Hidden Markov **Model (HMM).** Alternative predictions on the same sequence are also made.

Other gene prediction tools: GeneParser, PROCRUSTES and "Michael Zhang's Exon Finder (MZEF).

PRACTICAL EXPERIMENTS

(i) **Aim:** To carry out BLAST of a nucleotide sequence

Steps involved

(a) Open NCBI home page http://www.ncbi.nlm.nih.gov/

(b) The NCBI home page opens. Click on the text box showing "all databases" and change it to "Nucleotide".

(c) Type in the text box next to "for" , name of any protein or gene whose sequence you want to retrieve. e.g Human Insulin

(d) Click on GO

(e) Another page opens showing the number of Core Nucleotide, Expressed Sequence Tags and Genome Survey Sequence records.

(f) Click on the number next to Core Nucleotide records

(g) The resulting page shows a number of different entries for human insulin.

(h) Click on the entry which is NM_000599.

(i) The outcome is a page shoeing the entry in GenBank format. Since BLAST accepts only FASTA format changing format is important.

(j) To change the format click on the text box next to "Display" and select "Fasta".

(k) The new page that opens is our Query sequence in Fasta format (a part of it is shown below).

>gi|171460920|ref|NM_000599.3| Homo sapiens insulin-like growth factor binding protein 5 (IGFBP5), mRNA
GTAGTCTCTTTGGAAACTTCTGCAGGGGAAAAGAGCTAGGAAAGAGCTGCAAAGCAGTGTGGGCTTTTTC
CCTTTTTTTGCTCCTTTTCATTACCCCTCCTCCGTTTTCACCCTTCTCCGGACTTCGCGTAGAACCTGCGA
ATTTCGAAGAGGAGGTGGCAAAGTGGGAGAAAAGAGGTGTTAGGGTTTGGGGTTTTTTTGTTTTTGTTTT

(l) Copy this sequence by using right click option.

(m) Go back to NCBI homepage.

(n) On the homepage is an option for BLAST. Click on it and BLAST homepage will open.

(o) Click on Nucleotide Blast.

(p) A query window appears. Paste the query sequence you had copied by using right click option.

(q) After pasting the sequence scroll down to the block "Program Selection". Select Blastn. Click on BLAST.

(r) The result page gets displayed. The matches represent the homologous sequences found in the database. An example of a typical BLAST result is displayed below.

(ii) Aim: To carry out Multiple Sequence Alignment and build phylogenetic tree using ClustalW

Steps involved :

(a) In order to carry out multiple sequence alignment you need more than 2 sequences. Open NCBI home page. Select "Nucleotide database" for Actin proteins. Click on GO.

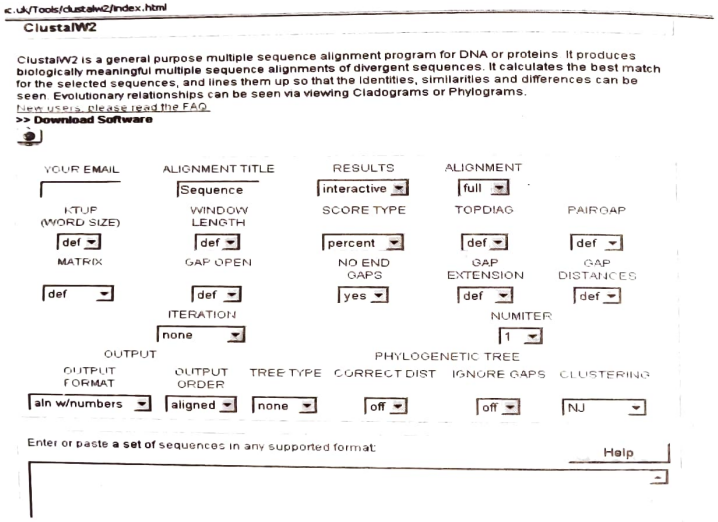
(b) Select the Core Nucleotide records. A number of entries for Actin in different organisms will be displayed.

(c) Retrieve 4 query sequences from 4 different organisms and past them in WordPad one by one.

(d) Save the WordPad file.

(e) Open page http://www.ebi.ac.uk/Tools/clustalw2/index.html

(f) The query window for ClustalW program will open. Scroll down at the bottom of the page. Click on "Browse" and select the WordPad file containing the sequences.



(g) Click on "Upload file". Once file gets uploaded click on "Run". One can also copy and past the sequences directly in the query window. The sequences should always be in Fasta format.

(h) The result page for ClustalW opens showing the multiple sequence alignment as well as phylogenetic tree of the given sequences. An e.g. of a typical Clustalw result is shown below.

(i) No error

Genomics

Genomics is the study of an organism's entire genome. The field includes intensive efforts to determine the entire DNA sequence of organisms and fine-scale genetic mapping efforts. Genomics was established by Fred Sanger when he first sequenced the complete genomes of a virus and a mitochondrion.

Basically there are 3 broad categories of genomics

(i) **Functional genomics**: It is a field of molecular biology that attempts to make use of the vast wealth of data produced by genomic projects (such as genome sequencing projects) to describe gene (and protein) functions and interactions. It focuses on the dynamic aspects such as gene transcription, translation, and protein-protein interactions, as opposed to the static aspects of the genomic information such as DNA sequence or structures. Functional genomics uses mostly high-throughput techniques to characterize the abundance gene products such as mRNA and proteins. Some typical technology platforms are: DNA microarrays and SAGE and for proteins two-dimensional gel electrophoresis and mass spectrometry.

(ii) **Structural Genomics**: It consists in the determination of the three dimensional structure of all proteins of a given organism, by experimental methods such as X-ray crystallography, NMR spectroscopy or computational approaches such as homology modelling. As opposed to traditional structural biology, the determination of a protein structure through a structural genomics effort often (but not always) comes before anything is known regarding the protein function.

(iii) **Comparative Genomics**: It is the study of relationships between the genomes of different species or strains. Comparative genomics is an attempt to take advantage of the information provided by the signatures of selection to understand the function and evolutionary processes that act on genomes. Gene finding/prediction is an important application of comparative genomics, as is discovery of new, non-coding functional elements of the genome. Identifying the mechanisms of eukaryotic genome evolution by comparative genomics is one of the important goals of the field.

Genome Size

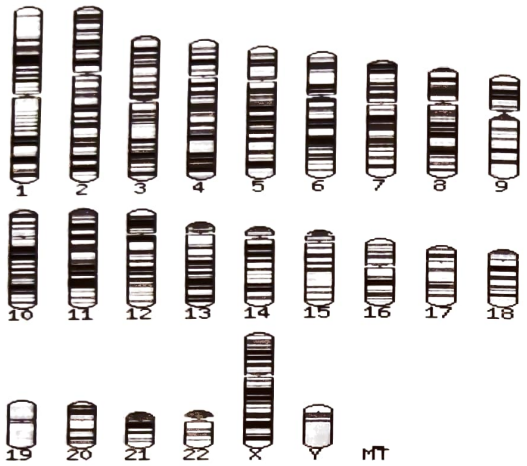
The genome of an organism is the complete set of genes specifying how its phenotype will develop (under a certain set of environmental conditions). In this sense, then, **diploid** organisms (like us) contain two genomes, one inherited from our mother, the other from our father. The table below presents a selection of representative genome sizes from the rapidly-growing list of organisms whose genomes have been sequenced:

Organism	Base Pairs	Number of Genes
Phi-X 174	5386	10
Mycoplasma genitalium	580,073	485
Helicobacter pylori	1,667,867	1,589
Haemophilus influenzae	1,830,138	1,738
Streptococcus pneumoniae	2,160,837	2,236
Bacillus subtilis	4,214,814	4,779
E. coli	4,639,221	4,377
Pseudomonas aeruginosa	6.3×10^6	5,570
Neurospora crassa	38,639,769	10,082
Caenorhabditis elegans	100,258,171	19,427
Drosophila melanogaster	122,653,977	13,379
Rice	3.9×10^8	37,544
Dogs	2.4×10^9	19,300
Humans	3.3×10^9	~20,500

Human Genome

The human genome is the genome of *Homo sapiens*, which is stored on 24 distinct chromosomes (22 autosomal + X + Y) containing an estimated 20,000 25,000 genes. The entire human genome occupies a

total of just over 3 billion DNA base pairs, and has a data size of approximately 750 Megabytes, which is slightly larger than the capacity of a standard Compact Disc. Surprisingly, the number of human genes seems to be less than a factor of two greater than that of many much simpler organisms, such as the roundworm and the fruit fly. However, human cells make extensive use of alternative splicing to produce several different proteins from a single gene, and the human proteome is thought to be much larger than those of the aforementioned organisms. Human genes are distributed unevenly across the chromosomes. Each chromosome contains various gene-rich and gene-poor regions.



Karyotype

The project will reap fantastic benefits for humankind, some that we can anticipate and others that will surprise us. Generations of biologists and researchers have been provided with detailed DNA information that will be key to understanding the structure, organization, and function of DNA in chromosomes. Genome maps of other organisms will provide the basis for comparative studies that are often critical to understanding more complex biological systems. Information generated and technologies developed are revolutionizing future biological explorations. The analysis of similarities between DNA sequences from different organisms is also opening new avenues in the study of the theory of evolution. In many cases, evolutionary questions can now be framed in terms of molecular biology; indeed, many major evolutionary milestones (the emergence of the ribosome and organelles, the development of embryos with body plans, the vertebrate immune system) can be related to the molecular level. Many questions about the similarities and differences between humans and our closest relatives (the primates, and indeed the other mammals) are expected to be illuminated by the data from this project.

Sequencing the Human Genome

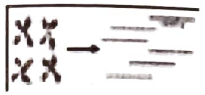
There are essentially two ways to sequence a genome.

(*i*) **BAC-to-BAC method**. The first to be employed in human genome studies, is slow but sure. The term 'BAC" is an acronym for 'Bacterial Artificial Chromosome', and in principle, it is used like a plasmid. The BAC-to-BAC approach, also referred to as the map-based method, evolved from procedures developed by a number of researchers during the late 1980s and 90s and that continues to develop and change.

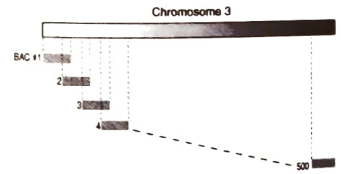
(*ii*) **Whole genome shotgun sequencing**. This brings speed into the picture, enabling researchers to do the job in months to a year. The shotgun method was developed by GNN president J. Craig Venter in 1996 when he was at the Institute for Genomic Research (TIGR).

BAC to BAC Sequencing

1. The BAC to BAC approach first creates a crude physical map of the whole genome before sequencing the DNA. Constructing a map requires cutting the chromosomes into large pieces and figuring out the order of these big chunks of DNA before taking a closer look and sequencing all the fragments.
2. Several copies of the genome are randomly cut into pieces that are about 150,000 base pairs (bp) long.



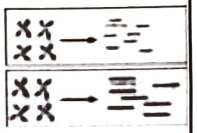
3. Each of these 150,000 bp fragments is inserted into a BAC-a bacterial artificial chromosome. A BAC is a man made piece of DNA that can replicate inside a bacterial cell. The whole collection of BACs containing the entire human genome is called a BAC library, because each BAC is like a book in a library that can be accessed and copied.
4. These pieces are fingerprinted to give each piece a unique identifi-cation tag that determines the order of the fragments. Fingerprinting involves cutting each BAC fragment with a single enzyme and finding common sequence landmarks in overlapping fragments that determine the location of each BAC along the chromosome. Then overlapping BACs with markers every 100,000 bp form a map of each chromosome.



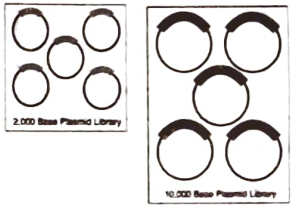
5. Each BAC is then broken randomly into 1,500 bp pieces and placed in another artificial piece of DNA called M13. This collection is known as an M13 library.
6. All the M13 libraries are sequenced. 500 bp from one end of the fragment are sequenced generating millions of sequences.

Whole Genome Shotgun Sequencing

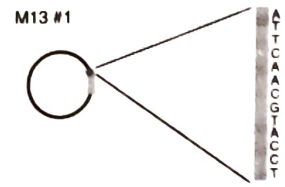
1. The shotgun sequencing method goes straight to the job of decoding, bypassing the need for a physical map. Therefore, it is much faster.
2. Multiple copies of the genome are randomly shredded into pieces that are 2,000 base pairs (bp) long by squeezing the DNA through a pressurized syringe. This is done a second time to generate pieces that are 10,000 bp long.



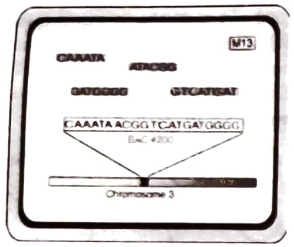
3. Each 2,000 and 10,000 bp fragment is inserted into a plasmid, which is a piece of DNA that can replicate in bacteria. The two collections of plasmids containing 2,000 and 10,000 bp chunks of human DNA are known as plasmid libraries.

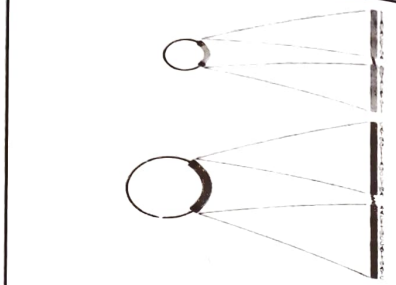


4. This step not needed in shotgun sequencing
5. This step not needed in shotgun sequencing
6. Both the 2,000 and the 10,000 bp plasmid libraries are sequenced. 500 bp from each end of each fragment are decoded generating millions of sequences. Sequencing both ends of each insert is critical for the assembling the entire chromosome.

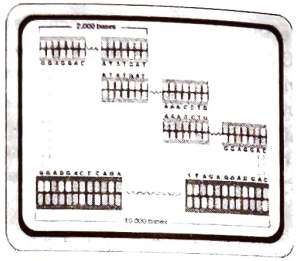


7. These sequences are fed into a computer program called PHRAP that looks for common sequences that join two fragments together.





7. Computer algorithms assemble the millions of sequenced fragments into a continuous stretch resembling each chromosome.



Methods of DNA Sequencing

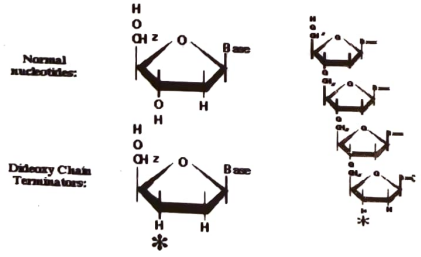
DNA sequencing is the process of determining the exact order of the bases A, T, C and G in a piece of DNA. In essence, the DNA is used as a template to generate a set of fragments that differ in length from each other by a single base. The fragments are then separated by size, and the bases at the end are identified, recreating the original sequence of the DNA.

DNA sequencing, first devised in 1975, has become a powerful technique in molecular biology, allowing analysis of genes at the nucleotide level. For this reason, this tool has been applied to many areas of research. For example, the polymerase chain reaction (PCR), a method which rapidly produces numerous copies of a desired piece of DNA, requires first knowing the flanking sequences of this piece. Another important use of DNA sequencing is identifying restriction sites in plasmids. Knowing these restriction sites is useful in cloning a foreign gene into the plasmid.

Type of DNA sequencing

(i) Dideoxy or Chain Termination Method:

(a) This technique utilizes 2',3'-dideoxynucleotide triphospates (ddNTPs), molecules that differ from deoxynucleotides by the having a hydrogen atom attached to the 3' carbon rather than an OH group. These molecules terminate DNA chain elongation because they cannot form a phosphodiester bond with the next deoxynucleotide.



(b) In order to perform the sequencing, one must first convert double stranded DNA into single stranded DNA. This can be done by denaturing the double stranded DNA with NaOH. A Sanger reaction consists of the following :

a strand to be sequenced (one of the single strands which was denatured using NaOH), DNA primers (short pieces of DNA that are both complementary to the strand which is to be sequenced and radioactively labelled at the 5' end), a mixture of a particular ddNTP (such as ddATP) with its normal dNTP (dATP in this case), and the other three dNTPs (dCTP, dGTP, and dTTP).

"G" tube: all four dNTP's, ddGTP and DNA polymerase

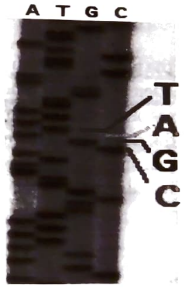
"A" tube: all four dNTP's, ddATP and DNA polymerase

"T" tube: all four dNTP's, ddTTP and DNA polymerase

"C" tube: all four dNTP's, ddCTP and DNA polymerase

(c) As the DNA is synthesized, nucleotides are added on to the growing chain by the DNA polymerase. However, on occasion a dideoxynucleotide is incorporated into the chain in place of a normal nucleotide, which results in a chain-terminating event.

(d) This reaction is performed four times using a different ddNTP for each reaction. When these reactions are completed, a polyacrylamide gel electrophoresis (PAGE) is performed. One reaction is loaded into one lane for a total of four lanes (Figure. The gel is transferred to a nitrocellulose filter and autoradiography is performed so that only the bands with the radioactive label on the 5' end will appear. In PAGE, the shortest fragments will migrate the farthest. Therefore, the bottom-most band indicates that its particular dideoxynucleotide was added first to the labeled primer.



BANDS ON THE GEL

(e) One reads the bases from the bottom up , one is reading the 5' to 3' sequence of the strand complementary to the sequenced strand. The sequenced strand can be read 5' to 3' by reading top to bottom the bases complementary to those on the gel.

(ii) Maxam-Gilbert sequencing

In 1976-1977, Allan Maxam and Walter Gilbert developed a DNA sequencing method based on chemical modification of DNA and subsequent cleavage at specific bases. However, with the development and improvement of the chain-termination method (see below), Maxam-Gilbert sequencing has fallen out of favour due to its technical complexity, extensive use of hazardous chemicals, and difficulties with scale-up.

In brief, the method requires radioactive labelling at one end and purification of the DNA fragment to be sequenced. Chemical treatment generates breaks at a small proportion of one or two of the four nucleotide bases in each of four reactions (G, A+G, C, and C+ T). Thus a series of labelled fragments is generated, from the radiolabelled end to the first 'cut' site in each molecule. The fragments are then size-separated by gel electrophoresis, with the four reactions arranged side by side. To visualize the fragments generated in each reaction, the gel is exposed to X-ray film for autoradiography, yielding an image of a series of dark 'bands' corresponding to the radiolabelled DNA fragments, from which the sequence may be inferred.

(iii) Dye Terminator Sequencing (Automated)

The major advantage of this method is that the sequencing can be performed in a single reaction, rather than four reactions as in the labelled-primer method. In dye-terminator sequencing, each of the four dideoxynucleotide chain terminators is labelled with a different fluorescent dye, each fluorescing at a different wavelength. This method is attractive because of its greater expediency and speed and is now the mainstayin automated sequencing. Its potential limitations include dye effects due to differences in the incorporation of the dye-labelled chain terminators into the DNA fragment, resulting in unequal peak heights and shapes in the electronic DNA sequence trace chromatogram after capillary electrophoresis.

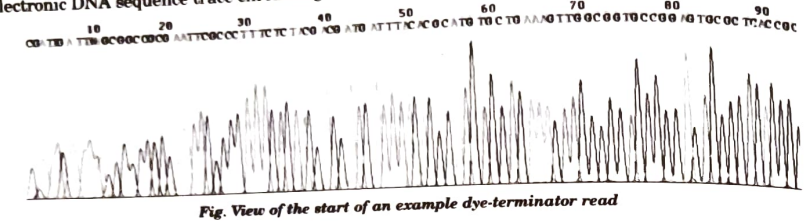


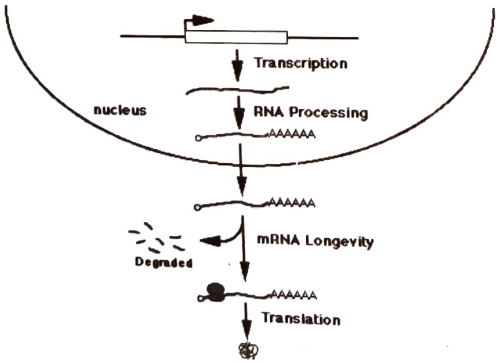
Fig. View of the start of an example dye-terminator read

Concept of Maps

- **Genetic Maps**: A genetic map is a representation of the genes on a chromosome arrayed in linear order with distances between loci expressed as percent recombination (map units, centimorgans). Also called a linkage map. Genome mapping is the creation of a genetic map assigning DNA fragments to chromosomes. Genetic mapping uses classical genetic techniques (e.g. pedigree analysis or breeding experiments) to determine sequence features within a genome.
- **Physical Maps**: A physical map describes the physical location of genes on chromosomes. A diagram or other representation that shows the physical location of genes or other genetic markers on chromosomes is also a physical map. Physical Mapping is the process of determining how DNA contained in a group of clones overlap without having to sequence entire DNA in the clones. Once the map is determined, we can use the clones as a resource to efficiently contain stretches of genome in large quantity. This type of mapping is more accurate than genetic maps. Physical maps can be divided into three general types: chromosomal or cytogenetic maps, radiation hybrid (RH) maps, and sequence maps.

Gene Expression

Gene expression is the process by which inheritable information from a gene, such as the DNA sequence, is made into a functional gene product, such as protein or RNA. Several steps in the gene expression process may be modulated, including the transcription (synthesis of RNA under the direction of DNA.) step and the post-translational modification of a protein.



Studying & Analysing Gene Expression

There are a number of techniques to study and analyse gene expression. Some important methods are as follows :

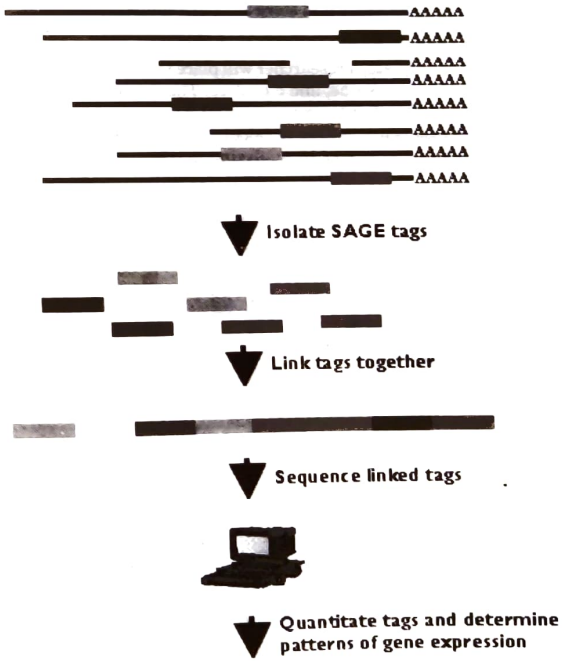
1. **Serial Analysis Of Gene Expression(SAGE) :** It is a method for comprehensive analysis of gene expression patterns. The original technique was developed by Dr. Victor Velculescu at the Oncology Center of Johns Hopkins University, and was published in the journal Science in 1995.

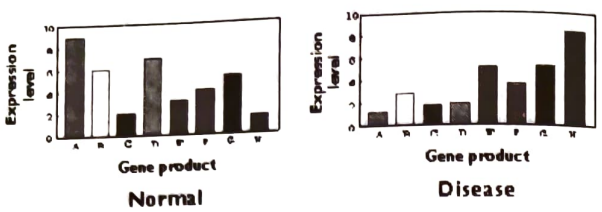
Three principles underlie the SAGE methodology :

1. A short sequence tag (10-14bp) contains sufficient information to uniquely identify a transcript provided that that the tag is obtained from a unique position within each transcript;
2. Sequence tags can be linked together to from long serial molecules that can be cloned and sequenced; and
3. Quantization of the number of times a particular tag is observed provides the expression level of the corresponding transcript.

Briefly, SAGE experiments proceed as follows:

1. Isolate the mRNA of an input sample (e.g. a tumor).
2. Extract a small chunk of sequence from a defined position of each mRNA molecule.
3. Link these small pieces of sequence together to form a long chain (or concatemer).
4. Clone these chains into a vector which can be taken up by bacteria.
5. Sequence these chains using modern high-throughput DNA sequencers.
6. Process this data with a computer to count the small sequence tags. Statistical methods can be applied to tag and count lists from different samples in order to determine which genes are more highly expressed. For example, a normal tissue sample can be compared against a corresponding tumor to determine which genes tend to be more (or less) active.





Microarrays

In the past several years, a new technology, called DNA microarray, has attracted tremendous interests among biologists. This technology promises to monitor the whole genome on a single chip so that researchers can have a better picture of the interactions among thousands of genes simultaneously. It can be called as a biochip, DNA chip, DNA microarray or gene array. Microarrays may be used to assay gene expression within a single sample or to compare gene expression in two different cell types or tissue samples, such as in healthy and diseased tissue.

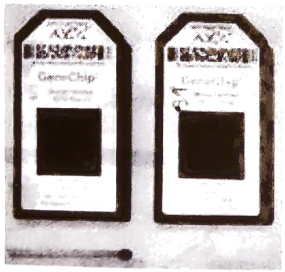
DNA Microarrays are small, solid supports onto which the sequences from thousands of different genes are immobilized, or attached, at fixed locations. The supports themselves are usually **glass microscope slides**, the size of two side-by-side pinky fingers, but can also be **silicon chips** or **nylon membranes**. The DNA is printed, spotted, or actually synthesized directly onto the support.

DNA Microarray Experiment

1. Prepare your DNA chip using your chosen target DNAs.
2. Generate a hybridization solution containing a mixture of fluorescently labeled cDNAs.
3. Incubate your hybridization mixture containing fluorescently labeled cDNAs with your DNA chip.
4. Detect bound cDNA using laser technology and store data in a computer.
5. Analyze data using computational methods.

After this hybridization step is complete, a researcher will place the microarray in a "reader" or "**scanner**" that consists of some lasers, a special microscope, and a camera. The fluorescent tags are excited by the laser, and the microscope and camera work together to create a digital image of the array. These data are then stored in a computer, and a special program is used either to calculate the red-to-green fluorescence ratio or to subtract out background data for each microarray spot by analyzing the digital image of the array.

The Colors of a Microarray



Two Affymetrix chips

In this schematic:

- **GREEN** represents **Control DNA**, where either DNA or cDNA derived from normal tissue is hybridized to the target DNA.
- **RED** represents **Sample DNA**, where either DNA or cDNA is derived from diseased tissue hybridized to the target DNA.
- **YELLOW** represents **a combination of Control and Sample DNA**, where both hybridized equally to the target DNA.

- **BLACK** represents areas where **neither the Control nor Sample DNA** hybridized to the target DNA. Each spot on an array is associated with a particular gene. Each color in an array represents either healthy (control) or diseased (sample) tissue. Depending on the type of array used, the location and intensity of a color will tell us whether the gene, or mutation, is present in either the control and/or sample DNA. It will also provide an estimate of the expression level of the gene(s) in the sample and control DNA.

Types of Microarrays

(a) Spotted Microarrays : In *spotted microarrays*, the probes are oligonucleotides, cDNA or small fragments of PCR products that correspond to mRNAs.

(b) **Oligonucleotide Microarrays :** the probes are short sequences designed to match parts of the sequence of known or predicted open reading frames.

(c) **Two-Color microarrays or Two-Channel microarrays :** These are typically hybridized with cDNA prepared from two samples to be compared (e.g. diseased tissue versus healthy tissue) and that are labeled with two different fluorophores(Fluorescent dyes like **Cy3 and Cy5)**

(d) **SNP Microarrays :** They are a type of DNA microarray that are used to identify genetic variation in individuals and across populations.

Companies producing microarrays:

- Affymetrix (GeneChip)
- Incyte Genomics
- Illumina
- Agilent Technologies
- Motorola (CodeLink bioarray system)

In Brief: Microarray Applications

Microarray type	Application
CGH(Comparative Genome Hybridisation)	Tumor classification, risk assessment, and prognosis prediction
Expression analysis	Drug development, drug response, and therapy development
Mutation/Polymorphism analysis	Drug development, therapy development, and tracking disease progression

Proteomics

The dream of having genomes completely sequenced is now a reality. The complete sequence of many genomes including the human one is known. However, the understanding of probably half a million human proteins encoded by less than 30'000 genes is still a long way away and the hard work to unravel the complexity of biological systems is yet to come. A new fundamental concept called proteome (PROTEin complement to a genOME) has recently emerged that should drastically help to unravel biochemical and physiological mechanisms of complex multivariate diseases at the functional molecular level. Proteomics can be defined as *the qualitative and quantitative comparison of proteomes under different conditions to further unravel biological processes*.

Proteome refers to all proteins expressed by a genome. The importance of proteomics lies in the fact that while genome reamains unchanged to a large extent, the proteins in any particular cell change drastically as genes are turned on and off in the environment.Applications of proteomics are:

- New drug protein target identification
- Validation of targets
- Pharmaco-proteomics
- Comparison of normal vs. disease sample as well as vs. treated samples etc.

The most important application of proteomics is believed to be discovery of new disease markers and drug targets which can help us make drugs to diagnose and treat cancer, diabetes type 2 etc.

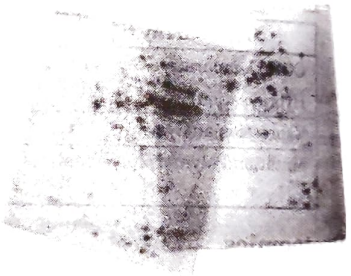
Studying Proteomics

Current research in proteomics requires first that proteins be resolved, sometimes on a massive scale. Protein separation can be performed using two-dimensional gel electrophoresis. Protein spots in a gel can be visualized using a variety of chemical stains or fluorescent markers. Proteins can often be quantified by the intensity of their stain. Once proteins are separated and quantified, they are identified, usually by in-gel digestion and subsequent mass spectrometry. For the in-gel digestion, individual spots are cut out of the gel and cleaved into peptides with proteolytic enzymes. These peptides are used for the identification of the protein by peptide mass fingerprinting or de novo sequencing. The peptide mass fingerprint relies on the specific pattern of peptide signals for a given protein in mass spectrometry, most often MALDI-TOF mass spectrometry. The pattern obtained in mass spectrometry is compared with database entries for the identification of the protein. Denovo sequencing of proteins uses electrospray ionization tandem mass spectrometry. Finally protein-protein interactions are studied using techniques like protein microarrys, yeast two hybrid assay etc.

Steps of a Typical Proteomics Experiment

1. **Protein Isolation :** this method basicaaly involves isolating the proteins from the cells.
2. **Protein purification :** The basic aim of purification is to get rid of unwanted substaces like detergents, chelating agents, reducing agents etc. A number of processes like dialysis, gel filtration, affinity chromatography and polyacrylamide gel electrophoresis can be used.
3. **Resolving proteins :** Proteins are resolved by the help of a technique called 2D PAGE/ two dimensional gel electrophoresis.It is a form of gel electrophoresis commonly used to analyze proteins. Mixtures of proteins are separated by two properties in two dimensions on 2D gels. 2-D electrophoresis begins with 1-D electrophoresis but then separates the molecules by a second property in direction 90 degrees from the first. The two dimensions that proteins are separated into using this technique can be isoelectric point, protein complex mass in the native state, and protein mass.

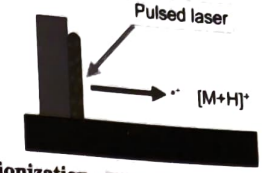
   To separate the proteins by isoelectric point is called isoelectric focusing(IEF). Thereby, a gradient of pH is applied to a gel and an electric potential is applied across the gel, making one end more positive than the other. The proteins applied in the first dimension will move along the gel and will accumulate at their isoelectric point. That is, the point at which the overall charge on the protein is 0 (i.e a neutral charge). After completion of the first dimension the complexes are destroyed by applying the denaturing SDS-PAGE in the second dimension, where the proteins of which the complexes are composed of are separated by their mass. Before separating the proteins by mass, they are treated with sodium dodecyl sulfate (SDS). The result of this is a gel with proteins spread out on its surface. These proteins can then be detected by a variety of means, but the most commonly used stains are silver, fluorescent and coomassie staining



2D-Gels (Coomassie stained)

4. **In-gel digestion :** The in-gel digestion primarily comprises of the four steps destaining, reduction and alkylation (R&A) of the cysteines in the protein, proteolytic cleavage(by trypsin) of the protein and extraction of the generated peptides. This step is the sample preparation step for mass spectrometric analysis.
5. **Peptide mass fingerprinting :** the unknown protein of interest is first cleaved into smaller peptides, whose absolute masses can be accurately measured with a mass spectrometer such as MALDI-TOF or ESI-TOF.[6] These masses are then *in silico* compared to either a database containing known protein sequences or even the genome.

   (*a*) MALDI –TOF : Matrix Assisted Laser Desorption /Ionization- Time Of Flight (MALDI-TOF) mass spectrometry has become a popular tool for large molecule analyses. Here a co-precipitate of an UV-light absorbing matrix and a biomolecule is irradiated by a nanosecond laser pulse. Most of the laser energy is absorbed by the matrix, which prevents unwanted fragmentation of the biomolecule. The ionized biomolecules are accelerated in an elctric field and enter the flight tube. During the flight in this tube, different molecules are separated according to their mass to charge ratio and reach the detector at different times. In this way each molecule yields a distinct signal. The method is used for detection and characterization of biomolecules, such as proteins, peptides, oligosaccharides and oligonucleotides, with molecular masses between 400 and 350,000 Da. It is a very sensitive method, which allows the detection of low (10^{-15} to 10^{-18} mole) quantities of sample with an accuracy of 0.1 - 0.01 %.



(*b*) ESI-MS : The **Electronspray ionization –mass spectrometery** instrument is the most adapted to satisfy the specific need in non-covalent and large supramolecular assemblies characterization. This mass spectrometer is also particularly well adapted for protein and peptide profiling and small molecule characterization. Electrospray ionization (ESI) is a technique used in mass spectrometry to produce ions. It is especially useful in producing ions from macromolecules because it overcomes the propensity of these molecules to fragment when ionized.

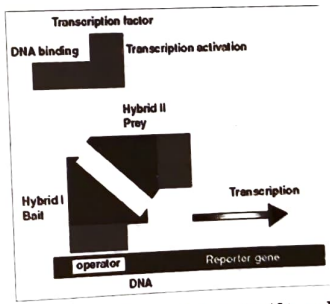
The first stage of tandem MS/MS isolates individual peptide ions, and the second breaks the peptides into fragments and uses the fragmentation pattern to determine their amino acid sequences. The sequences are used for a database search which gives satisfying results even with proteins not listed in the database by sequence similarity.

5. **Computational analysis :** The mass spectrometrical analysis produces a list of molecular weights which is often called peak list. The peptide masses are now compared to huge databases such as Swissprot, Genbank which contain protein sequence information. A comparison is made between the peak list of measured peptide masses and all the masses from the calculated peptides. The results are statistically analyzed.

Identifying Protein-protein & Protein-dna Interactions

Finding interactions between proteins involved in common cellular functions is a way to get a broader view of how they work cooperatively in a cell. **Protein-protein interactions** refer to the association of protein molecules and the study of these associations from the perspective of biochemistry, signal transduction and networks. There are a number of techniques to study these interactions.

1. **The Yeast two hybrid system**: the two-hybrid system is a molecular genetic tool which facilitates the study of protein-protein interactions. If two proteins interact, then a reporter gene (*e.g. gal1-lacZ* - the *beta*-galactosidase gene) is transcriptionally activated. The result is a colour reaction on specific media. This can be used to (1) study the interaction between two proteins which you expect to interact and/or (2) find proteins (prey) which interact with a protein you have already (bait).



2. Co-immunoprecipitation: A purification procedure to determine if two different molecules (usually proteins) interact. An antibody specific to the protein of interest is added to a cell lysis. Then the antibody-protein complex is pelleted usually using protein-G sepharose which binds most antibodies. If there are any protein/ molecules that bind to the first protein, they will also be pelleted. Identification of proteins in the pellet can be determined by western blot (if an antibody exist) or by sequencing a purified protein band.
3. Tandem affinity purification: The TAP method requires fusion of the TAP tag, either N- or C-terminally, to the target protein of interest. Starting from a relatively small number of cells, active macromolecular complexes can be isolated and used for multiple applications. Its simplicity, high yield, and wide applicability make the TAP method a very useful procedure for protein purification and proteome exploration.

There are a number of other methods for studying these interactions like: Fluorescence resonance energy transfer (FRET), Label transfer, Quantitative immunoprecipitation combined with knock-down (QUICK), SPINE (Strep-protein interaction experiment) etc.

Protein Microarrays

Protein arrays are rapidly becoming established as a powerful means to detect proteins, monitor their xpression levels, and investigate protein interactions and functions. protein microarray technology provides a robust way to study protein function in a rapid, economical, and system-wide fashion. Protein chips have the potential to function in many other applications including the study of protein–protein, protein–drug interactions, DNA-protein interactions, protein localization, antigen-antibody interactions, enzyme-substrate, and receptor-ligand interactions all of which may be amendable to array-type high-throughput screening.

Type of Protein Chips and Antibody Microarrays.

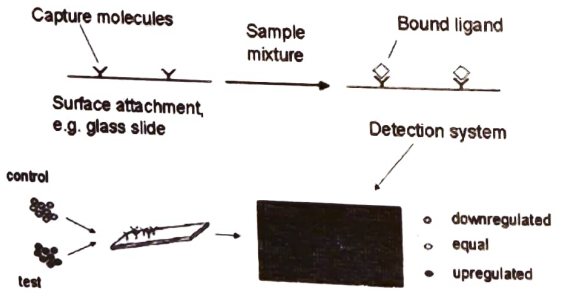
Protein Chips and Antibody Microarrays can be of two basic types: Glass Slides, Microwell/Nanowell. To attach proteins to a solid surface, the surface of the substrate has to be modified to achieve maximum binding capacity. The proteins are attached to the chip on a protein attachment layer. This layer is typically an organic film which varies with the nature of the application.

Protein microarray attachment methods

Requirements:	Strong binding, retention of function, high signal:noise, low background
Diffusion, e.g. polyacrylamide gel (Hydrogel), agarose	No protein modification, high capacity, low background, random orientation
Surface adsorption e.g. PVDF, Nitrocellulose Poly-L-lysine	Protein denaturation, high capacity, variable background, random orientation
Covalent e.g. Silanes (various) aldehyde	Protein modified, strong attachment, high sensitivity, random or uniform orientation
Affinity Biotin/Streptavidin, His/Ni-NTA	Protein modified, strong attachment, low background, random or uniform orientation

A mixture (e.g. of two tissue extracts) is applied to the array and the analytes of interest are captured by the specific ligand binders, followed by detection of binding. Similar to comparison of samples from normal and diseased tissues on DNA arrays or on 2D gels, reference and test samples can be labelled with Cy3 and Cy5 fluors, mixed, gel filtered to remove unbound dyes and then incubated on a chip of arrayed antibodies. Increased or decreased protein expression is assessed using a scanner and up- or down-regulated proteins can be identified from the ratios of the two dyes in the familiar 'traffic light' (red, yellow, green) system.

Antibody microarray principle



e.g. to compare protein expression, mix Cy3/Cy5 labelled samples

EXERCISE – I

MCQ TYPE QUESTIONS

1. Which of these is not a protein sequence database?
   (a) PIR (b) GenBank
   (c) PDB (d) COGs
2. Which of these is not a level of CATH?
   (a) Class (b) Homology
   (c) Topology (d) Alignment
3. A database of enzymatic pathways and biological chemicals is:
   (a) COGs (b) DDBJ
   (c) KEGG (d) SCOP
4. Human genome comprises of how many genes and base pairs?
   (a) ~37,544 and 3.3 x 108
   (b) ~20,500 and 3.9 x 109
   (c) ~19,300 and 3.3 x 109
   (d) ~19,300 and 3.9 x 108
5. Microsatellites belong to which category?
   (a) Transposons (b) Pseudogenes
   (c) Repeat elements (d) Junk DNA
6. Which of the following is not a step in Proteomics experiment?
   (a) Resolving (b) Purification
   (c) Digestion (d) None of these
7. Global Alignment uses _____ algorithm
   (a) Smith-Waterman algorithm
   (b) Needleman-Wunsch algorithm
   (c) Dot Plots
   (d) DALI
8. FASTA program was first described by
   (a) Lipmann and Pearson
   (b) Adachi and Hasegawa
   (c) Fitch and Margoliash
   (d) Kyte and Dolittle
9. BAC stands for
   (a) Bacteria Artifice Chromosome
   (b) Bacterial Artifice Chromatid
   (c) Bacterial Artificial Chromatid
   (d) Bacterial Artificial Chromosome
10. Which step is omitted in Whole Genome Sequencing?
    (a) Random shredding of genome
    (b) Making phage library
    (c) Computational analysis
    (d) None of these
11. To match protein query against translated (6 frame) nucleotide database you will use
    (a) TBLASTN (b) BLASTX
    (c) TBLASTX (d) BLASTP
12. Which stains can be used for staining gels in proteomics experiment?
    A. Coomasie B. Silver stain
    C. Fluorescent stain D. Radioactive stain
    (a) Only A & B (b) Only A & D
    (c) A, B & C (d) All of the above
13. Bioinformatics cannot analyse
    (a) Mathematical analysis
    (b) Statistical analysis
    (c) Biomedical analysis
    (d) Chemical analysis
14. Bioinformatics comprise of all except
    (a) DNA microarrays
    (b) Medical informatics
    (c) Agarose gel electrophoresis
    (d) Primer design
15. Bioinformatics research helps in
    (a) Forensic analysis (b) Dot plot hybridisation
    (c) Northern blotting (d) RFLP
16. Which of the following is not primary nucleotide sequence database?
    (a) DDBJ (b) OWL
    (c) GenBank (d) EMBL
17. GCG is
    (a) Protein sequencing tool
    (b) Compare two DNA or protein sequences
    (c) Compare multiple DNA or protein sequences
    (d) Nucleic acid sequencing tool
18. Clustal W is
    (a) Multiple sequence alignment tool
    (b) Protein secondary structure predicting tool
    (c) Data retrieving tool
    (d) Nucleic acid sequence analysis tool
19. Which is model organism database?
    (a) GOLD (b) SCOP
    (c) PROMISE (d) SGD
    (e) Yeast genome database at Stanford, USA
20. Which is data retrieving tool?
    (a) KEGG (b) EMBL
    (c) ENTREZ (d) PHD

21. Arrangement of α-helices and α-strands into discrete folding units is known as
(a) Primary structure (b) Secondary structure
(c) Super primary (d) Super secondary

22. Quaternary structure is
(a) Arrangement of separate moles, such as in protein-2 or protein nucleic acid interactions
(b) Arrangement of separate protein chains in a protein molecule with more than one subunit
(c) Overall fold of a protein sequence, formed by packing of its super secondary structure elements
(d) None of these

23. Motifs of protein sequences are
(a) Secondary databases
(b) Relational databases
(c) Primary databases
(d) Object oriented databases

24. Domains arise from
(a) Packing of 20 structure
(b) 30 structure
(c) Structural databases
(d) All of the above

25. BLASTX program is used for
(a) Translate DNA database
(b) Translate input sequence
(c) Translate both sequence
(d) Translate protein sequence

26. BLOSUM matrices are used for
(a) Multiple sequence alignment
(b) Phylogenetic analysis
(c) Pairwise sequence alignment
(d) All of these

27. BLAST and FASTA are used for
(a) Global similarity
(b) End free space alignment
(c) Local similarity
(d) Gap penalty

28. TFATSA program allows
(a) Protein query sequence to be compared to DNA databanks
(b) Protein query sequence to be compared to Protein databanks
(c) DNA query sequence to be compared to Protein databanks
(d) DNA query sequence to be compared to DNA databanks

29. One major difference between BLAST and FASTA is
(a) BLAST is word based method
(b) BLAST is local alignment tool
(c) BLAST can compare protein query sequence to protein databank
(d) It requires a pre-formatted search database

30. CLUSTAL family of the programs are used for
(a) Multiple sequence analysis
(b) Phylogenetic analysis
(c) Alignment of any sequence which reflect evolutionary history
(d) All of the above

31. In routine work, which language is used in bioinformatics
(a) Alphabetic-nucleotide
(b) Sentence-gene (euk)
(c) Words-gene (prok)
(d) All of these

32. Information of all known nucleotide and protein sequences are available on
(a) NCBI's Gene bank (b) DDBJ
(c) EMBL (d) All of these

33. PRINTS are software used for
(a) Detection of genes from genome sequence
(b) Identification of functional domains/motifs of proteins
(c) Detection of tRNA genes
(d) Prediction of function of a new gene

34. Phylogenetic relationship can be shown by
(a) Dendogram
(b) Data search tool
(c) Data retrieving tool
(d) Gene Bank

35. PDB is
(a) Composite database
(b) Databases for 2-D structure of biological macromolecule
(c) Primary database for macromolecules
(d) can be determined by gel electrophoresis

36. SCOP is
(a) Structural database, which identify structural and evolutionary relationships
(b) It is primary database
(c) SCOP database is a hierarchial classification of protein 2-D domain structures
(d) It is nucleotide sequence database

37. Structural genomics include
(a) Genetic mapping and physical mapping
(b) Physical mapping and sequencing
(c) Both (a) and (b)
(d) None of these

38. What is the primary goal of database?
(a) Minimizing data redundancy
(b) Achieving data independence
(c) Both (a) and (b)
(d) None of these

39. GeneBank and SwissProt are examples of
(a) Primary database (b) Composite database
(c) Secondary database (d) None of these

40. Archical databanks are also known as
(a) Primary database (b) Composite database
(c) Secondary database (d) None of these

41. Proteomics research can be categorized as
(a) Structural and Functional Prcteomics
(b) Structural, Functional and Comparative Proteomics
(c) Functional and Comparative Proteomics
(d) None of these

42. Expression of protein occur in
(a) Structural genomics
(b) Functional genomics
(c) Comparative genomics
(d) None of these

43. Chromosome evolution is a part of
(a) Structural genomics
(b) Functional genomics
(c) Comparative genomics
(d) None of these

44. BLAST's statistical theory was developed by
(a) Joseph Sambrook
(b) Samuel Karim and Steven Altschul
(c) Altschul et al
(d) None of these

45. Default format of file sequences present at GenBank is
(a) Fasta (b) Flat file format
(c) GCG format (d) XML format

46. Results of BLOSUM 62 nearly correspond to the results of
(a) PAM 160 (b) PAM 180
(c) PAM 250 (d) PAM 220

47. Which of the following BLAST is used for highly similar sequences?
(a) Mega BLAST (b) BLASTN
(c) BLASTP (d) None of these

48. Which of the following cannot be used for multiple alignments?
(a) Clustal W (b) Clustal X
(c) T-coffee (d) None of these

49. A BLAST like program which is used in Human Genome project for longer nucleotide sequence of > 95% similarity is
(a) BLAT (b) PSI BLAST
(c) BLASTZ (d) LAGAN

50. Which of the following can be concluded as result of gene duplication?
(a) Orthologs (b) Paralogs
(c) Xenologs (d) Homologs

51. Expasy stands for
(a) Expert Protein Analysis System
(b) Expressed Protein Analysis System
(c) Expression of Protein and Systematics
(d) Expressed Protein Analogy Server

52. All organisms that share a common ancestor belong to same
(a) Clade (b) Phylloclade
(c) Node (d) None of these

53. Which of the following is not a substitution model used in tree construction?
(a) Kimura (b) Jukes and Cantor
(c) Felsenstein (d) None of these

54. Tree evaluation is done by
(a) Boot strapping
(b) Permutation tests
(c) Likelihood ratio test
(d) All of these

55. In genome sequencing, which of the following is meant for calculating the quality of the sequence?
(a) Phred (b) Phrap
(c) Consed (d) None of these

56. Analytical microarrays are also known as
(a) Antibody microarrays
(b) DNA microarrays
(c) Functional microarrays
(d) All of these

57. Which of the following is a bibliographic database?
(a) Pubmed (b) OMIM
(c) GEO (d) Science

58. Which of the following is not the package of Phylogenetic analysis?
(a) Phylip (b) PAUP
(c) STAMP (d) None of these

59. The only character based tree building method among the following is
(a) NJ method
(b) Fitch Margoliash method
(c) Maximum likelihood method
(d) Minimum evolution mehod

60. Fasta format sequence file has which one of the following symbol as identifier?
(a) > (b) //
(c) : (d) #

61. Which of the following techniques are used for the determination of protein structure?
(a) X-ray crystallography
(b) NMR
(c) Both (a) and (b)
(d) None of these

62. Separation on mass spectrometry is based on
(a) Mass to charge ratio
(b) Charge to mass ratio
(c) Mass of the molecule
(d) None of these

63. Which of the following techniques can be used for charging the sample in MS?
(a) ESI (b) MALDI
(c) Both (a) and (b) (d) None of these

64. The first DNA genome sequenced is
(a) E.coli (b) Yeast
(c) Bacteriophage (d) Human

65. The study of gene products encoded by a genome and the time of their expression is known as:
(a) Genomics
(b) Proteomics
(c) Structural Genomics
(d) Genetic engineering

66. The Shot gun sequencing process work as
(a) The fragments are first aligned into contigs
(b) Random selected clones are selected until all clones in the genomic library are analyzed
(c) All the clones are selected serially in the genomic library
(d) All of these

67. Benefits of the genome sequencing
(a) They enable the determination of the complete genetic information present in the genome of various organisms
(b) Enable the relationships between genes
(c) They provide insight into genome organisation
(d) All of these

68. The one major reason for difference in genomic size in eukaryotes
(a) The presence of repetitive DNA
(b) Presence of more number of genes than prokaryotes
(c) Because of evolutionary hierarchy
(d) Because of conserved sequences

69. The single nucleotide polymorphism in beta-globin from Adenine to thymine leads to
(a) Emphysema
(b) Huntingtons disease
(c) Sickle cell anaemia
(d) Favism

70. The type of chip which is prepared by using cDNA is
(a) cDNA microarray (b) cDNA chip
(c) Both (a) and (b) (d) Genomic chip

71. The term used to describe the complement of mRNA transcribed from the genome of a cell
(a) Proteome
(b) Genome
(c) Functional genomics
(d) Transcriptome

72. Match the following

ORGANISMS	GENOME SIZE
I. E.coli	P. 3 x 109
II. S. Cerevisae	Q. 3 x 108
III. Human	R. 5 x 105
IV. Drosophila	S. 14 x 106

(a) I-Q; II-S; III-P; IV-R
(b) I-R; II-S; III-P; IV-Q
(c) I-R; II-P; III-S; IV-Q
(d) I-S; II-R; III-P; IV-Q

73. The achievements of microarray in the field of bioinformatics
(a) Statistical analysis
(b) Relation between probe and gene
(c) Data warehousing
(d) All of these

74. Single channel microarray is also known as
(a) One colour microarray
(b) One cDNA microarray
(c) Two cDNA microarray
(d) Two colour microarray

75. A disadvantage of chip-based assays is overcome by the use of
(a) Ball arrays (b) Seed shaped arrays
(c) Bead arrays (d) Disc shaped arrays

76. The qualitative and the quantitative comparison of proteomes under different conditions to further unravel biological process is
(a) Genomics
(b) Proteomics
(c) Structural genomics
(d) Functional genomics

77. The most common protein array is
(a) Cellular microarray
(b) Tissue microarray
(c) Antibody microarray
(d) Chemical compound microarray

78. An example of Protein functional array is
(a) Lysate array (b) Antibody array
(c) Capture array (d) Kinase array

79. DAPA is
(a) DNA array to protein array
(b) DNA array to programming array
(c) DNA array to pulse array
(d) DNA array to polymerase array

80. The set of parameters consisting of equilibrium bond angles, partial charge values, force constants is known as
(a) Variable (b) Force field
(c) Force (d) Molecular dynamics

81. Merck molecular force field is used to handle
(a) Bond angle (b) Bond length
(c) Functional group (d) Bond order

82. Consistent force field belongs to which class of force field?
(a) Class I (b) Class II
(c) Class III (d) Class IV

83. Proteome is larger than genome because of
(a) Alternative splicing
(b) Glycosylation
(c) Phosphorylation
(d) All of these

84. The measurement of the amount of mRNA in a sample that corresponds to a given gene is measured by
(a) Microarray
(b) Primer
(c) Probe
(d) Oligonucleotide sequence

85. The examination of the gene interactions with drugs is studied in
(a) Pharmacogenomics (b) Comparative genomics
(c) Chemogenomics (d) Structural genomics

86. I. The first complete nucleotide sequence of a viral RNA-genome (bacteriophage MS2) was discovered in 1976
II. The first DNA-genome project to be completed was the Phage ?-X174
III. The first bacterial genome to be completed was that of Haemophilus influenza.
Which among the following are true?
(a) I, II (b) II, III
(c) I, II, III (d) I, III

87. The study of the relationship of genome structure and function across different biological species or strains
(a) Structural genomics
(b) Functional genomics
(c) Comparative genomics
(d) Toxicogenomics

88. Genetic algorithms have been applied to
(a) NMR studies of proteins and peptides
(b) Conformational search problems
(c) Optimization of the overlap of pharmacophores
(d) All of these

89. NMR denotes
(a) Nuclear magnetic resonance
(b) Nuclear magnetic resource
(c) Nuclear molecular resonance
(d) Nuclear molecular resource

90. SAM indicates
(a) Significance analysis of microarrays
(b) Specific analysis of microarrays
(c) Signal analysis of microarrays
(d) Single analysis of microarrays

91. Microarray is
(a) 2D (b) 3D
(c) 1D (d) No dimension

92. CAMERA resource consist of the information about
(a) Eukaryotic genomics and metagenomics
(b) Microbes genomics and metagenomics
(c) Fish genome model
(d) Maize genomics and metagenomics

93. Wormbase deals
(a) Agrobacterium tumefaciens
(b) Caenorhabditis elegans
(c) Yeast
(d) Saccharomyces cerevisae

94. Method of determining the function of uncharacterized proteins translated from genomic or cDNA sequences is
(a) PIR (b) Pfam
(c) PROSITE (d) SWISS-MODEL

95. I. INTERPRO – Integrated documentation resource for protein families
II. In PSI-BLAST, PSI indicates position specific iterative
III. SCOP-denotes structural classification of proteins
Which of the above abbreviations are true?
(a) I, II (b) II, III
(c) I, III (d) all of these

96. I. Residue alignment score may be positive or negative
II. No gap penalty required
III. Score can increase or decrease or stay level between two cells of a pathway
Which of the above statements are true about Smith-Waterman algorithm?
(a) I, II (b) II, III
(c) I, III (d) I, II, III

97. Gap cost is calculated with the formulae in case if a is the opening gap cost, b is the extended gap cost and k is the gap length
(a) cost = a + bk (b) cost = b + ak
(c) cost = ka + b (d) cost = kb + a

98. Match the following

List – I	List – II
1. blastp	a. Translated nucleotide database translated nucleotide query
2. blastx	b. Protein database protein query
3. blastn	c. Protein database translated nucleotide query
4. tblastx	d. Nucleotide database nucleotide query

(a) 1-c; 2-b; 3-d; 4-a (b) 1-b; 2-c; 3-d; 4-a
(c) 1-b; 2-d; 3-c; 4-a (d) 1-a; 2-c; 3-d; 4-b

99. In CLUSTAL-W the GAPDIST indicates
(a) Penalty for gap separation
(b) Penalty for closing a gap
(c) Penalty for generating pair-wise gaps
(d) Penalty for multiple sequence gaps

100. Scores are derived from
(a) Mutation probability matrix
(b) Point associated matrix
(c) Position specific iterative
(d) Block substitution matrix

101. The BLOSUM 80 matrixes would be derived from a set of sequences having
(a) 80 % sequence identity
(b) 80 % sequence similarity
(c) 80 % sequence homology
(d) 80% sequence diversity

102. The closeness of sequence similarity is affected by
(a) Frames hit (b) Direct repeats
(c) Inverted repeats (d) All of these

103. The sequence which has remained essentially unchanged from ancestors is
(a) Conserved sequence (b) Common sequence
(c) Reserved sequence (d) Same sequence

104. I. Gene3D is a resource of form studying proteins and the component domains.
II. Gene3D takes the HMM's derived from SCOP
III. You can run FASTA files of up to 2.5 MB via the asynchronous server.
Which of the above sentences are true about Gene3D?
(a) I (b) II
(c) I, III (d) III

105. Which of the below sentences are true about FuncNet?
I. FuncNet brings together an ensemble of algorithms developed and hosted at various sites
II. FuncNet was funded by the European Union's EMBRACE and ENFIN networks.
III. FuncNet is an open architecture on which multiple prediction algorithms can be queried in parallel in order to provide higher-quality results.
(a) I, II (b) I, II, III
(c) II, III (d) I, III

106. I. All non-proteins, models, and structures with greater than 30% "C-alpha only" are excluded from CATH
II. The major levels of hierarchy : Class, Architecture, Topology (fold family)
III. The CATH database is a hierarchical domain classification of protein Structures in the Protein Data Bank
Which of the above sentences are true about CATH ?
(a) I, II
(b) I, II
(c) II, III
(d) I, II, III

107. Match the following

I. Horizontal line	P. Size of motif
II. Vertical line	Q. % identity
III. Height of yellow block	R. Identity
IV. Width of yellow blcok	S. Sequence

In BLAST Graphscan image, match the lines with their indications.

Code :

	I	II	III	IV
(a)	S	R	Q	P
(b)	S	P	Q	R
(c)	S	C	P	Q
(d)	R	S	Q	P

108. Which of the below sentences are true about Pfam ?
I. Pfam is a database of three parts
II. Pfam-B smaller families is taken from PRODOM
III. Pfam used to view domain organization of proteins
(a) I, II (b) I, II
(c) II, III (d) I, II, III

109. I. Print database is a compendium of protein motif fingerprints
II. The bulk of the database entries are of latter type
III. Composite fingerprints encode multiple motifs
Which of the above sentences are true about PRINTS ?
(a) I, II, III
(b) I, III
(c) II, III
(d) I, II

110. Match the following

Structures	Function
I. Domain	P. Set of motifs
II. Motif	Q. Independent structural unit
III. Fingerprint	R. Function of translated protein
IV. Prosite alignment	S. Conserved element of sequence

Code :

	I	II	III	IV
(a)	Q	S	P	R
(b)	S	Q	P	R
(c)	R	S	P	Q
(d)	Q	R	P	S

111. Match the following

Protein structure database	description
I. CATH	P. Database of comparative protein models
II. SCOP	Q. Server for protein structure models
III. SWISS-MODEL	R. Protein structure classification
IV. ModBase	S. Structure classification of protein

Code :

	I	II	III	IV
(a)	R	S	Q	P
(b)	Q	S	P	R
(c)	R	S	P	Q
(d)	S	R	Q	P

112. Match the following

Domain Database	Description
I. SignalP	P. Protein domain families
II. ProDom	Q. Server for signal peptide prediction
III. Pfam	R. University of California
IV. DPI	S. Sanger institutes

Code :

	I	II	III	IV
(a)	Q	P	S	R
(b)	S	P	Q	R
(c)	R	P	S	Q
(d)	Q	P	R	S

113. A Web server that integrates a database of allergenic proteins with various computational tools
(a) SDAP (b) PDB
(c) EBI (d) ProDom

114. In genetic algorithms, genetic operator is analogous to

(a) Reproduction (b) Biological crossover
(c) Both (a) and (b) (d) Transcription

115. Three categories : SSS (Sequence Search Services), MSA (Multiple Sequence Alignment) and BSA (Biological Sequence Analysis) was classified by

(a) EMBL (b) DDBJ
(c) EBI (d) NCBI

116. A computer algorithm that identifies similar protein 3-dimensional structures

(a) VecScreen (b) VAST
(c) CD search (d) CDART

117. A system for quickly identifying segments of a nucleic acid sequence that may be of vector origin

(a) VecScreen (b) BLAST
(c) Splign (d) Primer-BLAST

118. Tool for aligning a query sequence to GenBank sequences included on microarray

(a) e-PCR (b) ORF finder
(c) GEO-BLAST (d) 4Genome ProtMap

119. Prosplign is associated with

(a) Exons (b) Promoter
(c) Introns (d) All of these

120. A tool for comparing genomes on the basis of the protein sequences they encode

(a) Genome protmap (b) Dot plot
(c) Taxplot (d) BLAST

121. A public repository that archives and freely distributes tandem mass spectrometry peptide and protein identification data generated by the scientific community

(a) Peptidome (b) CD search
(c) Blink (d) Proteome

122. Which of the below sentences are true about sequence alignment ?

I. Each path represents an alignment
II. Only ONE possible transitions transitions are allowed from each cell
III. In the spreadsheet demonstration, second phase, of recursively tracing the paths through the table to find the high-scoring alignments, is done using spreadsheet formulas.

(a) I
(b) III
(c) II, III
(d) I, II, III

123. Which of the below sentences are true about artificial neural networks ?

I. For microarray gene expression data, the Pearson correlation coefficient is used to measure similarity between vectors of expression
II. An artificial neural network is a set of interconnected artificial neurons
III. Perceptron is an example of artificial neuron

(a) I, II (b) I, II, III
(c) II, III (d) I, III

124. Array Express is the micro array database setup by

(a) EMBL (b) EBL
(c) NCBI (d) SWISSPROT

125. Match the databases with their examples

Databases	Example
I. Structural database	P. PubMed
II. Mutant database	Q. BioCyc
III. Pathway database	R. ATIDB
IV. Literature database	S. PDB

Code :

	I	II	III	IV
(a)	S	R	Q	P
(b)	R	S	Q	P
(c)	S	R	P	Q
(d)	S	P	Q	R

126. DALI structural alignment method splits the input structure into

(a) Nanopeptide fragments
(b) Dipeptide fragments
(c) Hexapeptide fragments
(d) Pentapeptide fragments

127. In MAMMOTH-based structure alignment methods, the similarity score between two heptapeptides is calculated using a

(a) Unit-vector RMS method
(b) SSM
(c) RAPIDO
(d) SSAP

128. Sequerome is a web-based

(a) Sequence profiling tool
(b) Sequence alignment tool
(c) Structure prediction tool
(d) All of these

129. I. Stem loc is a program for Pairwise DNA structural alignment
II. Stem loc implements constrained versions of the Sankoff algorithms
III. T-Coffee (Tree-based Consistency Objective Function For alignment Evaluation) is a multiple sequence alignment

Which of the sentences are true about multiple sequence alignment ?

(a) I, II (b) I, III
(c) II (d) II, III

130. 74% of protein sequences have at least one match of Pfam. This number is called the

(a) Sequence homology (b) Sequence distance
(c) Sequence coverage (d) Sequence match

131. Match the following tools with their description

Tool	Description
I. PPsearch	P. Protein repeat detection
II. Phobius	Q. Discover patterns in unaligned protein sequence
III. Pratt	R. Predicts transmembrane topology
IV. Rader	S. Protein motif search

Code :

	I	II	III	IV
(a)	S	R	Q	P
(b)	R	S	B	P
(c)	S	R	P	B
(d)	S	P	Q	R

132. The good evidence of common ancestry for two sequences requires

(a) 5% identity only (b) 25% identity only
(c) 15% identity only (d) 35% identity only

133. Match the following

Words	Definition
I. Analogous	P. Divergent function
II. Homologous	Q. Conserved function
III. Orthologous	R. Convergent evolution
IV. Paralogous	S. Common ancestry

Code :

	I	II	III	IV
(a)	R	S	Q	P
(b)	Q	S	P	R
(c)	S	R	Q	P
(d)	R	S	P	Q

134. Which of the following sentences are true ?

I. Local alignment is not critical
II. Insertion and deletion at the end of alignment contribute a weight of zero
III. In biological application global alignment is far more meaningful than local similarity

(a) I, II, III (b) I, II
(c) II (d) II, III

135. Which of the following sentences are true ?

I. FASTA uses the dynamic programming algorithm to compute optimal alignment
II. FASTA doesn't miss cases of significant homology
III. BLAST focuses on no gap alignments of a certain fixed length

(a) I, II, III (b) I, III
(c) II, III (d) III

136. Scoring matrices used for computing alignment scores are based on observed

(a) Substitution rates
(b) Distance
(c) Transition and transversion
(d) Insertion and deletion

137. In the scoring matrix, score reflects

(a) Distance between the sequence
(b) Frequency of occurrence of aminoacid in nature
(c) Number of gaps
(d) Number of matches and mismatches

138. PAM 250 indicates

(a) 250% mutation (b) 250% similarity
(c) 7.5% similarity (d) 7.5% mutation

139. In scoring matrix the disadvantages are

I. Evolutionary rates doesn't very within a sequence
II. Environment changes over evolutionary time
III. Each position has a unique 3D environment

Which of the statements are true disadvantages of scoring matrix ?

(a) I, II (b) II, III
(c) I, III (d) I, II, III

140. Match the PAM to BLOSUM equivalent

PAM	BLOSUM
I. PAM100	P. BLOSUM 80
II. PAM120	Q. BLOSUM90
III. PAM160	R. BLOSUM52
IV. PAM200	S. BLOSUM60

Code :

	I	II	III	IV
(a)	Q	P	S	R
(b)	R	P	S	Q
(c)	Q	R	S	P
(d)	Q	S	P	R

141. I. Specificity is the fraction of the true matches that are correctly predicted as matches by the pattern

II. Sensitivity is the fraction of the sequence predicted as matches that really are true matches

III. Sensitivity is used to report all homologous hits

Which of the above sentences are true about sequence dtection efficiency ?

(a) III (b) II, III

(c) I, II (d) I, II and III

142. Match the parameters with their description.

PARAMETER	DESCRIPTION
I. True positives	P. Not reported and homologous
II. False positives	Q. Not reported and not homologous
III. True negatives	R. Not homologous
IV. False negatives	S. Homologous

Code :

	I	II	III	IV
(a)	S	R	Q	P
(b)	R	S	Q	P
(c)	S	P	Q	R
(d)	S	R	P	Q

143. I. Local alignment need to use gaps where as global alignment doesn't need to use

II. MSA can be looked as an extension of the Pairwise alignment

III. The first step in MSA is Pairwise alignment of all sequences

Which of the above sentences are true ?

(a) I, II, III (b) II, III

(c) I, III (d) III

144. Which of the following sentences are true about sum pair method ?

I. Sum of pair's method is a dynamic programming method

II. You need to align three or more sequences in sum of pair's method

III. The SP function scores each position in the nucleic acid

(a) I, II, III (b) II, III

(c) I, II (d) III

145. Arrange the following steps in proper order in process working of phylogenetic tree.

The phylogenetic tree process works as follows

I. Align the sequences in an orderly fashion

II. Compute a tree by single linkage clustering by using methods like UPGMA OR nearest neighbour

III. Compute all Pairwise distances between given sequences

(a) I, II, III (b) II, III, I

(c) III, II, I (d) I, III, II

146. Match the node with the taxonomic unit in a tree.

Node	Representing in tree
I. Branch	P. Number of changes in branch
II. Topology	Q. Common ancestor
III. Root	R. Branching pattern
IV. branch length	S. Difference between decent and ancient

Code :

	I	II	III	IV
(a)	S	R	Q	P
(b)	Q	R	S	P
(c)	R	S	Q	P
(d)	S	R	P	Q

147. Which of the above sentences are true with their functions ?

I. A dendogram reflects evolutionary relationship

II. Cladistic methods use each alignment position as evolutionary information to build a tree

III. In distance method insertion and deletion are given lesser weight than replacements

(a) I, II (b) II, III

(c) II (d) III

148. I. Neighbour joining method is a slowest tree building method

II. Neighbour joining method uses empirical substitution scoring method

III. Neighbour joining method tests a single tree

Which of the above sentences are true about neighbour joining method ?

(a) I, II (b) II, III

(c) II, III (d) I, III

149. Match the following parsimony with their occurence.

Parismony	Occurs
I. Camin-sokal	P. (1 –> loss of an allele >> 0 - 1)
II. Dollo	Q. (1 – 0 = 0 – 1)
III. Wagner	R. (1 – 0 >> 0 –)
IV. Polymorphism	S. (0 – 1)

Code :

	I	II	III	IV
(a)	S	P	Q	R
(b)	S	R	Q	P
(c)	S	R	P	Q
(d)	R	S	Q	P

150. Which of the following sentences are true about Jacknife ?

I. If you get a different tree with each sample, then you can conclude that no tree is strongly supported by the data

II. Jacknife resampling has largely been replaced by bootstrap resampling

III. In Jacknife the sub-replicates are of a larger size than the original data set

(a) I, II (b) II, III

(c) III (d) I, III

151. Which of the following sentences are true about phylogenetic analysis ?

I. Maximum parsimony uses only relevant sites

II. Distance methods are powerful than character state approaches

III. The results of maximum parsimony drastically change for its sensitivity to codon bias and unequal rates of evolution

(a) I, II

(b) I, III

(c) II, III

(d) I, II, III

152. General time reversible models include

(a) Transition and transversion

(b) Insertion and deletion

(c) CLADE and Node

(d) BLAST and FASTA

NUMERICAL TYPE QUESTIONS

1. According to jukes-cantor model assumption, the frequency of each base in DNA sequences

2. The Kimura method which distinguishes AT/GC transversion is of

3. In $D = -\ln(1 - P - 0.2\,p^2)$ the value of p should not exceed

EXERCISE – II

(QUESTIONS FROM PREVIOUS GATE EXAMS)

MCQ TYPE QUESTIONS

2016

1. Which one of the following is **NOT** an algorithm for building phylogenetic trees?

(a) Maximum parsimony

(b) Neighbor joining

(c) Maximum likelihood

(d) Bootstrap

2015

2. Identify the file format given below:

>Pl; JMFD

Protein X – *Homo sapiens*

MKALTARQQEVFDLIRDHISRTLRQQGDWL

(a) GDE

(b) FASTA

(c) NBRF

(d) GCG

3. How many rooted and unrooted phylogenetic trees, respectively, are possible with four different sequences ?

(a) 3 and 15 (b) 15 and 3

(c) 15 and 12 (d) 12 and 3

2014

4. The algorithm for BLAST is based on

(a) Dynamic Programming

(b) Hidden Markov Model

(c) k-tuple analysis

(d) Neural Network

5. The amino acid substitution matrices in decreasing order of stringency for comparing protein sequences are

(a) PAM250, PAM120, PAM100

(b) PAM100, PAM120, PAM250

(c) PAM250, PAM100, PAM120

(d) PAM120, PAM250, PAM100

2013

6. The suitable substitution matrix to align closely related sequences is

(a) PAM 250 or BLOSSUM 80
(b) PAM 40 or BLOSSUM 80
(c) PAM 120 or BLOSSUM 40
(d) PAM 250 or BLOSSUM 40

7. Determine the correctness or otherwise of the following **Assertion (A)** and **Reason (R)**.

Assertion: UPGMA method produces ultrametric tree.

Reason: Sequence alignment is converted into evolutionary distances in UPGMA method.

(a) Both (A) and (R) are true and (R) is the correct reason for (A)
(b) Both (A) and (R) are true and (R) is not the correct reason for (A)
(c) (A) is true but (R) is false
(d) (A) is false but (R) is true

8. Match the entries in the **Group I** with the entries in **Group II.**

Group I	Group II
P. Threading	1. Gene duplication
Q. FASTA	2. Fold prediction
R. Profile	3. HMM
S. Paralogs	4. k-tuple

(a) P-2, Q-1, R-3, S-4
(b) P-2, Q-4, R-3, S-1
(c) P-3, Q-4, R-2, S-1
(d) P-1, Q-4, R-3, S-2

2012

9. An example of a derived protein structure database is

(a) Pfam (b) SCOP
(c) GEO (d) Prosite

10. An example of a program for constructing a phylogenetic tree is

(a) phylip (b) phrap
(c) prodom (d) PHDsec

11. Synteny refers to

(a) gene duplication from a common ancestor
(b) a tree representation of related sequences
(c) the extent of similarity between two sequences
(d) local conservation of gene order

12. While searching a database for similar sequences, E value does NOT depend on the

(a) sequence length
(b) number of sequences in the database
(c) scoring system
(d) probability from a normal distribution

2011

13. The most widely used program for multiple sequence alignment is

(a) BLAST
(b) FASTA
(c) CLUSTAL
(d) Chime

2010

14. Program used for essentially local similarity search is

(a) BLAST
(b) RasMol
(c) ExPASY
(d) SWISS-PROT

15. Accession number is a unique identification assigned to a

(a) single database entry for DNA/Protein
(b) single database entry for DNA only
(c) single database entry for Protein only
(d) multiple database entry for DNA/Protein

2009

16. The method used for prediction of three dimensional structure of a protein from known structure(s) of one or more related proteins is

(a) Multiple sequence alignment
(b) Homology modeling
(c) Phylogeny
(d) Docking

17. Match the item in Group I with an appropriate description in Group 2:

Group 1	Group 2
P. UPGMA	1. Protein sequence database
Q. CLUSTAL	2. Phylogenetic analysis
R. SWISS-PROT	3. 3-D structure visualization
S. RasMOl	4. Multiple sequence alignment

(a) P-4, Q-1, R-2, S-3
(b) P-2, Q-4, R-1, S-3
(c) P-2, Q-3, R-1, S-4
(d) P-2, Q-1, R-4, S-3

2008

18. In bioinformatics, the term 'BLAST' refers to

(a) database retrieval tool
(b) computational tool for sequence homology searching and alignment
(c) computational tool to view genomic sequences
(d) computational tool to view protein structures

19. Match the terms in group 1 with their possible explanations in group 2

Group 1	Group 2
P. Orthologs	1. A cell or an organsim having foreign gene
Q. Paralogs	2. The complement of a protein expressed by a genome
R. Proteome	3. Genes from different spe-cies related to each other
S. Transgenic	4. Genes from same species related to each other

	P	Q	R	S
(a)	2	4	1	3
(b)	4	3	2	1
(c)	3	4	2	1
(d)	1	2	3	4

2007

20. Match the methods available on world wide web in group 1 for performing the jobs listed in group 2

Group 1	Group 2
P. Boxshade	1. Searching family data base
Q. BCM launcher	2. Finding alignments
R. Prosite	3. Displaying alignments
S. PSI-BLAST	4. Searching for multiple alignments

(a) P-1, Q-3, R-2, S-4
(b) P-2, Q-3, R-2, S-4
(c) P-3, Q-4, R-1, S-4
(d) P-3, Q-2, R-1, S-4

2006

21. For prediction of three dimensional structure of protein

(P) homology modeling tries many possible alignments
(Q) threading first identifies homologues
(R) threading evaluates many rough models
(S) homology modeling optimizes one model

(a) Q, S (b) P, Q
(c) R, S (d) Q, R

NUMERICAL TYPE QUESTIONS

2015

1. The determinant of the matrix $\begin{bmatrix} 3 & 0 & 0 \\ 2 & 5 & 0 \\ 6 & -8 & -4 \end{bmatrix}$ is ____.

2. Consider the following infinite series :

$1 + r + r^2 + r^3 + ... \infty$

If $r = 0.3$, then the sum of this infinite series is ____.

3. Consider the following multiple sequence alignment of four DNA sequences.

A	C	T	A
A	C	T	G
A	G	T	C
A	G	C	T

Shannon's entropy of the above alignment is ____

4. An isolated population on an island has the following genotypic frequencies :

Genotype	AA	Aa	aa
Frequency	0.3	0.4	0.3

Assuming that there are only two alleles (A and a) for the gene, the genotypic frequency of AA in the next generation will be _____.

2014

5. In an affine gap penalty model, if the gap opening penalty is -20, gap extension penalty is -4 and gap length is 8, the gap score is _____.

ANSWERS

EXERCISE – I

MCQ Type Questions

1. (b)	2. (d)	3. (c)	4. (c)	5. (c)	6. (c)	7. (b)	8. (a)	9. (d)	10. (a)
11. (a)	12. (d)	13. (d)	14. (c)	15. (b)	16. (b)	17. (d)	18. (a)	19. (e)	20. (c)
21. (d)	22. (b)	23. (a)	24. (c)	25. (b)	26. (a)	27. (c)	28. (c)	29. (c)	30. (d)
31. (a)	32. (d)	33. (b)	34. (a)	35. (c)	36. (c)	37. (c)	38. (c)	39. (a)	40. (c)
41. (b)	42. (b)	43. (c)	44. (b)	45. (b)	46. (c)	47. (a)	48. (d)	49. (b)	50. (d)
51. (a)	52. (a)	53. (c)	54. (a)	55. (b)	56. (d)	57. (a)	58. (c)	59. (c)	60. (a)
61. (c)	62. (a)	63. (c)	64. (c)	65. (b)	66. (b)	67. (d)	68. (a)	69. (c)	70. (c)
71. (d)	72. (b)	73. (d)	74. (a)	75. (c)	76. (b)	77. (c)	78. (d)	79. (a)	80. (b)
81. (c)	82. (b)	83. (d)	84. (a)	85. (a)	86. (c)	87. (c)	88. (d)	89. (a)	90. (a)
91. (a)	92. (c)	93. (b)	94. (c)	95. (d)	96. (c)	97. (a)	98. (b)	99. (a)	100. (a)
101. (a)	102. (d)	103. (a)	104. (c)	105. (b)	106. (b)	107. (a)	108. (c)	109. (a)	110. (a)
111. (a)	112. (a)	113. (a)	114. (c)	115. (c)	116. (b)	117. (a)	118. (c)	119. (c)	120. (c)
121. (a)	122. (a)	123. (b)	124. (b)	125. (a)	126. (c)	127. (a)	128. (a)	129. (d)	130. (c)
131. (a)	132. (b)	133. (a)	134. (c)	135. (b)	136. (a)	137. (b)	138. (a)	139. (b)	140. (a)
141. (a)	142. (a)	143. (b)	144. (c)	145. (c)	146. (a)	147. (c)	148. (b)	149. (b)	150. (a)
151. (b)	152. (a)								

Numerical Type Questions

1. 0.25 2. 3 parameter 3. 0.75

EXERCISE – II

MCQ Type questions

1. (d)	2. (c)	3. (b)	4. (c)	5. (b)	6. (b)	7. (b)	8. (b)	9. (b)	10. (a)
11. (d)	12. (d)	13. (c)	14. (a)	15. (a)	16. (b)	17. (b)	18. (b)	19. (c)	20. (d)
21. (c)									

Numerical Type Questions

1. (–60) 2. (1.4285) 3. (3.80-3.82 bits) 4. (0.25) 5. (– 52 to – 52)

EXPLANATIONS

EXERCISE – I

MCQ TYPE QUESTIONS

1. GenBank is a nucleotide database having information about the sequences of DNA fragments as well as genomic DNA in FASTA format. Protein information repository (PIR), Protein Data Bank (PDB) and Cluster of Orthologous Genes (COG) are protein databases.
2. CATH classification of proteins comprises of Class, Architecture, Topology and Homology as the four different levels of identification of various proteins.
3. Kyoto's Encyclopaedia of Genes and Genomes (KEGG) is a database for metabolic pathways and molecules of biological nature. Most metabolic pathways are enzymatic in nature and the database comprises of information on the biochemistry of the enzymes, substrates, intermediate molecules and all other chemicals (some may be inhibitory compounds) that are related or influencing the metabolic construction.
4. As a result of the Human Genome Project concluded in 2001, major achievements and significant features of the human genome came to the fore. The human genome comprises of approximately 20,000 genes in total which have functional significance. Overall there are around 3.2 billion base pairs present in a human genome of which 98 % are repeats and only 2% genes are coding for functional proteins.
5. Microsatellites are repeat elements found in the genome. They belong to the category of simple sequence repeats or tandem repeats. The repeats are mainly of 2-6 bp long and mostly the repeat elements are dinucleotides such as (AG), (GT), (CG) etc. Which are repeated a number of times within the genome. They arise mainly due to errors in replication such as replication slippage, mispairing of nucleotides etc.

6. Digestion is a step not involved in proteomics experimentation. It is mainly involved in genomics study for the understanding of genes and genomes of various organisms.
7. Global alignment uses Needleman-Wunsch algorithm for the purpose of pair-wise alignment of sequences globally. The algorithm involves a match score, a mis-match score and a gap score. Gap scores are determined using gap penalties to obtain a good alignment result.
8. FASTA program was first developed by William Pearson and his co-workers in the 1980s.
9. BAC is Bacterial artificial chromosome which is a very good vector for carrying large inserts. BAC vector is used in whole genome sequencing strategies for the purpose of generation of library of reads from the genome of a complex higher order organism.
10. Random shredding and cleaving of the entire genome is not followed because breaking a large genome of a higher organism into smaller fragments randomly would result in the formation of a number of overlapping reads having repeats in them. This would result in a lot of ambiguity.
11. When the database to be searched using a query is of a nucleotide and the query sequence is of a protein, it is better to search the nucleotide database using a translated function that would convert the amino acid sequences into their respective nucleotide codes. This translation is achieved through tBLASTn where the protein query is translated into a nucleotide sequence and then searched against a nucleotide database.
12. In any proteomics study, Coomasie Brilliant Blue (CBB), Silver stain (Ag), Fluorescent stain (Horse Radish Peroxidase) and Radioactive stain (used in Autoradiography) are used for various staining purposes based on the need of the experiment.
13. Chemical analyses are achieved through chemical reactions undergone in vitro under sterile and artificial conditions. Bioinformatics study mainly revolves around the computational approach towards studying and analysing the various processes of life.
14. Bioinformatics is majorly and strictly restricted to computational study of life and living organisms involving various mathematical and statistical tools for the purpose of achieving meaningful conclusions. But Agarose gel electrophoresis involves wet lab analysis of macromolecules mainly DNA for separation according to molecular mass and size.
15. Dot plot hybridization is mainly involved in analysis of sequence alignment of a pair of residues and revealing the relationship in terms of phylogeny. Research in bioinformatics enables a lot of help.
16. OWL is a secondary database. It is not a primary nucleotide database. EMBL, DDBJ and GenBank are primary nucleotide databases.
17. GCG is a nucleic acid sequencing tool used to sequence DNA and RNA sequences of organisms.
18. Clustal W is a multiple sequence alignment tool. It takes input in the form of FASTA for query sequences and returns alignment result. The alignment results are in the form of a text file showing base by base matches of the sequences. Also, cladogram and phylogram are generated which are useful for phylogenetic relationships.
19. For model organisms, yeast genome database is a database that is present in Stanford, USA.
20. Entrez is a data retrieving tool of NCBI (National Centre for Biotechnology Information). It is like a search engine that displays links to various tools and databases against a particular query. It is a gateway that allows text-based searches for a wide variety of data, including annotated genetic sequence information, structural information, as well as citations and abstracts, full papers, and taxonomic data.
21. α-helices and α-sheets are secondary structures of proteins. Arrangement of these secondary structures into fine and discrete folding units giving rise to folds or turns within a secondary structure of protein forms super secondary structure.
22. Quaternary structures are formed from intermolecular hydrogen bonding of various secondary structures or chains of protein giving rise to a 3-dimensional structure of protein. The main energy of stabilisation of this structure is through hydrophobic bonding. The segments forming the quaternary structures are known as subunits.
23. Databases for storage of motifs, profiles and folds or threads come under secondary databases. Secondary databases contain computationally processed or manually curated information, based on original information from primary databases. Translated protein sequence databases containing functional annotation belong to this category.
24. Domains are certain regions within the protein structures that are formed through conservation of sequences through evolution. Domains can be ascertained from regions of homology in sequences of various organisms of related species or genera. These can be found out from structural databases of proteins. Because, structural databases would assign and annotate functional relevance to sequences through structure prediction and so the domains can be visualized.
25. BLASTX uses nucleotide sequences as queries and translates them in all six reading frames to produce translated protein sequences, which are used to query a protein sequence database.

26. The BLOSUM matrices are actual percentage identity values of sequences selected for construction of the matrices. For example, BLOSUM32 indicates that the sequences selected for constructing the matrix share an average identity value of 62%. Other BLOSUM matrices based on sequence groups of various identity levels have also been constructed. In the reversing order as the PAM numbering system, the lower the BLOSUM number, the more divergent sequences they represent.

27. Local alignments of sequences are done using BLAST and FASTA. Local alignment of sequences is done through selecting regions within the sequences and aligning them locally to find out the similarity and identity profile of the two sequences.

28. TFASTA is a translated FASTA program that is used to analyze DNA query sequences against protein databases for finding out various functional and structural features of the nucleic acid sequence under question.

29. BLAST uses a substitution matrix to find matching words, whereas FASTA identifies identical matching words using the hashing procedure. By default, FASTA scans smaller window sizes. Thus, it gives more sensitive results than BLAST, with a better coverage rate for homologs. However, it is usually slower than BLAST. The use of low-complexity masking in the BLAST procedure means that it may have higher specificity than FASTA because potential false positives are reduced. BLAST sometimes gives multiple best-scoring alignments from the same sequence; FASTA returns only one final alignment.

30. CLUSTAL family of programs are responsible for multiple sequence alignment of various nucleotide and protein sequences. There are two versions of CLUSTAL- CLUSTAL W (online tool) and CLUSTAL X (stand-alone tool). Both phylograms and cladograms can be generated using CLUSTAL which enables tracing of evolutionary relatedness and ancestral history.

31. English alphabets are used in bioinformatics to an5alyze and denote nucleotides and protein sequences and data.

32. GenBank of NCBI, DDBJ and EMBL are primary databases containing information on sequences, an7notation and all other details of nucleotides and proteins. They are the major information repository in the world.

33. PRINTS is a protein fingerprint database containing ungapped, manually curated alignments corresponding to the most conserved regions among related sequences. This program breaks down a motif into even smaller non-overlapping units called fingerprints, which are represented by unweighted PSSMs.

34. Dendogram: is a type of tree showing phylogenetic relationships between organisms based on their sequences. Dendograms are constructed based on the rate of conservation of 0.5 rRNA in the organisms.

35. Protein Data Bank (PDB) is a primary redundant database for macromolecules, primarily proteins containing major information and details about structure, function, molecular significance, ligand binding sites etc.

36. SCOP is structural classification of proteins database for the characterisation of secondary structures of proteins. It is divided into four main classes of class, architecture, topology and homology. Mainly the α helices and the β–sheets of proteins are characterised in this database.

37. Structural genomics involve studying the genetic constitution of organisms and the significance of the structural basis of genes and genomes. Thus it involves genetic mapping, physical mapping and sequencing for integrated and systemized study.

38. The chief objective of the development of a database is to organize data in a set of structured records to enable easy retrieval of information. Also, limiting repetition of data or data redundancy and achieving data independence are essential for a functional database.

39. Both SwissProt and GeneBank are examples of primary databases that store primary information obtained through various experimental protocols.

40. Archical databanks store information about various phylogenetic and evolutionary relationships established through various tools and databases. This secondary information is responsible for incorporation of archical data in secondary database.

41. Research in the field of proteomics can be majorly categorised into structural, functional and comparative proteomics because of the wide range of study involved in studying the proteome of an organism.

42. Protein expression occurs when genome translates into a proteome to give functional aspect to the nucleotide sequences. So protein expression occurs in functional genomics.

43. Evolution is a gradual and constantly changing process that ceases to stop. Studying evolution in any respect requires comparative study amongst various organisms to ascertain traits of similarity and dissimilarity. Comparative genomics is an approach to study chromosome and DNA evolution.

44. The BLAST program was developed by Stephen Altschul of NCBI in 1990 and has since become one of the most popular programs for sequence analysis. The statistical theory encircling BLAST was developed by Samuel Karim and Steven Altschul.

45. GenBank is a relational database. However, the search output for sequence files is produced as flat files for easy reading. The resulting flat files contain three sections – Header, Features, and Sequence entry.

46. PAM 250 and BLOSUM 62 have correspondingly similar results.

47. If two sequences are highly similar in terms of percentage of the same nucleotide bases, then Mega BLAST is used for obtaining local similarity search and alignment.

48. CLUSTAL W, CLUSTAL X, and T-coffee are involved in multiple sequence alignment of various sequences fed as input query.

49. T is a fast local alignment algorithm that aligns short sequences of forty bases with more than 95% similarity. It produces a graphical output that shows the sequence percent identity. This BLAST like program was used in human genome project for the purpose of achieving better similarity search results for closely related species.

50. Homologs are a result of gene duplication events. Some homologs result in orthologs and others result in paralogs.

51. ExPASY (www.expasy.ch/tools/) is a comprehensive proteomics web server with a suite of programs for searching peptide information from the SWISS-PROT and TrEMBL databases.

52. Clade is a group of taxa on a phylogenetic tree that are descended from a single common ancestor. They are also referred to as being monophyletic.

53. Jukes Cantor and Kimura model are the models used for tree construction based on substitution method.

54. Boot strapping is a method for evaluation of newly constructed trees.

55. Phrap (www.phrap.org/) is a UNIX program for sequence assembly. It takes Phred base-call les with quality scores as input and aligns individual fragments in a pairwise fashion using the Smith–Waterman algorithm. The base quality information is taken into account during the pairwise alignment.

56. Analytical microarrays involve antibody microarrays, DNA microarrays and functional microarrays for the purpose of analysis of RNA gene expression.

57. Pubmed is a literature database that is used for the purpose of research articles, literature review articles and research journals along with other research documentations.

58. STAMP (www.compbio.dundee.ac.uk/Software/Stamp/stamp.html) is a UNIX program that uses the intermolecular approach to generate protein structure alignment. The main feature is the use of iterative alignment based on dynamic programming to obtain the best superposition of two or more structures. So it is not a phylogenetic analysis tool.

59. Maximum likelihood method is a character based method for the purpose of tree building.

60. FASTA format has a special identifier which characterizes it. This is the '>' sign that is present before the sequence file.

61. Both X-ray crystallography and NMR are used for the determination of 2-D and 3-D structures od protein from their crystal structure prepared in vitro.

62. The main principle of separation of protein and peptides in a mass spectrometer is on the basis of mass to charge ratio.

63. Both Electro Spray Ionization (ESI) and Matrix Assisted Laser Desorption Ionization (MALDI) are used for charging of samples and making them ready for subjecting to high energy magnetic fields.

64. Bacteriophage was the first living organism whose genome was sequenced.

65. Proteome refers to the entire set of expressed proteins in a cell. In other words, it is the full complement of translated product of a genome. Proteomics is simply the study of the proteome. More specifically, it involves simultaneous analyses of all translated proteins in a cell. It encompasses a range of activities including large-scale identification and quantification of proteins and determination of their localization, modifications, interactions, and functions.

66. The shot gun strategy of whole genome sequencing involves random selection of clones from phage library and sequencing those clones to generate reads and contigs for obtaining results of whole genome sequence reads.

67. Genome sequencing is an essential methodology for the study of genes, their functions and expressions, structural significance and also medical relevance.

68. The major difference between size of genome between prokaryotes and eukaryotes is due to the high repeat content of eukaryotic genomic DNA as compared to the highly conserved and unique prokaryotic genome.

69. Sickle cell anaemia is an example of single nucleotide polymorphism resulting in mutation in the 5th amino acid, beta-globin changing the base from adenine to thymine and so glutamic acid is replaced by valine.

70. cDNA overlaid on a grid surface to be used as a chip in microarray technology is called cDNA microarray and the chip is known as cDNA chip.

71. Transcriptome is the complement of mRNA obtained through transcription of genome in a cell.

72. The genome sizes are E. coli: 5 x 105 bp

    S. cerevisae: 14 x 106 bp

    Human: 3 x 109 bp

    Drosophila: 3 x 108 bp

73. Microarray technology has contributed majorly in the field of bioinformatics in terms of statistical development, data warehousing and mining, relationship between probe and gene and chip based advancements with DNA and mRNA.

74. Single channel microarray or single chip microarray is also known as one dimensional microarray and gives intensity in a single colour and so is known as one colour microarray.

75. Glass or gel based beads are formed to overlay DNA or RNA so as to enhance the efficiency of interaction of the probe with the gene. This fixes the problem of chip based arrays.

76. Both qualitative and quantitative analysis and study of proteomes under different conditions to achieve better understanding of metabolic pathways and functions of proteins is involved in proteomics.

77. Antibody-antigen relationship is the most common way of ascertaining immunological tests to find out any kind of antigenic diseases. So, the most common way of ascertaining this is to build an in situ protein array with coating of antibody on it. This is called antibody microarray.

78. Kinase array is a type of functional array because kinase is an enzyme that asserts functional relevance to proteins under study.

79. DAPA is DNA array to Protein Array and this is done because DNA molecules have higher shelf life than protein molecules and so the longevity of the chip could be enhanced if a protein chip which can be obtained through transcription and translation of DNA in situ could be formed.

80. Force field are set of parameters and functions that comprises of bond lengths, bond angles, force constants, partial charges and equilibrium constants.

81. Functional groups of proteins and other macromolecules are handled using Merck molecular force field functions.

82. Class II force fields comprise of Consistent force field functions.

83. A proteome is always larger than a genome because the complement of genome formed results in polyadenylation, phosphorylation, glycosylation and alternative splicing leading to a larger size.

84. Microarray is the technique of measurement of amount of gene in a given sample of gene transcript consisting of complements mRNA.

85. Pharmacogenomics is the study of various interactions of genes and genomes with various ligands showing drug ability and drug viability.

86. All three are correct.

87. Comparative genomics is the study of inter-relationships between genomes of various species and organisms depicting structure, function and physiology of the organisms.

    Genetic algorithms have been applied to a variety of chemical problems including of distance constraints, systematic searching of conformational space for large structures

88. Genetic algorithms are at the forefront of bioinformatics study. They are involved in NMR study of macromolecules, pharmacophore optimization and searching conformers of various proteins.

89. NMR is the abbreviation for Nuclear magnetic resonance.

    SAM uses permutations of repeated measurements to estimate the False Discovery Rate

90. SAM is the significance analysis of microarrays.

91. Microarray is two-dimensional study of mRNA transcripts to find out gene content of the sample.

92. CAMERA resource possesses information about fish genome model.

93. Wormbase deals with C. elegans, a very important model organism.

94. PROSITE is the method to determine function of various proteins that are uncharacteristically translated from cDNA to mRNA. PROSITE (www.expasy.ch/prosite/) is the first established sequence pattern database and is still widely used. It primarily uses a single consensus pattern or "sequence signature" to characterize a protein function and a sequence family.

95. All the abbreviations are true.

96. Smith-Waterman algorithm is developed for calculation of scoring function of scoring matrices involved in local alignment of sequences. There is no gap penalty (default 0) that is involved in this algorithm. There are scores for matches and mis-matches of nucleotide bases.

97. The Gap cost is given by a function of cosine:

    Cos (t) = a + bk

98. BLAST is a family of programs that includes BLASTN, BLASTP, BLASTX TBLASTN, and TBLASTX. BLASTN queries nucleotide sequences with a nucleotide sequence database. BLASTP uses protein sequences as queries to search against a protein sequence database. BLASTX uses nucleotide sequences as queries and translates them in all six reading frames to produce translated protein sequences, which are used to query a protein sequence database. TBLASTN queries protein sequences to a nucleotide sequence databasewith the sequences translated in all six reading frames. TBLASTX uses nucleotide sequences, which are translated in all six frames, to search against a nucleotide sequence database that has all the sequences translated in six frames.

99. GAPDIST in CLUSTAL-W indicates the penalty score that is invoked for separation of gaps.

    Matrices with scoring schemes based on observed situation rates are superior to simple identity scores, or scores based solely on side chain moiety similarity. The two commonly used types of scoring matrices are the PAM matrices and the BLOSUM matrices. PAM (Percentage of Acceptable point Mutations per 108 years) matrices are based on global alignments of closely related proteins. The PAM 1 is the matrix calculated from comparisons of sequences with no more than 1 % divergence. A PAM matrix is specific for a particular evolutionary distance, but may be used to generate matrices for greater evolutionary distances by multiplying it repeatedly by itself. However, at large evolutionary distances the information present in the matrix is essentially degenerated. It is rare that a PAM matrix would be used for an evolutionary distance any greater than 256 PAMs.

    Scores for convergence or divergence between or amongst various sequences are derived from Mutation probability matrix where probability of occurrence of a mutation within a sequence in a particular region is defined as the parameter for scoring.

101. BLOSUM 80 indicates that the sequences selected for constructing the matrix share an average identity value of 80%. The BLOSUM score for a particular residue pair is derived from the log ratio of observed residue substitution frequency versus the expected probability of a particular residue.

    Dot plots are one of the oldest ways of comparing two sequences. They compare two sequences by organizing one sequence on the X-axis, and another on the Y-axis, of a plot. When the residues of both sequences match at the same location on the plot, a dot is drawn at the corresponding position. The closeness of the sequences in similarity will determine how close the diagonal line is to what a graph showing a curve demonstrating a direct relationship is. Low-complexity regions in the sequence with only a few amino acids, which in turn, causes redundancy within that small or limited region. These regions are typically found around the diagonal, and may or may not have a square in the middle of the dot plot.

102. Sequence similarity is affected by repeats and mutations within sequences. Repeats can be direct, tandem, inverted, simple and mutations are also responsible for less similarity.

103. Conserved sequences show little or no variations at all in terms of overall composition of the nucleotide sequences or protein sequences. Those regions code for the same protein (may or may not be functional) and are responsible for conserved motifs and domains in various organisms. These regions can be traced back through phylogenetic tree analysis to find out the ancestors of several related species. The regions remain conserved throughout evolution because of absence of any kind of mutation within the concerned sequences.

104. I. Gene3D is a resource of form studying proteins and the component domains.

    II. Gene3D takes the HMM's derived from CATH.

    III. You can run FASTA files of up to 2.5MB via the asynchronous server.

105. I. FuncNet brings together an ensemble of algoritms developed and hosted at varisous sites

    II. FuncNet was funded by the European Union's EMBRACE and ENFIN networks.

    III. FuncNet is an open architecture on which multiple prediction algorithms can be queried in parallel in order to provide higher-quality results.

106. I. All non-proteins, models, and structures with greater than 30% "C-alpha only" are excluded from CATH.

    II. There are four major levels in this hierarchy : Class, Architecture, Topology (fold family) and Homologous family.

    III. The CATH database is a hierarchical domain classification of protein structures in the Protein Data Bank.

113. SDAP is a Web server that integrates a database of allergenic proteins with various computational tools that can assist structural biology studies related to allergens. SDAP is an important tool in the investigation of the cross-reactivity between known allergens, in testing the FAO/WHo allergenicity rules for new proteins, and in predicting the IgE-Binding potential of genetically modified food proteins. Using this Internet service through a browser, it is possible to retrieve information related to an allergen from the most common protein sequence and structure databases (Swissprot, PIR, NCBI, PDB), to find sequence and strctural neighbors for an allergen, and to search for the presence of an epitope other the whole collection of allergens.

114. In genetic algorithms, crossover is a genetic operator used to vary the programming of a chromosome or chromosomes from one generation to the next. It is analogous to reproduction and biological crossover, upon which genetic algorithms are based.

115. Basic bioinformatics services are classified by the EBI into three categories: SSS (Sequence Search Services), MSA (Multiple Sequence Alignment) and BSA (Biological Sequence Analysis). The availability of these service-oriented bioinformatics resources demonstrate the applicability of web based bioinformatics solutions, and range from a collection of standalone tools with a common data format under a single, standalone or web-based interface, to integrative, distributed and extensible bioinformatics workflow management systems.

116. Vector alignment search tool computer algorithm that identifies similar protein 3-dimensional structures. Structure neighbors for every structure in MMDB are pre-computed and accessible via links on the MMDB Structure Summary pages. These neighbors can be used to identify distant homologs that cannot be recognized by sequence comparison alone.

117. VecScreen system for quickly identifying segments of a nucleic acid sequence that may be of vector origin. VecScreen searches a query sequence for segments that match any sequence in a specialized non-redundant vector database (UniVec).

118. GEO BLAST Tool for aligning a query sequence (nucleotide or protein) to GenBank sequences included on microarray or SAGE platforms in the GEO database.

119. A utility for computing alignment of proteins to genomic nucleotide sequence. It is based on a variation of the Needleman Wunsch global alignment algorithm and specifically accounts for introns and splice signals. Due to this algorithm, Prosplign is accurate in determining splice sites and tolerant to sequencing errors.

120. Taxplot tool for comparing genomes on the basis of the protein sequences they encode. To use Taxplot, one selects a reference genome and two species for comparison. Pre-computed BLAST results are then used to plot a point for each predicted protein in the reference genome, based on the best alignment with proteins in each of the two genomes being compared.

121. Peptidome is a public repository that archives and freely distributes tandem mass spectrometry peptide and protein identification data generated by the scientific community. Includes identified proteins, identified peptides for protein identification, mass spectra that support identifications, and supporting documentation.

122. I. Each path represents an alignment
II. Only three possible transitions are allowed from each cell
III. In the spreadsheet demonstration, the first phase of filling in the table is done using spreadsheet formulas. The second phase, of recursively tracing the paths through the table to find the high-scoring alignments, is done using a recursive script function.

123. I. For microarray gene expression data, the Pearson correlation coefficient is used to measure similarity between vector of expression values for either a given gene in a set of samples (Eisen 1998), or for a set of genes in a given sample.
II. An artificial neural network is a set of interconnected artificial neurons.
III. A simple type of artifical neuron is the perceptron, in which each input is multiplied by a weight, and the weighted sum of the inputs is compared to a threshold.

124. The microarray data can be accessed through ArrayExpress whihc is a public database for microarray based gene expression data. ArrayExpress is the microarray database setup by the EBI. It exchanges information with the NCBI and DDBJ micro array database every week.

126. A common and popular structural alignment method is the DALI, or distance alignment matrix method, which breaks the input structures into hexapeptide fragments and calculates a distance matrix by evaluating the contact patterns between successive fragments.

127. MAMMOTH-based structure alignment methods decompose the protein structure into short peptides (heptapeptides) which are compared with the heptapeptides of another protein. The similarity score between two heptapeptides is calculated using a unit-vector RMS (URMS) method.

128. Sequerome is a web-based Sequence profiling tool for integrating the results of a BLAST sequence-alignment report with external research tools and servers that perform advanced sequence maniulations, and allowing the user to record the steps of such an analysis. Sequerome is a web based Java tool that acts as a front-end to BLAST queries and provides simplified access to web-distributed resources for protein and nucleic acid analysis.

129. I. Stemloc is a program for Pairwise RNA structural alignment
II. Stemloc implements contrained versions of the Sankoff algorithms
III. T-Coffee (Tree-based Consistency Objective Function For alignment Evaluation is a multiple sequence alignment).

130. 74% of protein sequences have at least one match to Pfam. This number is called the sequence coverage.

132. The good evidence of common ancestry for two sequences requires 25% identity

134. I. Local alignment is critical
II. Insertion and deletion at the end of alignment contribute a weight of zero
III. In biological application local alignment is far more meaningful than global similarity

135. I. FASTA uses the dynamic programming algorithm to compute optimal alignment
II. FASTA misses very few cases of significant homology
III. BLAST focuses on no gap alignments of a certain fixed length

136. Scoring matrices used for computing alignment scores are based on observed substitution rates derived from the substitution frequencies.

137. The score is calculated from the frequency of the occurrence of a match of the two individual aminoacids in evolutionary related sequences.

139. In scoring matrix the disadvantages are
I. Evolutionary rates vary within a sequence
II. Environment changes over evolutionary time
III. Each position has a unique 3D environment

141. I. Sensitivity is the fraction of the true matches that are correctly predicted as matches by the pattern
II. Specificity is the fraction of the sequence predicted as matches that really are true matches
III. Sensitivity is used to report all homologous hits

143. I. Global alignment need to use gaps where as local alignment doesn't need to use
II. MSA can be looked at as an extension of the pairwise alignment
III. The first step in MSA is pairwise alignment of all sequences

144. I. Sum of pair's method is a dynamic programming method
II. You need to align three or more sequences in sum of pair's method
III. The SP function scores each position in the protein

145. The phylogenetic tree process works as follows
I. Compute all pairwise distances between given sequences
II. Computer a tree by single linkage clustering by using methods like UPGMA OR nearest neighbour
III. Align the sequences in an orderly fashion

147. I. A dendogram not necessarily reflects evolutionary relationship
II. Cladistic methods use each alignment position as evolutionary information to build a tree
III. In distance method insertion and deletion are given larger weight than replacements

148. I. Neighbour joining method is a FASTEST tree building method
II. Neighbour joining method uses empirical substitution scoring method
III. Neighbour joining method tests a single tree

150. I. If you get a different tree with each sample, then you can conclude that no tree is strongly supported by the data
II. Jacknife resampling has largely been replaced by bootstrap resampling
III. In Jacknife the sub-replicates are of a smaller size than the original data set

151. I. Maximum parsimony uses only relevant sites
II. Character state approaches are powerful than distance methods
III. The results of maximum parsimony drastically change for its sensitivity to codon bias and unequal rates of evolution

152. General time reversible models include transition and transversion

NUMERICAL TYPE QUESTIONS

1. According to jukes-cantor model assumption, the frequency of each base in DNA sequences is 0.25
2. The Kimura method which distinguishes AT/GC transversion is of 3 parameter
3. According to maximum likelihood,

$$\text{In } D = -\ln(1 - p - 0.2 + p^2)$$

the value of p should not exceed 0.75

EXERCISE – II

MCQ TYPE QUESTIONS

1. Bootstrapping is a resampling analysis that involves taking columns of characters out of your analysis, rebuilding the tree, and testing if the same nodes are recovered.
2. GDE → Start with # $ @ etc
FASTA → Start with > Description
NBRF → Start with > P1;
GCG → Start with Anotation lines

3. Here $n = 4$

$$\text{Number of rooted trees (NR)} = \frac{(2n-3)!}{2^{n-2}(n-2)!}$$

$$\text{Number of unrooted trees (NU)} = \frac{(2n-5)!}{2^{n-3}(n-3)!}$$

$\therefore$ NR = 15 and NU = 3

4. The BLAST program sets a size k for k-tuple subwords. The program then looks for diagonals in the comparison matrix between query and search sequence along which many k-tuples match. This program also aims at identifying core similarities for later extension. The core similarity is defined by a window with a certain match density on DNA or with an amino acid similarity score above some threshold for proteins. Independent of the exact definition of the core similarity, BLAST rests on the precomputation of all strings which are in the given sense similar to any position in the query. n BLAST nomenclature this set of strings is called the neighborhood of the query. The code to generate this neighborhood is in fact exceedingly fast.

5. Substitution matrices like PAM100, PAM120 and PAM250 are constructed by observing the frequencies of amino acid replacements in large samples of protein sequences. For a given replacement, the PAM value is proportional to the natural log of the frequency with which that replacement was observed to occur. One PAM unit is defined as the amount of sequence divergence corresponding to a 1% amino acid replacement rate. For closely-related sequences, it is appropriate to use a PAM100 matrix, in which PAM units have been extrapolated to 100% replacement. In other words, 100% of the positions show at least one replacement. For most database searches, a PAM250 matrix is preferred, since larger databases will tend to have sets of more distantly-related sequences. So, decreasing order of stringency of point accepted mutation matrices would be: PAM100 > PAM120 > PAM250.

6. Point accepted mutation (PAM) matrix is used for species having deletion or insertion at any particular point in the nucleotide sequence. Block substitution matrix (BLOSUM) is used for species whose sequences have undergone substitution. PAM 40 means that 40 % of the sequences are mutated and 60% are conserved while BLOSUM 80 means that 80% of the sequences are conserved and rest 20% are mutated. So, PAM 40 and BLOSUM 80 can be used for less divergent or closely related species for performing the alignment.

7. The unweighted pair-group method with arithmetic mean (UPGMA) is a popular distance analysis method. UPGMA is the simplest method for constructing trees. The great disadvantage of UPGMA is that it assumes the same evolutionary speed on all lineages, i.e. the rate of mutations is constant over time and for all lineages in the tree. This is called a 'molecular clock hypothesis'. This would mean that all leaves (terminal nodes) have the same distance from the root. In reality the individual branches are very unlikely to have the same mutation rate. Sequence alignment is converted into evolutionary distances. UPGMA is "ultrametric", meaning that all the terminal nodes (i.e. the sequences/taxa) are equally distance from the root. In molecular terms, this means that UPGMA assumes a molecular clock, i.e. all lineages are evolving at a constant rate.

8. Threading is the method of three dimensional structure predictions of proteins by generating threads that help to determine folds in a protein. FASTA is a program for rapid alignment of pairs of protein and DNA sequences. Rather than comparing individual residues in the two sequences, FASTA instead looks for matching sequence patterns or words, called k-tuples, and then attempts to build a local alignment based upon these word matches. Profile HMMs turn a multiple sequence alignment into a position-specific scoring system suitable for searching databases for remotely homologous sequences. Profile HMM analyses complement standard pairwise comparison methods for large-scale sequence analysis. Paralogs are genes related by duplication within a genome. Orthologs retain the same function in the course of evolution, whereas paralogs evolve new functions, even if these are related to the original one.

9. Protein structural database: Primary database: PDB Secondary database: SCOP, CATH

10. A program for constructing phylogenetic tree is PHYLIP. It is a Phylogeny Inference Package. It is a free computational phylogenetic package which has programs for inferring evolutionary tree constructions.

11. Synteny can be defined as the local conservation of gene order. It is the condition of two or more genes being located on the same chromosome whether or not there is demonstrable linkage between them.

12. The Expect value (E) is a parameter that describes the number of hits one can "expect" to see by chance when searching a database of a particular size. It decreases exponentially as the Score (S) of the match increases. Essentially, the E value describes the random background noise. Virtually identical short alignments have relatively high E values. This is because the calculation of the E value takes into account the length of the query sequence and also depends upon the number of sequences in the database. These high E values make sense because shorter sequences have a higher probability of occurring in the database purely by chance.

Thus the E value does not depend on the probability from a normal distribution.

13. The Clustal series of programs are widely used in molecular biology for the multiple alignment of both nucleic acid and protein sequences and for preparing phylogenetic trees. The popularity of the programs depends on a number of factors, including not only the accuracy of the results, but also the robustness, portability and user-friendliness of the programs. New features include NEXUS and FASTA format output, printing range numbers and faster tree calculation. Although, Clustal was originally developed to run on a local computer, numerous Web servers have been set up, notably at the EBI (European Bioinformatics Institute).

14. BLAST (Basic Local Alignment Search Tool) is a program which is used for local similarity search. BLAST is an algorithm for comparing primary biological sequence information, such as the amino-acid sequences of different proteins or the nucleotides of DNA sequences. A BLAST search enables a researcher to compare a query sequence with a library or database of sequences, and identify library sequences that resemble the query sequence above a certain threshold. Different types of BLASTs are available according to the query sequences like nBLAST (nucleotide BLAST), pBLAST (protein BLAST) etc.

15. An accession number is a unique identifier assigned to a particular genome or protein sequence to uniquely identify it in a database. It is assigned to the entire sequence record (whether DNA or protein) when the record is submitted to GenBank. The NCBI Entrez System is searchable by accession number using the Accession [ACCN] search field.

16. Homology modeling, also known as comparative modeling of protein, refers to constructing an atomic-resolution model of the "target" protein from its amino acid sequence and an experimental three-dimensional structure of a related homologous protein (the "template"). Homology modeling relies on the identification of one or more known protein structures likely to resemble the structure of the query sequence, and on the production of an alignment that maps residues in the query sequence to residues in the template sequence. It has been shown that protein structures are more conserved than protein sequences amongst homologues, but sequences falling below a 20% sequence identity can have very different structure.

17. • **UPGMA** (Unweighted Pair Group Method with Arithmetic Mean) is a simple agglomerative or hierarchical clustering method. Inbioinformatics, UPGMA is used for the creation of phenetic trees (phenograms). In a phylogenetic context, UPGMA assumes a constant rate of evolution (molecular clock hypothesis).
    - **CLUSTAL** is a general purpose multiple sequence alignment program for DNA or proteins.
    - **SWISS-PROT** is a curated protein sequence database which strives to provide a high level of annotation (such as the description of the function of a protein, its domain structure, post-translational modifications, variants, etc), a minimal level of redundancy and a high level of integration with other databases.
    - **RasMol** is a computer program written for molecular graphics visualization intended and used primarily for the depiction and exploration of biological macromolecule structures

18. In bioinformatics, Basic Local Alignment Search Tool, or BLAST, is an algorithm for comparing primary biological sequence information, such as the amino-acid sequences of different proteins or the nucleotides of DNA sequences. A BLAST search enables a researcher to compare a query sequence with a library or database of sequences, and identify library sequences that resemble the query sequence above a certain threshold.

19. Orthologs, or orthologous genes, are genes in different species that originated by vertical descent from a single gene of the last common ancestor. Paralogous sequences provide useful insight into the way genomes evolve. The genes encoding myoglobin and hemoglobin are considered to be ancient paralogs.

20. • **BOXSHADE** is a program for pretty-printing multiple alignment output. The program itself doesn't do any alignment, you have to use a multiple alignment program like ClustalW or Pileup and use the output of these programs as input for BOXSHADE. Of course, you can also use manually aligned sequences (in a proper format).

- **BCM Launcher**, i.e. Baylor College of Medicine Search Launcher is an on-going project to organize molecular biology-related search and analysis services available on the WWW by function by providing a single point-of-entry for related searches (e.g., a single page for launching protein sequence searches using standard parameters). It helps in finding alignments.
- **PROSITE** is a database of protein families and domains. It is based on the observation that, while there is a huge number of different proteins, most of them can be grouped, on the basis of similarities in their sequences, into a limited number of families. Proteins or protein domains belonging to a particular family generally share functional attributes and are derived from a common ancestor. PROSITE currently contains patterns and profiles specific for more than a thousand protein families or domains. Each of these signatures comes with documentation providing background information on the structure and function of these proteins.

21. Protein threading, also known as fold recognition, is a method of protein modeling (i.e. computational protein structure prediction) which is used to model those proteins which have the same fold as proteins of known structures, but do not have homologous proteins with known structure. It differs from the homology modeling method of structure prediction as it (protein threading) is used for proteins which do not have their homologous protein structures deposited in the Protein Data Bank (PDB), whereas homology modeling is used for those proteins which do. Threading works by using statistical knowledge of the relationship between the structures deposited in the PDB and the sequence of the protein which one wishes to model. Homology modeling, also known as comparative modeling of protein, refers to constructing an atomic-resolution model of the "target" protein from its amino acid sequence and an experimental three-dimensional structure of a related homologous protein (the "template"). Homology modeling relies on the identification of one or more known protein structures likely to resemble the structure of the query sequence, and on the production of an alignment that maps residues in the query sequence to residues in the template sequence.

NUMERICAL TYPE QUESTIONS

1. $3\begin{vmatrix}5 & 0\\ -8 & -4\end{vmatrix} - 2\begin{vmatrix}0 & 0\\ -8 & -4\end{vmatrix} + 6\begin{vmatrix}0 & 0\\ 5 & 0\end{vmatrix}$

$= 3(-20 + 0) - 2(0 - 0) + 6(0 - 0)$

$= -60$

2. sum $= a\dfrac{(1-r^n)}{1-r}$ a = 1, r = 0.3 $n \to \infty$

$= 1\dfrac{(1-0.3^n)}{(1-0.3)} = \dfrac{1}{0.7} = 1.4285$

3. $H(p) = -\sum P_i \log_2 P_i$

$P_1 = 0$

$P_2 = 0.5$

$P_3 = 0.25$

$P_4 = 0.25$

$\sum P_i = 1$

$H(p) = 0 + (-0.5 \log_2 0.5) + (-0.25 \log_2 0.25) + (-0.25 \log_2 0.25)$

$= 3.80$ bits

4. $f(AA) = 0.3, f(Aa) = 0.4, f(a,a) = 0.3$

($\therefore$ Allele frequency in next generation = p

$= f(AA) + \dfrac{1}{2}f(Aa) = 0.3 + 0.2 = 0.5$)

and genotype frequency = p^2

$= 0.5 \times 0.5 = 0.25$

5. Gap score = 2 × Gap opening penalty + Gap extension penalty – Gap length

Here, Gap opening penalty = –20, Gap extension penalty = –4 and gap length = 8

Therefore, Gap score = $2 \times (-20) - 4 - 8 = -52$

■■

General Aptitude

Part I : Verbal Ability

CHAPTER 1

English Grammar

ERRORS IN USE OF ARTICLES

ARTICLES

Indefinite (A, An) — **Definite (The)**

Use of 'An'

1. Before words beginning with vowel sounds [a, e, i, o, u are called vowels, others are consonants].
   e.g. an apple, an egg, an owl.
2. Before words beginning with silent 'h' but sounds as vowel.
   e.g. an hour, an honourable man, an heir, an honest man.
3. F, H, L, M, N, R, S, X are letters that are not vowels but begin with vowel sound 'M' has the sound of 'em'. So, 'an' is used before abbreviations beginning with vowels of these letters.
   e.g. an M.L.A., an R.A.F., an N.C.C. officer, an F.I.R., an X-ray, an H.E. school, an S.P.

Use of 'A'

1. In the sense of one.
   e.g. He couldn't speak a word to save himself.
   With 'one' (since 'one' begins with sound of 'w')
   e.g. a one-man show, a one-rupee note.
2. Before words beginning with consonant sound
   e.g. a boy, a box, a dog.
3. With Vowel letters having consonant value.
   e.g. a university, a unique article, a euphenism, a unit, a European language
4. With units and rate (per).
   e.g. He earns rupees five hundred a month.
5. In exclamatory expressions before singular countable nouns.
   *e.g. What **a** pretty girl !*
6. When two subjects or articles are thought of as a single unit.
   *e.g. He was ready with **a** cup and saucer.*
7. With certain expressions of quantity.
   *e.g. **a** lot of , **a** dozen, **a** great deal of, **a** couple.*
8. With a person's name to indicate that the person is perhaps unknown to the person addressed.
   *e.g. **A** Mr. roy is at the door.*
9. With a special meal (to celebrate something or in someone's honour).
   *e.g. I called my friends to **a** lunch to celebrate my success.*
10. To make a common noun of a proper noun.
    e.g. This man is 'a second Newton'.
    (This phrase means 'a philosopher as great as Newton')

Use of 'The'

1. When we speak of a particular person or thing already referred to.
   *e.g. I dislike **the** follow.*
2. When a singular noun represents a whole class.
   e.g. The mango is considered the king among fruits.
3. With name of
   (*i*) gulfs, rivers, oceans, islands and mountains e.g. the Himalayas, the Indian ocean, the Persian Gulf, the Red sea, the Andaman islands, the Brahmaputra river.
   (*ii*) Certain books
   e.g. the Vedas, the Puranas, the Bible, the Ramacharitmanas.
   (*iii*) Musical instruments
   e.g. the flute, the violin, the tabla, the trumpet.
   (*iv*) The inventions
   e.g. I hate the telephone for its constant ringing.
   (*v*) Parts of body
   *e.g. He was wounded in **the** leg.*
   (*vi*) Religious groups
   e.g. the Sikhs, the Hindus, the Parsees.
   (*vii*) Names enforcing law
   e.g. the Police, the Navy, the Air Force.
   (*viii*)Political parties
   e.g. the Congress, the Janata Pary, the B.J.P.
   (*ix*) Aeroplanes, ships, trains etc
   e.g. the Makalu (aeroplane), the Vikrant (Ship), the Rajdhani express (train).
   (*x*) Before names of an empire, dynasty or historical event
   e.g. the Gupta dynastry, the Old Stone Age, the First World War, the American Revolution.
   (*xi*) Clubs, foundations etc.
   e.g. the Lion's Club, the Ford Foundation.
   (*xii*) Before common nouns denoting unique things
   e.g. the sun, the sky, the earth, the worl.
   (*xiii*) With superlatives
   e.g. *He is **the** best boy in the class.*
   (*xiv*) With ordinals
   *e.g. He took **the** first taxi that came his way.*
   (*xv*) Before the comparative degree
   e.g. ***The more** they get, **the** more they want.*
   (*xvi*) Before an adjective when the noun is understood
   *e.g. **The** poor would favour him.*

ERRORS IN USE OF NOUNS

1. Nouns which are used in singular form

(i) Scenery, information, furniture, advice, machinery, stationery, news, poetry, business, mischief, fuel, issue, repair, bedding.
*e.g. **The scenery** of this place is worth seeing.*

(ii) Physics, Mathematics, Economics, Classics, Ethics, Athletics, innings, gallows.
*e.g. **Economics** is a very interesting subject.*

(iii) Brick, bread, fruit, word (as 'promise')
*e.g. Let me buy some **fruit**.*

(iv) Words like dozen, score, hundred, thousand, million when preceded by a numeral.
*e.g. He bought ten **dozen** oranges.*

(v) Expressions as a ten-rupee note, a two-hour journey, a four-mile walk, a Five-year plan, a six-man committee etc.
e.g. A ten-rupee note is lying there.

2. Nouns used only in plural form

(i) Cattle, police, poultry, people, gentry, peasantry, artillery.
*e.g. The **police have** caught the thief.*

(ii) Scissors, trousers, stockings, spectacles, shorts, alms, remains, riches, goods, measles.
e.g. My scissors are very sharp.

3. Nouns used both as singular and plural in the same form

(i) Dear, sheep, fish, apparatus, wages.
*e.g. The **wages** of sin **is** death.*
*The **wages** of the workers **have** been raised.*
*I saw **a sheep** grazing in the field.*
***Sheep are** sold cheaper than goat.*

(ii) Collective nouns as jury, public, team, audience, committee, government, audience, congregation, orchestra.
*e.g. The **team are** looking quite fit.*
*The **team has** not turned up yet.*

4. Use of Collective Nouns

- **crowd** of people
- **herd** of cattle/cows
- **team** of players
- **flight** of birds
- **bouquet** of flowers
- **Shoal/school** of fish
- **bundle** of sticks
- **army** of soldiers
- **flock** of sheep
- **crew** of sailors
- **swarm** of bees
- **garland** of beads
- **gang** of thieves
- **library** of books
- **mob** of angry people
- **fleet** of ships/cruisers
- **pack** of wolves
- **pack** of cards
- **loaf** of bread
- **heap** of corn
- **stock** of grain
- **class** of pupils
- **bunch** of grapes
- **herd** of lions
- **pile** of books
- **hoarde** of nomads
- **block** of flats
- **file** of papers

5. One of or **any of** is followed by plural words.
*e.g. I want **one of** the **books** kept on the table.*
***Any of** these **tools** may serve the purpose.*

6. Plural nouns are used with fractions and decimal over 1.
*e.g. It took us one and a half **hours**.*

ERRORS IN USE OF PRONOUNS

Pronouns are words used to replace nouns or noun groups already mentioned.

Nominative/ Vocative case (comes before verb)	*Accurative/ Objective case (comes after verb)*	*Possessive case*	*Reflexive Pronoun*
I	Me	My	Myself
We	Us	Our	Ourselves
He	Him	His	Himself
She	Her	Her	Herself
They	Them	Their	Themselves
Who	Whom	Whose	—

1. The pronoun *'One'* must be followed by *'one's'*.
e.g. One must do one's duty to one's country.

2. When 'one' means 'one in number', the pronoun for it is third person singular pronoun (he, she, it).
e.g. *One of them has given up one's studies.* (×)
One of them has given up his studies. (✓)

3. 'Everyone' or 'Everybody' must be followed by 'his'.
e.g. Everyone should love his country.

4. Each, every, anyone, anybody must be followed by the singular pronoun of their person.
e.g. Anyone can do this if he tries.

5. 'Let' is followed by pronoun in the objective case.
e.g. Let him go.

6. 'But' and 'except' are followed by pronoun in the objective case.
e.g. Everyone attended the party except him.
'such as' is followed by pronoun in the subjective case.
e.g. I have no liking for such a man as he.

7. Verbs like enjoy, avail, pride, resign, apply, acquit, assert, absent are followed by reflexive pronouns.
e.g. He absented himself from the class.
We enjoyed ourselves at the party.

8. Reflexive pronouns are never used with verbs keep, conceal, quality, spread, rest, stay.
e.g. I stayed away from my class.
He qualified in the test.

9. When first, second and third person singular pronouns (I, you and He) are used together, they are placed in the order : You, he and I.
e.g. You he and I are neighbours.
In case of plural pronouns, 'we' comes first, then 'you' and then 'they'.
e.g. We, you and they must work together.
But if we have only two persons including first, then first person pronoun is written first.
e.g. I and Sanjeev have done this job.

10. 'Who' denotes subject and 'whom' denotes object.
e.g. Who do you think did the job ?

11. 'Whose' is used for persons and 'which' for lifeless objects.
e.g. This is the table which I was talking about.

12. 'Which' conveys additional information and 'that' explains a certain thing.
e.g. *I will tell you the first thing which I remember.*

13. The following expressions usually take 'that' in place of 'who' or 'which'.
Only, Any, It is, All, Superlatives.
e.g. He is the only man that can do it.
Any man that listens to you is a fool.

14. *'Each other'* is used for two; *'one another'* for more than two.
e.g. Rahul and Renu love each other.

15. The complement of the verb *to be*, when it is expressed by a pronoun, should be in Nominative case.
e.g. It was he who did it.

16. When the same person is the subject and object, it is necessary to use reflexive pronouns.
e.g. I cut me shaving this morning. (×)
I cut myself shaving this morning. (✓)

17. When a pronoun is the object of a verb or preposition it should be in objective case.
e.g. These books are for you and I. (×)
These books are for you and me. (✓)
Between him and me there is an understanding.

18. The Relative Pronoun should be placed as near as possible to the antecedent.
e.g. I have read Shakespeare's works who was a great dramatist. (×)
I have read the works of Shakespeare who was a great dramatist.(✓)

19. The case of the pronoun following 'than' and 'as' is decided by mentally supplying the verb and completing the sentence.
e.g. She is taller than I (am)
I love more than (I love) him.

ERRORS IN USE OF PREPOSITIONS

A preposition is a word used with a noun or pronoun to show its relation to some other word in a sentence.

PREPOSITIONS OF TIME

1. At is used
(a) with a definite point of time
e.g. I usually get up at 5 o'clock.
(b) with festivals
e.g. He will come at Holi.

2. In is used :
(a) with the parts of the day, (with **noon**, use **at**), months, seasons and years
e.g. He takes a walk in the afternoon.
(b) with the future tense referring to the period in which action may take place
e.g. You must be careful in future.

3. On is used : with days and dates
e.g. My brother will arrive on Monday.

4. By refers to the latest time at which an action will be over
e.g. The examination will be over by 5 p.m.

5. For is used with perfect continuous tense showing the duration of action
e.g. I have been here for three years.

6. Since is used with the point of time when action begins and continues :
e.g. He has been ill since last Tuesday.

7. From refers to the starting point of action.
e.g. This water came from a spring.

PREPOSITIONS OF POSITION

1. At refers to an exact point.
e.g. The tourist stayed at the Tourist Hotel.
He studied at Oxford.

2. In refers to larger areas.
e.g. He lives in Bombay.

3. Between is used for two persons or two things.
e.g. Share these sweets between him and me.

4. Among is used with more than two persons or things but before the word which starts with a consonant letter.
e.g. Divide the sweets among the three boys.

5. Amongst is also used with more than two persons or things but before the word which starts with a vowel letter.
e.g. Divide the sweets amongst us.

6. Above is used for **higher than.**
e.g. The sun rose above the horizon.

7. Under is used for **vertically below.**
e.g. It is shady under the trees.

8. Below is used for **lower than.**
e.g. When the sun sets it goes below the horizon.

9. Over is used for **vertically above.**
e.g. There is an aircraft coming over.

10. Beneath means a lower position.
e.g. the ground was slippery beneath her.

PREPOSITIONS OF DIRECTION

1. **To** is used to express motion from one place to another.
   e.g. We walked to the river and back.
2. **Towards** refers to direction.
   e.g. He saw me running towards him.
3. **Into** denotes motion towards the inside of something.
   e.g. She fell into a ditch.
4. **At** refers to aim.
   e.g. He aimed at the bird.
5. **For** denotes direction.
   e.g. I will leave for Pune today.
6. **Against** shows pressure.
   e.g. She cleaned the edge of her knife against the plate.
7. **Off** refers to separation.
   e.g. He was wiping sweat off his face.
8. **From** refers to the point of departure.
   e.g. The man parted from his friends.

OTHER USES OF PREPOSITION

1. **About** shows nearness.
   e.g. His father is about to retire.
2. **Along** stands for **in the same line.**
   e.g. She led them along the corridor.
3. **After** refers to sequence.
   e.g. She came after me.
4. **Across** means from one side.
   e.g. He blew a cloud of smoke across the table.
5. **Before** stands for **in front of.**
   e.g. A thief was brought before the judge.
6. **Behind** means **at the the back of.**
   e.g. She sat down behind the hedge.
7. **Beyond** means **on the farther side of.**
   e.g. This is beyond his power.
8. **Beside** means **by the side of.**
   e.g. *I sat down beside my wife.*
9. **Besides** means **in addition to.**
   e.g. He is guilty of five killings and more besides.

WORDS FOLLOWED BY PREPOSITIONS

A

- abound with
- abstain from
- according to
- accordance with
- accuse of
- accustomed to
- acquaint with
- acquit of
- adverse to
- affiliate to/with
- afflict with
- afraid of
- agree to/on
- aim at
- allegation
- about
- allude to
- alternate with
- alternative to
- angry with
- apply to (body)
- appreciation of
- apprise of
- approximate of
- arraign against
- arrest in
- arrive at/in
- ashamed of
- aspire to/after/at
- associate with
- assure of
- attribute to

B, C

- beset with
- beware of
- blink at
- boast of
- break off
- bungle over
- careful of
- cause of
- centre on
- charge with
- clamour for
- clamp on
- coerce into
- collide with
- comment on
- compatible with
- complain of
- comply with
- compliment on
- composed of
- concur in
- condole with
- conducive to
- confer on
- confirmation of
- conform to
- congratulate on
- conscious of
- contiguity with
- converge on
- convict of
- co-opt to
- cope with
- correspond with
- converge on
- convict of
- co-opt to
- cope with
- correspond with
- cost of
- credit with
- criticism of
- cure of

D

- decide on
- debar from
- delight in
- deliberate on
- denude of
- depart to
- departure for
- depend on/upon
- deprive of
- derive from
- derogate from
- derogatory to
- desist from
- despair of
- detrimental to
- devoid of
- difficulty in
- diffident of
- disabuse of
- disagree with
- disgusted with
- dispose of
- dissociate from
- divest of
- divide into
- dressed in

E

- effort in (—ing)
- election to
- embroil in
- emphasis on
- employ in
- enamoured of
- encroach on
- endow with
- enlarge on
- engage in
- essential to
- expostulate with
- extend to

F, G, I

- fed up with
- focus on
- forbid to
- fraught with
- fritter in
- full of
- give to
- grapple with
- good at
- identical with
- impose on
- incur on
- indict for
- indifferent to
- indulge in
- infatuate with
- infect with
- infest with
- information on
- insensible to
- insist on
- interest in
- intimate to
- intoxicate with
- intrigue with
- intrude on
- investigation of
- involve in

J, K, L

- jealous of
- jeer at
- jump at
- just to
- knock at
- lash at
- lay siege to
- leave for
- live on (income)
- long for

O, P

- oblivious of
- operate on
- opinion on
- overcome with
- pay a call on
- persevere in
- persist in
- plan for
- plead for
- pleased with
- prefer to
- preference for
- present with
- prevent from
- priority to
- probe into
- prohibit from
- proud of

R, S

- recoil from
- reconcile to
- remand to
- remonstrate with
- repent of
- replete with
- reply to
- research on
- resolve on/to
- sail for
- satiate with
- satisfied with
- sentence to
- sick of
- side with
- sparing
- square with
- stickler for
- subversive of
- succeed in
- sue for
- superior to
- supply with
- sure of
- surprised at
- susceptible to
- suspect of
- sympathy for

T, U, V, W

- testify to
- thank for
- throw at
- tinker at
- tire of
- trace to
- trample on
- translate into
- try to (do)
- unequal to (the task)
- vie with
- vote for (a candidate)
- wait for (or await)
- ways of
- wish for

SOME SPECIAL CASES

- absolve from (blame) / of (sin)
- admit to (hospital, place, class)
- appear for (examination)/at
- begin on (not from Sunday)
- blow down (the roof)/off (steam)/over (pass off)
- buy in (market, store)/at (shop)
- come/go by (bus, train)
- come into (conflicit with)
- compare with (one man height with another's)/to (one man's height to a building)
- consist of (to denote the substance)/in (define the subject)
- deal with (somebody)/ in (goods)
- die of (disease)/from (other causes)
- differ from (to be different from)/with (to disagree)
- due to (used after a noun, not after a verb)
- entrust to (somebody a job)
- entrust (somebody) with a job)
- fill with (enthusiasm)/ in (a form)
- fire at (when not hit)/on (when injured to killed)
- glad of (the news)/at (having arrived safe)
- hostility to/in hand (being attended to)/ on hand (in possession of)
- ingratiate (oneself) with
- inquire into (matter) / of (person)/about. often (thing)
- invest with (power/in (blank)
- key of (the door)/to (a problem)
- level (charge) against.
- look at (consider evidence)/to (somebody for aid)
- meeting (called) for (Sunday)
- motive in (doing)/for (an act)
- neglect of (duty)/in (doing a thing)
- nominated to (board)
- noted for (good things)
- owing to (used after a verb, not after a noun)
- part from (people)/with (things)
- preventive for priority
- put out (light)/off (trip)
- Shoot at (when injured)/in (leg, head)
- take up (job)/to (a hobby)
- taste of (sensation)/for (liking)
- tear off (remove)/up (to pieces)
- vest with (The P.M. is vested with power to ...)/ in (Powers are vested in the P.M. to ...)

ERRORS IN USE OF CONJUNCTIONS

1. **Scarcely or hardly** is followed by **when.**
   e.g. Hardly had I slept when the telephone rang.
2. **Though** is followed by **yet.**
   e.g. Though he worked hard yet he failed.
3. **No sooner** is followed by **than.**
   e.g. No sooner did we reach there than it began to rain.
4. **Not only** is followed by **but also.**
   e.g. Not only did he help her, but also dropped her home safely.
5. **Lest** is negative and so should not be followed by **not**; it is followed by **should.**
   e.g. Work hard lest you should fail.
   If **would** or **may** is used in place of **should**, then **else** should be used in place of **lest.**
   e.g. Give him water, else he may die.
6. **Both** is complemented by **and**, not by **as well as.**
   e.g. Both Amit and Satish are good at Science.

7. **So.....as** is used in negative sentences, whereas **as** .....**as** is used in affirmative sentences.
   e.g. He is not so tall as his brother.
8. **Other** is followed by **than**.
   e.g. He has no other claim than his wealth.
9. The word **reason** is not followed by **because,** but by **that.**
   e.g. The reason why he didn't go was that his mother was ill.
10. **Because** denotes reason.
    in order that denotes purpose.
    e.g. He went to the doctor because he was ill.
11. Words such as regard, describe, define, treat, mention, depict, portray are followed by **as**.
    e.g. I regard her as my sister.
12. **As** and **since** are also used to express reason.
    e.g. As he was not there, I spoke to his brother.
13. **Neither** is followed by **nor** and both are followed by same auxiliary verb.
    e.g. Neither Ravi nor Raju was seen.
14. **Either** is followed by **or**.
    e.g. Either you are or he is to blame.
15. **Unless, until, if not, so that** should not be followed by **not.**
    e.g. Wait here until I come.
16. **If** is used in conditional series
    Whether is used in uncertainty.
    e.g. I don't know whether he was present.
17. To express **time before** use **until** or **till** and to express **how long** use **as long as.**
    *e.g. Work **as long as** you live.*
18. **Such** is followed by **as.**
    e.g. We talked about such subjects as the weather.
    Such is followed by **that** if we emphasize degree of something by mentioning the result.
    e.g. The extent of the disaster was such that not a single man could survive.
19. **Like** is followed by pronoun; **as** is followed by a clause.
    e.g. He runs like me.

ERRORS IN SUBJECT–VERB AGREEMENT

1. Singular subject must have singular verb.
   e.g. He writes; I write.
2. Plural subject must have plural verb.
   e.g. They write; We write.
3. Two subjects joined by **and** will always take a plural verb.
   e.g. The doctor and nurse work together.
   The doctor and nurses work together.
   The doctors and nurse work together.
4. Two singular subjects joined by **or** or **nor** will take a singular verb.
   e.g. A doctor or a nurse is working in the hospital.
5. A singular subject and a plural subject joined by **or** or **nor** will take a singular or plural verb depending on which subject is nearer the verb.
   e.g. Neither Deepak nor his friends are joining the tour.
   Neither his friends nor Deepak is joining the tour.
6. If the subject is singular and the predicate is plural, the verb must agree with its subject and not its predicate.
   e.g. Physical conditioning and mental attitude are winning combination.
   The winning combination is physical conditioning and mental attitude.
7. Indefinite pronouns such as
   someone, somebody, each, nobody, anyone, anybody, one, no one, everyone, everybody, either, neither, etc. always take a singular verb.
   e.g. Each of my friends calls me once a month.
8. Indefinite pronouns which indicate more than one (several, few, both, many) always take plural verbs
   e.g. Both of the books required careful reading.
9. Collective nouns (fleet, army, committee, crowd) are singular when the group works together as a unit and hence take singular verbs.
   e.g. The jury has reached its verdict.
10. Collective nouns are plural when the members of the group are acting individually and hence take plural verbs.
    e.g. The jury have argued for five hours.
11. Some words (such as news, measles, mumps etc.) end in —*s* but represent a single thing. These words need singular verbs.
    e.g. The 7 o'clock news is about to begin.
12. Some words (such as scissors, trousers, spectacles, shorts etc.) end in —*s* and seem to represent a single thing, but they are two parts to that single thing. These words take plural verbs.
    e.g. The scissors are on the table.
13. Words (such as politics, ethics, athletics etc.) that end in —ies are usually singular and hence take singular verbs.
    e.g. Mathematics is his favourite subject.
    But these words are singular when they refer to a study, science or practice. If these words have modifiers with them, they become plural and hence take plural verbs.
    e.g. His politics are somewhat divided.
14. Title of the books need singular verbs.
    e.g. Great Expectations is a good book.
15. Some nouns in the plural form represent an amount, a fraction, or an element of time. These nouns are considered singular and hence take singular verbs.
    e.g. Sixty minutes is enough to finish this work.
16. If two subjects are joined together by **as well as** the verb will act according to the first subject.
    e.g. Students as well as the teacher are playing.

17. The subject **Many a .....** is always followed by the singular verb.
    e.g. Many a man was drowned in the sea.
18. If two subjects are joined together by **with** the verb will act according to the subject.
    e.g. The principal together with his students was seeing the final match.
19. If subject is **The number of ...** use a singular verb.
    e.g. The number of books is very small.
20. If the subject begins with **A number of ....** (= many), use a plural verb.
    e.g. A number of books are missing.
21. If two subjects express one idea, use a singular verb.
    e.g. Bread and butter is wholesome food.
22. When adjectives such as **much, less, little** and **more** are used as nouns, they must have a singular verb.
    e.g. A little of good habits makes our life happy.

ERRORS IN THE USE OF TENSES

The changed forms of a verb that indicate time of the action are called **tenses** of the verb.

1. When the verb in the Principal Clause is in the Past tense, the verbs of the Subordinate Clauses should be in the Past tense.
   e.g. He said that he had finished his work.
2. But a past tense in the Principal Clause may or may not be follwoed by the Past tense in the sub-ordinate clause if the latter expresses unviersal or habitual truth.
   e.g. The teacher said that the earth revolves round the sun.
3. Any tense may be used in the sub-ordinate clause if it gives a comparison by using the word **than.**
   e.g. The teacher liked Anil better than he liked me.
4. Any tense can be used when the subordinate clause is in a quotation.
   e.g. I said, "I am going to Delhi today".
5. The Present Perfect Tense (subject + has/have + V_3) cannot be used when an expression of Past time (yesterday, last night, ago etc.) is used.
   e.g. We wrote to you yesterday about his mother.
6. In conditional sentences the subordinate clause, beginning with if, is generally in Present or Past tense but not in future tense.
   e.g. If I go to Delhi, I shall see the Red Fort.
7. With the phrases **as if** and **as though**, Past tense and plural form of the verb should be used.
   e.g. He behaves as if he were a millionaire.
8. Past Perfect tense (subject + had + V_3) is used when a sentence refers to two past actions and one of them occurs earlier than the other.
   e.g. When my uncle came to India I had passed M.A.
9. Words like **usually, generally, often, whenever** etc. are used in Present Indefinite tense.
   e.g. I usually go to play cricket.
10. If the action began in the past and is still continuing in the present, use Present Perfect tense (Subject + has/have + been V_1 + ing).
    e.g. I have been reading in this class for two months.
11. Do not use the Future or the Present tense after such expressions as **suppose that, it is high time, it is time, as if** etc.
    e.g. It is high time that you went home.

GERUND & INFINITIVE

Use of Gerund (V_1 + ing)

1. When an action is being considered in general sense, gerund is used as subject.
   e.g. Swimming is his favourite pastime.
2. Gerund is used as subject in short prohibitions
   e.g. Smoking is prohibited.
3. Verbs such as help, stop, detest, avoid, finish, dread, mind, prevent, dislike, risk, deny, recollect, no good, no use, resent, delay, postpone, defer, enjoy, forgive, pardon, excuse, suggest etc. are followed by the gerund.
   e.g. It is no use saying.
4. A gerund is placed after a preposition :
   e.g. She is tired of walking.
   But the prepositions **except** and **but** are followed by an infinitive.
5. A gerund, showing physical activity, is put after **come** or **go**
   e.g. He is to go shooting.
6. Nouns, followed by a gerund, are put in the possessive case.
   e.g. He hates his brother's drinking everyday.
7. If there is a sense of dislike, hesitation, risk etc. in a sentence, use gerund.
   e.g. I dislike reading cheap novels.
8. Adjective before gerund are put in the possessive case.
   e.g. I do not mind your going away.
9. **With a view to** is always followed by a gerund and not an adjective.
   e.g. We go to school with a view to studying.

INFINITIVE (TO + V_1)

1. Verbs such as learn, remember, promise, swear, consent, neglect, refuse, purpose, try, endeavour, attempt, fail, care, hope, decide, hesitate, prepare, determine, undertake, manage, arrange, seem, forget, agree, regret etc. are followed by infinitive:
   e.g. We endeavour to invite such persons.
   We promise to do this work soon.
2. Verbs such as order, tell, invite, oblige, allow, permit, compel, teach, instruct, advise, tempt, encourage, urge, request, show, remind, forbid, warm etc. are followed by object and infinitive
   e.g. The teacher instructed the students to go.
3. Verbs or expressions like—will, can, do, must, may, let are followed by infinitive without **to**
   e.g. Let him do this work.

4. Expressions like would rather, would sooner, rather than, sooner than, had better are followed by infinitive without **to**
   e.g. You had better tell him.
5. **See** can be used with infinitive or without it
   e.g. Go to see him.
6. The infinitive is used after adjectives like delight, angry, glad, astonished
   e.g. I was astonished to find him.
7. The verb know is never directly followed by the infinitive. It is followed by a conjunction and then the infinitive
   e.g. Do you know to play the harmonium ? (×)
   Do you know how to play the harmonium ?(✓)
8. **...had better** is always followed by the infinitive without to
   e.g. You had better to stop taking the medicine which has harmful side-effects. (×)
   You had better stop taking the medicine which has harmful side-effects.(✓)
9. Infinitive is not used with **to** after verbs of perception like-to see, to hear, to feel, to watch etc.
   e.g. I saw him cross the street.
10. Avoid using split-infinitive :
    e.g. He planned to not go on a vacation this year.(×)
    He planned not to go on a vacation this year.(✓)

AUXILIARY VERBS

These are helping verbs.

1. **May** implies permission, doubt or possibility.
   e.g. It may rain tonight.
2. **Might** is the past form of **may** but it does not necessarily represent past time.
   Often it implies more doubt than **may.**
   e.g. If the clouds are salted, the rains might come.
3. **Can** is used to express ability.
   e.g. He can do this work.
4. **Could** is the past form of **can** but it does not necessarily represent past time. Often it implies a more uncertain condition than **can.**
   e.g. She could refuse, but she never does.
5. **Should** is the past form of **shall**. It is used in sub-ordinate clauses after **in case** and sometimes after **if.**
   e.g. I shall get some money in case brother comes.
   It is also used in past sentences with **so that** and **in order that :**
   e.g. He turned the stereo down very low so that he should not disturb him.

- **Should** can also be used in sub-ordinate clauses when we are expressing the idea that something must be done or is important. The fact is that this happens after verbs like command, order, request, insist, suggest, advise etc. & after adjectives like important, vital, essential, necessary, eager, anxious, concerned etc.
  e.g. I am anxious that nobody should be hurt.

- **Should** is also used in sub-ordinate clauses in sentences where we express personal reactions to events. We express our reactions with words like— amazing, interesting, shocked, sorry, normal, natural, it's a shame etc.
  e.g. I am sorry you should think I did it on purpose.

6. **Must** is followed by the infinitive without **to.**
   e.g. I must get up at five tomorrow.
   Must is used to give strong advice or orders :
   e.g. I really must stop drinking.
   Must is used to say that we are sure about something :
   e.g. I am in love—that must be nice.
   Must is used after a past reporting verb.
   e.g. I felt there must be something wrong.
7. **Should** and **Ought** have similar meanings, but **ought** is followed by **to. Ought to** has a more objective force and is used when we are talking about laws, duties and regulations.
   e.g. We ought to see her tomorrow.
   Should and **ought to** can also be used to talk about strong probability.
   e.g. He has bought thirty pints of whisky—that ought to be enough.
   To talk about things which did not happen, although they were supposed to, we use
   should and **ought to** with the perfect infinitive :
   e.g. The taxi should have arrived at 8.30.
8. **Do** is used to make question and negative forms of ordinary verbs.
   e.g. Do you know Rajat ?
   Do is used in question-tags and short answers
   e.g. Does he know I am here ? —Yes, he does.
   Do is used before an imperative to make the request more persuasive
   e.g. Do accompany us.
   Do is not used in questions which have **who, what or which** as their subject
   e.g. Who said that ?
   What happened ?
   Which one's that ?
   But if **who, what** or **which** is the object of the sentence, **do** is used
   e.g. Who did you see ?
   Which department do you want ?
   Do is also used with an —**ing** form when we want to talk of activity that takes a certain time or that is repeated. In this case we find determiners like —the, my, some, much, etc.
   e.g. I usually do most of my washing on Sundays.

ERRORS IN USE OF ADJECTIVES

Word qualifying a noun or pronoun is called an **adjective.**

1. The adjectives ending in —ior (prior, junior, senior, superior, inferior, posterior) take 'to' and not 'than' after them.
   e.g. He is senior to me.

2. Some adjectives like unique, ideal, perfect, extreme, complete, universal, infinite, perpetual, chief, entire, round, impossible are not compared.
   e.g. It is the most unique book. (×)
   It is a unique book (✓)
3. Comparative degree is used in comparing two things or persons.
   e.g. It is the better of the two books.
   Superlative degree is used in comparing more than two things or persons.
   e.g. He *is the best* of the three boys.
4. Double comparatives and double superlatives must not be used.
   e.g. He is more wiser than his brother. (×)
   He is wiser than his brother. (✓)
5. When we compare two qualities in the same person or thing, the comparative ending —er is not used.
   e.g. You are wiser than old. (×)
   You are more wiser than old. (✓)
6. When two adjectives in superlative or comparative degree are used together, the one formed by adding 'more' or 'most' must follow the other adjective.
   e.g. He is more intelligent and wiser than his brother. (×)
   He is *wiser* and *more intelligent* than his brother. (✓)
7. When two adjectives with differing degrees of comparison are used they should be complete in themselves.
   e.g. He is as wise, if not wiser than his brother. (×)
   He is as wise as, if not wiser than his brother. (✓)
8. When two changes happen together, comparative degree is used in both.
   e.g. The higher you go, the cooler you feel.
9. When comparative degree is used in superlative sense, it is followed by '*any other*'.
   e.g. Kapil is better than any bowler. (×)
   Kapil is better than any other bowler. (✓)
10. Compound adjective formed by adding 'worth' is placed after the noun it qualifies.
    e.g. This is a worth seeing sight. (×)
    This is a sight worth seeing. (✓)
11. When two or more comparatives are joined by **'and'**, they must be in the same degree.
    e.g. Russel was one of the wisest and most learned men of the world.
12. When there are two objects of comparison, then to avoid repetition of noun, 'that' is used for singular noun and 'those' for plural noun.
    e.g. The climate of Ranchi is better than Gaya. (×)
    The climate of Ranchi is better than that of Gaya. (✓)
13. If comparison is made by using 'other', 'than' is used instead of 'but'.
    e.g. He turned out to be no other than my old friend.
14. **Likely, certain** and **sure** are followed by **'to'.**
    e.g. He is likely to win.

CONFUSED ADJECTIVES

1. **Beautiful** is used for woman; **handsome** for man.
   e.g. He is a handsome youth.
   She is a beautiful girl.
2. **Less** refers to quantity, **fewer** denotes number.
   e.g. He takes no less than a litre of milk.
   They have fewer books than I have.
3. **Last** is the final one; **Latest** is last upto the present.
   e.g. Z is the last letter of the alphabet.
   This is the latest edition of the book.
4. **Each** is used for one of two or more things; **every** is used for more than two things, taken as a group.
   e.g. Each of the two boys was wrong.
   He read every book I gave him.
5. **Older** refers to persons or things.
   e.g. This tree is older than that.
   Elder refers to persons only.
   e.g. He is my elder brother.
6. **Little** means 'not much'.
   A little means 'at least some'.
   e.g. He slept little.
   He slept a little.
7. **Farther** means 'more distant'.
   Further means 'additional'.
   e.g. Mumbai is farther than Delhi.
   I shall get further information.
8. **Latter** means the second of two things.
   e.g. *Keats and Byron are romantic poets, but I prefer the latter.*
   Later refers to time.
   e.g. She came to school later than I.

ERRORS IN USE OF ADVERBS

Adverbs are words that add information about the verb.

1. Adverbs of manner, place and time are usually placed after the verb or object.
   e.g. He was running slowly.
   Reena does her work carefully.
2. Adverbs of frequency (e.g. never, often, usually, always, rarely, etc.) and other adverbs (like already, almost, just, quite, nearly, hardly) are normally put between subject and verb. If there is more than one word in the verb, they are put after first word.
   e.g. He often goes to Delhi.
   I quite agree with you.
   But if verb is 'am', 'is' and 'are', adverb is placed after the verb.
   e.g. I am *never* late for school.
3. The adverb *enough* is placed after the adjective.
   e.g. She is cunning enough to tackle him.
4. 'Ever' is sometimes incorrectly used for 'never'.
   e.g. He seldom or ever tells a lie. (×)
   He seldom or never tells a lie. (✓)

5. Adverb 'not' shouldn't be used with words having negative meaning.
   e.g. The teacher forbade me not to go. (×)
   The teacher forbade me to go. (✓)
6. The word 'only' should be placed immediately before the word it modifies.
   e.g. Hari answered only two questions.
7. An adverb should not be used before an infinitive.
   e.g. He quickly did the job. (×)
   He did the job quickly. (✓)
8. The auxiliaries *have to* and *used to* come after the adverb.
   e.g. He often used to go to cinema.

SOME MORE TIPS OF USAGE

1. Avoid the use of **lots, a lot** and **a whole lot** in the sense of **much** or **a great deal.**
   e.g. He expects to earn a lot of money on his sale of farm products. (×)
   He expects to earn a great deal of money on his sale of farm products. (✓)
2. **Plenty** is a noun and is always followed by **of.**
   e.g. He has plenty of room in his old house.
3. **Due to** always modifies a noun and not a verb. Hence, no sentence should begin with **due to,** it must be used only after some form of the verb **to be**.
   e.g. His death was due to natural causes.
4. **Alright** is incorrect, use **all right.**
   e.g. I think it is a quite alright if you stay. (×)
   I think it is quite all right if you stay. (✓)
5. If the gender is not determined, use the pronoun of the masculine gender.
   e.g. If anybody has got the book let her return it (×)
   If anybody has got the book let him return it (✓)
6. When two nouns are closely connected, apostrophe —'s will be written after second only.
   e.g. Ravi's and Shikha's mother is ill. (×)
   Ravi and Shikha's mother is ill. (✓)
7. **Both** and **as well as** cannot be used together in the same sentence because both convey the same sense.
   e.g. Both Pinki as well as Pooja are beautiful (×)
   Both Pinki and Pooja are beautiful. (✓)
8. **Supposing** and **if** cannot be used together in the same sentence.
   e.g. Supposing if he fails, what will he do ?(×)
   If he fails, what will he do ? (✓)
9. **Endure** suggests **suffering,** usually in silence and **tolerate,** means to allow it with some degree of approval.
   e.g. He endured the pain without complaint.
   He will not tolerate laziness.
10. Avoid the use of **had** with **ought.**
    e.g. You had not ougth to do it. (×)
    You ought not do it. (✓)
11. **As** is used when one compares things or persons of equal or about equal size or quality, **so** is used when one compares things or persons which are unequal.
    e.g. He weighs as much as his father.
    He does not weigh so much as his father. (×) (✓)
12. Avoid using the double comparative and double superlative.
    e.g. *This pen is the most costliest in this store.* (×)
    This pen is the costliest in this store. (✓)
13. **Cent per cent** and **word by word** are wrong uses. The real terms are **hundred percent** and **word for word.**
14. When two comparatives are used in a sentence for proportion, **the** is used before both of them.
    e.g. The higher we go, the cooler it is.
15. **Do the needful** is incorrect, write **do what is necessary.**
16. Never say **family members,** say **members of the family.**
    e.g. His family members have gone to Mumbai. (×)
    The members of his family have gone to Mumbai (✓)
17. **Else** is always followed by **but** and never by **than.**
    e.g. It is nothing else but your pride which makes you say such a thing.
18. **All of** is used in conversation, not in standard written English.
    e.g. She gave all of her antiques to the museum (×)
    She gave all her antiques to the museum (✓)
19. **Talking terms** is wrong, use **speaking terms.**
    e.g. He is not on talking terms with his brother (×)
    He is not on speaking terms with his brother (✓)
20. Verbs like resemble, recommend, comprise, order, accompany, reach, join, shirk, attack, emphasise, discuss, fear, succeed, resist, request, assist, benefit, afford etc. are not followed by any preposition when used in the active voice.
    e.g. He emphasised on the need for discipline. (×)
    He emphasised the need for discipline. (✓)
    She will accompany with you to Chennai. (×)
    She will accompany you to Chennai. (✓)
21. The phrase **type of, sort of** and **kind of** do not follow **a** or **an.**
    e.g. What kind of a friend are you ? (×)
    What kind of friend are you ? (✓)
22. A full hour number follows **o'clock,** but in fraction it does not follow.
    e.g. He left this place by 10.40 o'clock. (×)
    He left this place by 10.40. (✓)
23. **Yet** means up to the present time something that has not happened.
    e.g. His brothers are not married even yet.

EXERCISE – I

MCQ TYPE QUESTIONS

Directions (Q. 1 – 20)
In this section you are required to spot errors in sentences. Read each sentence to find out whether there is any error in any of the parts. Errors, if any, are only in one of the parts. No sentence has more than one error. When you detect an error, in any of the parts of the sentence, choose the corresponding alphabet.

1. (a) The scenery here is not good
   (b) I have lost my furnitures
   (c) We have received no information
   (d) He told his mother this news
   (e) No error.
2. (a) Please excuse the trouble
   (b) He took pains over his work
   (c) In India there are many poor
   (d) Scouts wear shorts
   (e) No error.
3. (a) I spent the holidays with my family
   (b) Gone him some blotting paper
   (c) Write this in your note book
   (d) Gone my kind regards to all
   (e) No error.
4. (a) We had a good play of football
   (b) We saw a play
   (c) We like acting
   (d) He is a tall man
   (e) No error.
5. (a) The boy was wearing a new suit
   (b) He took offence at this
   (c) Please put your sign here
   (d) Please put your signature here
   (e) No error.
6. (a) He is my cousin brother
   (b) We are all brothers
   (c) I spent the middle of the day working
   (d) He had a large amount of money
   (e) No error.
7. (a) He got into bad company
   (b) One of my servant tells me
   (c) Some of my servants tell me
   (d) The front of the house
   (e) No error.
8. (a) I had a bathe in the sea
   (b) I have hurt a toe
   (c) Each of these days play games
   (d) None of us went
   (e) No error.
9. (a) A man should work hard
   (b) A man should not waste his time
   (c) A boy should not waste his time
   (d) Here is my cup; please fill it
   (e) No error.
10. (a) Have you a pencil ? I have not got one
    (b) Is he coming ? Yes I think so
    (c) He enjoyed during the holidays
    (d) I asked for my pencil but he did not give it to me
    (e) No error.
11. (a) The boy who does best he will get a prize
    (b) Whoever does best will get a prize
    (c) Who did this ? I
    (d) He and I are brothers
    (e) No error.
12. (a) I went for a walk with some friends
    (b) He is wiser than I
    (c) The master tested the boy if he could read English
    (d) I shall see whether the brakes work well
    (e) No error.
13. (a) Everyone is frightened when he sees a tiger
    (b) None of us has seen him
    (c) People strave when they have no money
    (d) The size of the shoe should be the same as this shoe
    (e) No error.
14. (a) Everyone knows this
    (b) Every man knows this
    (c) These all mangoes are ripe
    (d) He hold the book in both hands
    (e) No error.
15. (a) I have no any friends
    (b) Neither man has come
    (c) All idle man should do some work or other
    (d) Shakespeare is greater than any other poet
    (e) No error.
16. (a) Open your book at page six
    (b) He is in class ninth
    (c) This article costs ten rupees
    (d) He came a second time
    (e) No error.
17. (a) King George VI
    (b) He is older than I
    (c) He is better than I
    (d) Raipur is Hotter thanSimla
    (e) No error.
18. (a) He is worse than I
    (b) A horse is more useful than a car
    (c) He gets a small salary
    (d) I our library the number of books is less
    (e) No error.
19. (a) He is the more clever of the two
    (b) He is the cleverer of the two
    (c) From the three he is more clever
    (d) Of the two plans this is the better
    (e) No error.

20. (a) He is becoming strong
(b) There is a very good teacher in that class
(c) He will spend the rest of his life here
(d) This is a sight worth seeing
(e) No error.

Directions (Q. 21 – 30)
Each question below has a sentence, from the choices provided, identify the one which best restates the given sentence and mark its number as the answer.

21. Uneasy lies the head that wears a crown.
(a) Uneasily lies the head that wears a crown.
(b) Uneasy is the head wearing a crown.
(c) The head which wears the crown lies uneasy.
(d) The head which is wearing the crown lies uneasy.

22. Had I been a little early, I would not have missed the train.
(a) Had I been a little early, I would not have missed the train.
(b) Had I been a little early than now, I would not have missed the train.
(c) Had I been a little earlier, I would not have missed the train.
(d) Had I only been a little early than now, I would not have missed the train.

23. She is as intelligent as her sister if not more.
(a) She is more intelligent than her sister.
(b) She is as intelligent as her sister if not more intelligent.
(c) She and her sister are equally intelligent.
(d) She is less intelligent than her sister.

24. You will not succeed unless you are optimistic.
(a) You will not be successfull unless you are optimistic.
(b) You will not succeed until you are an optimist.
(c) You will not succeed unless you don't be optimistic.
(d) You will not succeed unless you don't be an optimistic.

25. The students were advised to follow the instructions of the examiner.
(a) The instructions of the examiner were followed by the students.
(b) The students followed the examiner's instructions.
(c) The examiner's instructions were advised to be followed by the students..
(d) It has been advised to the students that they should follow the examiner's instructions.

26. Despite being ill, he attended the classes.
(a) Despite of being ill, he attended the classes.
(b) Despite his illness, he attended the classes.
(c) Inspite of his being ill, he attended classes.
(d) The classes were attended by him inspite of being ill.

27. That man is aggressive by nature is a hard fact of life and no one can deny it.
(a) That man is aggressive by nature is a hard fact of life which none can deny.
(b) That man is aggressive by nature is a hard fact of life and no one can deny it.
(c) That man is aggressive by nature is a hard fact of life and not one can deny.
(d) That man is aggressive by nature is hard for anyone to deny.

28. He is as old as I.
(a) He is as old as me.
(b) He and I are equally old.
(c) He and I are equally older.
(d) Both he and I are of the same age.

29. A man becomes wiser with age and experience.
(a) A man as he is aged and experienced becomes wiser.
(b) A man with age and experience becomes wiser.
(c) A man as he is aged and experienced becomes the wiser.
(d) With age and experience, a man becomes wiser.

30. Hazards of life cannot be negated but they can be quite effortlessly evaded.
(a) Hazards in life can be quite effortlessly evaded and completely nullified.
(b) Hazards of lilfe cannot be avoided but they can be made ineffective.
(c) To be made ineffective, hazards of life must be avoided.
(d) It may not be possible to nullify the hazards of life but they can quite easily be avoided.

Directions(Q. 31 –40)
Each sentence is broken into four parts a, b, c, d. Mark the part which has an error. Ignore errors of punctuation.

31. (a) Every man, woman
(b) and child
(c) in the house on fire
(d) have been saved.

32. (a) One of the
(b) best lawyers in town
(c) have been
(d) hired.

33. (a) I request you (b) kindly to
(c) comc to me (d) immediately.

34. (a) My friend's mother
(5) is the principal
(c) of a
(d) girl's college.

35. (a) To succeed in these tests
(b) it is absolutely necessary
(c) for us
(d) tc aim for speed and accuracy.

36. (a) The aeroplane is a
(b) powerful instrument of war
(c) and their peacetime role
(d) is just as important.

37. (a) The depletion of ozone in the atmosphere
(b) is a reminder to all of us
(c) both in the developed and developing world
(d) that they cannot continue to use contraptions which give out noxious gases.

38. (a) Scotsmen are notorious
(b) for their thrift
(c) and he is the butt
(d) of many exaggerated jokes.

39. (a) A mixed economy
(b) is it in which
(c) the public sector
(d) and the private sector co - exist.

40. (a) Although the beaver's hind feet
(b) are webbed for swimming
(c) their front feet
(d) are small and handlike.

Directions(Q. 41 – 45)
Given below are five sentences, each of which may or may not have errors.
Mark (a), if there is only one error.
Mark (b), if there are two errors.
Mark(c), if there are more than two errors.
Mark (d), if there is no error.

41. "This feels like one big dream," says a weary American woman to her husband, as they sit in a plane flying home.

42. The worst mistake a non - profit organization can make is to take all the money it is given and be beholden to doing things on someone else's terms.

43. Although most hotels in Bhutan has a distinctly local flavour, some pander with the perceived tastes of Western tour groups - and can be rather bland.

44. The crustaceans find sanctuary in the jords, often clinging to the walls in large, gregarious groups.

45. In these tropical lowlands the Maya built an agrarian civilization that eventually supported one of the highest population densities in the pre - industrial America.

Directions (Q. 46–50)
Each of the following questions has two sentences A and B.
Mark (a), if you think sentence A has an error.
Mark (b), if you think sentence B has an error.
Mark (c), if you think both sentences A and B have errors.
Mark (d), if you think neither sentence has an error.

46. A. The boss himself or his secretary answer the phone on Saturday.
B. Neither the quality nor the prices have changed.

47. A. The members re-commended that all delinquents be fined.
B. A stone lying in one position for a long time may gather moss.

48. A. Until I received that letter, I was hoping to have had a visit from Krishnan.
B. Follow the main road for a mile; then you need to take the next road at the left.

49. A. The people to watch closely are the ones ruling behind the political scene.
B. Give the tools to whoever can use them best.

50. A. The magazine has been published continuous since 1951, and it has the funniest cartoons you can possibly imagine.
B. I feel glumly when I read his nightmarish tales.

EXERCISE – II

(QUESTIONS FROM PREVIOUS GATE EXAMS)

MCQ TYPE QUESTIONS

2015

1. Choose the statement where underlined word is used correctly
(a) The minister insured the victims that everything would be all right.
(b) He ensured that the company will not have to bear any loss.
(c) The actor got himself ensured against any accident.
(d) The teacher insured students of good results

2. The following question presents a sentence, part of which is underlined. Beneath the sentence you find four ways of phrasing the underline part. Following the requirements of the standard written English, select the answer that produces the most effective sentence.
Tuberculosis, together with its effects, ranks one of the leading causes of death in India.
(a) ranks as one of the leading causes of death
(b) rank as one of the leading causes of death
(c) has the rank of one of the leading causes of death
(d) are one of the leading causes of death

3. Choose the statement where underlined word is used correctly.
(a) When the teacher eludes to different authors, he is being elusive
(b) When the thief keeps eluding the police, he is being elusive
(c) Matters that are difficult to understand, identify or remember are allusive
(d) Mirages can be allusive, but a better way to express them is illusory

4. Fill in the blank with the correct idiom/phrase.
That boy from the town was a _____ in the sleepy village.
(a) Dog out of herd (b) Sheep from the heap
(c) Fish out of water (d) Bird from the flock

5. Select the appropriate option in place of underlined part of the sentence.
Increased productivity necessary reflects greater efforts made by the employees.
(a) Increase in productivity necessary
(b) Increase productivity is necessary
(c) Increase in productivity necessarily
(d) No improvement required

6. Ram and Shyam shared a secret and promised to each other that it would remain between them. Ram expressed himself in one of the following ways as given in the choices below. Identify the correct way as per standard English.
(a) It would remain between you and me
(b) It would remain between me and you
(c) It would remain between you and I
(d) It would remain with me

7. What is the adverb for the given word below?
Misogynous
(a) Misogynousness (b) Misogynity
(c) Misogynously (d) Misogynous

8. In the following sentence certain parts are underlined and marked P, Q and R. One of the parts may contain certain error or may not be acceptable in standard written communication. Select the part containing an error. Choose D as your Answer: if there is no error.
The student corrected all the errors (P) that the instructor marked (Q) on the answer book (R)
(a) P (b) Q
(c) R (d) No Error

9. Choose the statement where underlined word is used correctly.
(a) The industrialist had a personnel jet.
(b) I write my experience in my personnel diary.
(c) All personnel are being given the day off.
(d) Being religious is a personnel aspect.

10. Out of the following four sentences, select the most suitable sentence with respect to grammer and usage:
(a) Since the report lacked needed information, it was of no use to them.
(b) The report was useless to them because there were no needed information in it.
(c) Since the report did not contain the needed information, it was not real useful to them.
(d) Since the report lacked needed information, it would not had been useful to them.

11. Select the alternative meaning of the underlined part of the sentence.
The chain snatchers took to their heels when the police party arrived.
(a) took shelter in a thick jungle
(b) open indiscriminate fire
(c) took to flight
(d) unconditionally surrendered

2014

12. Which of the following options is the closest in meaning to the word underlined in the sentence below?
In a democracy, everybody has the freedom to disagree with the government.
(a) dissent (b) descent
(c) decent (d) decadent

13. 'Advice' is _____.
(a) a verb (b) a noun
(c) an adjective (d) both a verb and a noun

14. Which of the options given below best completes the following sentence?
She will feel much better if she _____.
(a) will get some rest
(b) gets some rest
(c) will be getting some rest
(d) is getting some rest

15. While trying to collect (I) an envelope from under the table, (II) Mr. X fell down (III) and was losing consciousness. (IV)
Which one of the above underlined parts of the sentence is NOT appropriate?
(a) I (b) II
(c) III (d) IV

16. Which of the following options is the closest in meaning to the phrase underlined in the sentence below?
It is fascinating to see life forms cope with varied environmental conditions.
(a) Adopt to
(b) Adapt to
(c) Adept in
(d) Accept with

17. In a press meet on the recent scam, the minister said, "The buck stops here". What did the minister convey by the statement?
(a) He wants all the money
(b) He will return the money
(c) He will assume final responsibility
(d) He will resist all enquiries

2013

18. The professor (I) ordered to (II) the students to go (III) out of the class. (IV)
Which of the above underlined parts of the sentence is grammatically incorrect?
(a) I (b) II
(c) III (d) IV

19. Choose the grammatically INCORRECT sentence:
(a) He is of Asian origin.
(b) They belonged to Africa.
(c) She is an European.
(d) They migrated from India to Australia.

20. Choose the grammatically CORRECT sentence:
(a) Two and two add four.
(b) Two and two become four.
(c) Two and two are four.
(d) Two and two make four.

21. All engineering students (I) should learn mechanics, (II) mathematics and (III) how to do computation. (IV)
Which of the above underlined parts of the sentence is not appropriate?
(a) I (b) II
(c) III (d) IV

2012

22. Choose the most appropriate alternative from the options given below to complete the following sentence :
Suresh's dog is the one _____ was hurt in the stampede.
(a) that (b) which
(c) who (d) whom

23. Choose the grammatically **INCORRECT** sentence :
(a) They gave us the money back less the service charges of Three Hundred rupees.
(b) This country's expenditure is not less than that of Bangladesh.
(c) The committee initially asked for a funding of Fifty Lakh rupeesr, but later settled for a lesser sum.
(d) This country's expenditure on educational reforms is very less.

24. One of the parts (A, B, C, D) in the sentence given below contains an ERROR. Which one of the following is **INCORRECT** ?
I requested that he should be given the driving test today instead of tomorrow.
(a) requested that
(b) should be given
(c) the driving test
(d) instead of tomorrow

ANSWERS

EXERCISE – I

MCQ Type Questions

1. (b)	2. (a)	3. (e)	4. (a)	5. (c)	6. (a)	7. (b)	8. (c)	9. (e)	10. (c)
11. (a)	12. (c)	13. (d)	14. (c)	15. (a)	16. (b)	17. (d)	18. (d)	19. (c)	20. (a)
21. (c)	22. (a)	23. (c)	24. (a)	25. (d)	26. (b)	27. (a)	28. (d)	29. (d)	30. (d)
31. (d)	32. (c)	33. (b)	34. (d)	35. (d)	36. (c)	37. (d)	38. (c)	39. (b)	40. (c)
41. (a)	42. (d)	43. (b)	44. (d)	45. (d)	46. (a)	47. (d)	48. (c)	49. (d)	50. (c)

EXERCISE – II

MCQ Type Questions

1. (b)	2. (a)	3. (b)	4. (c)	5. (c)	6. (a)	7. (c)	8. (b)	9. (c)	10. (a)
11. (c)	12. (a)	13. (b)	14. (b)	15. (d)	16. (b)	17. (c)	18. (b)	19. (c)	20. (d)
21. (d)	22. (a)	23. (d)	24. (b)						

EXERCISE – I

MCQ TYPE QUESTIONS

21. Uneasily does not denote a state, which is required in the context. 'Wearing a crown' denotes that the head is doing it, 'is wearing' denotes a continuous action, which is impossible, as the head cannot do it. Hence choice (*c*) restates the given sentence correctly.

22. All the statements have the correct combination of tenses. So, we need not check it. Let us find the errors in other statements. 'Early than now' and 'earlier' suggest that I reached early; in that case, I did not miss the train. So, choices (*b*), (*c*), and (*d*) do not state the same idea as in the question statement. Hence correct choice is (*a*).

23. The given statement states that both she and her sister are equally intelligent. This is given in statement. Hence correct choice is (*c*).

24. 'Unless' denotes condition, 'until' denotes time. Choice (*b*) has adverb clause of time.With unless, another negative is not used. 'Unless ..... don't' is wrong. So, Choices (*c*) and (*d*) are incorrect. Hence correct choice is (*a*).

25. Choices (*a*) and (*b*) state the same idea [(*a*) – passive voice; (*b*) Active voice] that the students followed the instructions, which differs from the question statement. Instructions cannot be advised. So, choice (*c*) is meaningless. Choice (*d*) states the context of the given statement aptly.

26. Despite means inspite of. Despite of is a wrong usage. 'His' need not be used in choice (*c*), as the sentence is about one person. Choice (*d*) means that the classes were ill. Hence correct choice is (*b*).

27. Choice (*a*) best restates the given sentence, though choice (*b*) is the repetition of the given statement. 'Not one can deny' in choice (*c*) and for any one to deny' in choice (*d*) make them incorrect. Hence correct choice is (*a*).

28. In this type of comparison, the pronouns are in subjective case 'Me', which is in objective case is wrongly used in choice (*a*). Equally is not used with 'old'. Of the same age is the correct expression. Hence correct choice is (*d*).

29. Choice (*a*) and (*b*) mean that the prerequisite for wisdom is age and experience, which is incorrect. Just by growing old and working for a long period, a person cannot improve his wisdom. 'The wiser' is used only when we compare wisdom with some other quality. Choice (*d*) is the correct restatement of the given one. Hence correct choice is (*d*).

30. The given statement says that one cannot completely nullify the hazards of life but they can be evaded quite easily. Choice (*a*) which states that hazards cannot be avoided is wrong (*b*) states that hazards can be made ineffective and hence this is also wrong (*c*) does not convey the meaning as stated and hence it too is incorrect. (*d*) correctly restates the idea and is our answer.

31. Every man refers to a singular noun, therefore.... has been saved.

32. has

33. drop 'kindly'

34. Girl's implies possessive noun, whereas the college is meant for girls.

35. aim at, aim for is wrong usage.

36. delete their - add : its

37. delete they - add : we

38. delete he is - add : they are.

39. delete it - add : that

40. delete their - add : its.

41. One sits 'on' a plane.

42. There are no errors.

43. Hotels 'have' and some pander 'to....'

44. There are no errors in this sentence.

45. There are no errors.

48. (A) hoping to have a visit;
(B) .... then take the next road to

50. (A) continuously; (B) glum.

EXERCISE – II

MCQ TYPE QUESTIONS

1. *Insured*–the person, group, or organization whose life or property is covered by an insurance policy.
Ensured–to secure or guarantee

3. Elusive: Difficult to answer.

4. From the statement, it appears that boy found it tough to adapt to a very different situation.

8. The is not required in 'Q'.

22. The correct answer is 'that'. In the given sentence 'was hurt in the stampede' determines which dog belongs to Suresh.

■■

CHAPTER **2**

Sentence Completion

WHAT ARE SENTENCE COMPLETION PROBLEMS?

Sentence completion problems test your vocabulary skills as well as your reading ability. Unlike the reading comprehension questions, which require you to read long passages, these problems contain a single sentence expressing a complete idea that can be understood without any additional information. Each sentence contains one or two blanks, indicating that one or two words are missing. You are then presented with five words or phrases, or five pairs of words or phrases if there are two blanks in the sentence. From these choices, you need to select the words or phrases that fit into the blanks to best complete the sentence. To make the correct choice, you will need to be able to understand the main idea of the sentence and the logical structure of the sentence. Next, you will need to know the definitions of the words in the answer choices.

BASIC STRATEGIES FOR SENTENCE COMPLETION PROBLEMS

Strategy 1

The first strategy is to break the sentence down into the main clause and introductory clause or phrase in order to identify the relationship it expresses. For example, consider the following:

In a hopsital setting, hand washing is primarily a_____measure.

(*a*) prophylactic (*b*) pseudocyetic
(*c*) protensive (*d*) propaedeutic

The main clause declares that hand washing is primarily a measure. However, preceding the declaration is a condition or constraint expressed in an introductory phrase, "a hospital setting." This phrase sets the contextual field within which the main clause is to be understood. If you generalize from your knowledge, it would seem that any measure promoted in a hospital should result in good health. Only one of the answers would result in this. That is answer (*a*) in which *prophylactic* means "prevents disease."

Strategy 2

A second strategy is to attend to sets, whether a set of synonyms or antonyms or special word groups. For example, consider the following:

Even when a_____ mother was made available to the infant chimpanzee, the infant detected the_____and reacted with high anxiety.

(*a*) hedonistic.. contrivance
(*b*) vagrant.. apostasy
(*c*) veteran.. ruse
(*d*) pretender.. interference

Nothing in a common sense association supports the pairing of *hedonistic* and *contrivance*, *vagrant* and *apostasy*, *veteran* and *ruse*, or *pretender* and *interference*. The only set of related words is *surrogate* (meaning substitute) and *substitution*.

Strategy 3

Pay attention to the words that control the figurative field of the sentence. Consider the following passage:

In seventeenth-century France, private and public opinion portrayed the physician as a vulture of greed, circling the sick and dying, battening off the suffering of the afflicted, and _____the remains of his prey.

(*a*) spending (*b*) devouring
(*c*) retreating from (*d*) decrying

Clearly the correct answer is (*b*), devouring. The key words that control the figurative field are *vulture*, *circling*, and *battening*. Read the items carefully and make sure that the completed sentence is meaningful, that it makes sense.

EXERCISE – I

MCQ TYPE QUESTIONS

Directions for Q. 1 to 41 : *In each of the following sentences, a part/parts of the sentence is left unfinished. Beneath each sentence, four different ways of completing the sentence are indicated. Choose the best alternative among the four.*

1. For some of our ruling politicians, social justice has apparently come to mean that _____.
(*a*) they have a right to plunder public money from the treasury
(*b*) they have a right to the plunder of public money from the treasury
(*c*) theirs is the right to plunder public money from the treasury
(*d*) the right is theirs to plunder public money from the treasury

2. Some critics believe that Satyajit Ray never quite came back to the great beginning he made in his path-breaking film Pather Panchali. _____ have endured decades of well-travelled bad prints to become a signpost in cinematic history.
(*a*) The bizarre history of its misty origins
(*b*) Its haunting images
(*c*) Its compelling munificence
(*d*) The breathtaking awe it inspires

3. _____ , the more they remain the same.
(*a*) The less the dynamism
(*b*) The more things change
(*c*) The more pronounced the transformation
(*d*) The more the merrier

4. Revenues are likely to register a dramatic increase if _____ on the hundred million people who are said to comprise the risingIndian middle class.
   (a) a flat tax is collected
   (b) a flat tax is placed
   (c) taxes at a flat rate are charged
   (d) tax is imposed at a flat rate
5. I am an entertainer. _____ I have to keep smiling because, deep in my heart, laughter and sorrow have an affinity.
   (a) Even if I have tears in me
   (b) Despite conditions of extreme adversity
   (c) Inspite of misery around me
   (d) Although I have yet to make it big
6. The stock market is probably _____ And the way the markets are plunging says a lot about investor confidence.
   (a) an ideal indication of the health of public sentiment
   (b) the least imperfect mechanism for judging the quantity of the sentiment of the public.
   (c) the best indicator of public sentiment
   (d) the best barometer to assess the sentiment of the public
7. Home is _____ one has yet to improve.
   (a) a discovery over which
   (b) a certain over which
   (c) an invention on which
   (d) an institution upon which
8. The highest reward for a man's toil is not what he gets for it but what _____.
   (a) he makes out of it.
   (b) he gets for others.
   (c) he has overcome through it.
   (d) he becomes by it.
9. An act of justice closes the book on a misdeed; an act of vengeance _____.
   (a) reopens the first chapter
   (b) writes an epoilogue
   (c) writes one of its own
   (d) opens new books
0. The trail is the thing , not the end of the trail. Travel too fast and you miss _____.
   (a) all you are travelling for.
   (b) all the sights you are supposed to see.
   (c) the very excitement of your travel.
   (d) all the enjoyments of travel.
   _____ that in this apparent mess, two things not be interfered with.
   (a) It is important.
   (b) it is of cardinal important.
   (c) It should be urgently under-stood.
   (d) It cannot be emphasised.
12. A country's wealth is its people. But instead of drawing out the strengths of the people, instead of drawing out their talent, this use of religion debases, degrades and depresses than _____.
   (a) in greater and greater measure
   (b) further
   (c) beyond reasonable limit
   (d) more and more
13. Wines that yielded a good commercial profit _____ in the same limited areas of France as of now.
   (a) seem to have been produced
   (b) appear to have a remarkable semblance
   (c) bear a significant similarity in terms of production to those grown
   (d) appear to have been similarly produced
14. This is about _____ a sociological analysis can penetrate.
   (a) the other limits that
   (b) just how far into the subject that
   (c) the relative distance that
   (d) as far as
15. The safest general characterisation of the European philosophical tradition as it has developed up to now, with all its diverse proponents, is that it consists of a_____Plato.
   (a) series of footnotes to
   (b) set of prologues to
   (c) collection of chapters on
   (d) string of commentatries to
16. The ideas that these companies used seem so simple with _____ that their competitors will now _____ themselves for not thinking of them first.
   (a) advanced technology, hit
   (b) analysis , abuse
   (c) passage of time, curse
   (d) hindsight, kick
17. No doubt, it was our own government but it was being run on borrowed ideas, using _____solutions.
   (a) worn out
   (b) second hand
   (c) impractical
   (d) appropriate
18. Overall, all the recent policy changes by the government only amount to a _____ in the sugar industry.
   (a) superficial attemp at liberalisation
   (b) drop in the ocean
   (c) large change
   (d) small regulating authority

19. In pusuance of their decision to resist what they saw as anti-labour policies, the company employees, union launched agitation to _____.
   (a) show their virility
   (b) reaffirm their commitment to the company.
   (c) bring down the government
   (d) demonstrate their strength
20. Scientists have found _____ evidence that Neanderthals, now an extinct species of humans _____ Europe, were cannibals.
   (a) grisly, living in
   (b) incontrovertible, a cross
   (c) chilling inhabiting
   (d) proper, in
21. The telephone symbolises that awkard _____ in all communication technologies ; while it _____ to bring us together, it keeps us apart.
   (a) paradox, needs (b) irony, intends
   (c) paradox , tries (d) irony wishes
22. The world is fast moving into a period of 'civilisational clash' in which the primary identification of people and countries will not be _____ as during the Cold War, but _____.
   (a) ideological, technological
   (b) cultural, ethnic
   (c) political, economic
   (d) ideological, cultural
23. In our country, the challenges are to raise _____ incomes to reduce poverty, and to _____inefficient public sector enterprises.
   (a) farm, liberalise
   (b) middle-class, privatise
   (c) workers, take over
   (d) rural, restructure
24. The interest generated by the Soccer World Cup is _____ compared to the way cricket _____ the nation.
   (a) milder, fascinates
   (b) lukewarm, electrifies
   (c) tepid, inspires
   (d) unusual, grips
25. Though one eye is kept firmly on the _____, the company now also promotes _____contemporary art.
   (a) present, experimental
   (b) future, popular
   (c) present, popular
   (d) market, popular
26. The law prohibits a person from felling a sandalwood tree, even if it grows on one's own land, without prior permission from the government. As poor people cannot deal with the government, this legal provision leads to a rip-roaring business for _____, who care neither for the _____, nor for the trees.
   (a) middlemen, rich (b) the government, poor
   (c) touts, rich (d) touts, poor
27. It will take some time for many South Koreans to _____ the conflicting images of North Korea, let alone to _____ what to make of their northern cousins.
   (a) reconcile, decide (b) understand, clarify
   (c) make out, decide (d) reconcile, understand
28. In these bleak and depressing times of _____ prices, non-performing governments and _____ crime rates, Sourav Ganguly has given us, Indians, a lot to cheer about.
   (a) escalating, increasing
   (b) spiralling, booming
   (c) spiralling, soaring
   (d) ascending, debilitating
29. The manners and _____ of the nouveau riche is a recurrent _____ in the literature.
   (a) style, motif (b) morals, story
   (c) wealth, theme (d) morals, theme
30. Many people suggest _____and still others would like to convince people not to buy pirated cassettes.
   (a) to bring down audio-cassette prices to reduce the incidence of music piracy, others advocate strong legal action against the offenders,
   (b) bringing down audio-cassette prices to reduce the incidents of music piracy, others are advocating strong legal action against offenders,
   (c) bringing down audio-cassette prices to reduce the incidence of music piracy, others advocate strong legal action against offenders,
   (d) audio-cassette prices to be brought down to reduce incidence of music piracy, others advocate that strong legal action must be taken against offenders,
31. The ancient Egyptians believed_____so that when these objects were magically reanimated through the correct rituals, they would be able to function effectively.
   (a) that it was essential that things they portrayed must have every relevant feature shown as clearly as possible
   (b) it was essential for things they portray to have had every relevant feature shown as clearly as possible,
   (c) it was essential that the things they portrayed had every relevant feature shown as clearly as possible,
   (d) that when they portrayed things, it should have every relevant feature shown as clearly as possible

32. Archaeologists believe that the pieces of red-ware pottery excavated recently near Bhavnagar and _____shed light on a hither to dark 600-year period in the Harappan history of Gujarat.
(a) estimated with a reasonable certainty as being about 3400 years old,
(b) are estimated reasonably certain to be about 3400 years old
(c) estimated at about 3400 years old with reasonable certainty,
(d) estimated with reasonable certainty to be about 3400 years old,

33. The genocides in Bosnia and Rwanda, apart from being mis-described in the most sinister and _____manner as 'ethnic cleansing', were also blamed, in further hand-washing rhetoric, on something dark and interior to _____ and perpetrators alike.
(a) innovative; communicator
(b) enchanting; leaders
(c) disingenuous; victims
(d) exigent; exploiters

34. As navigators, calendar makers, and other ____of the night sky accumulated evidence to the contrary, ancient astronomers were forced to_____that certain bodies might move in circles about points, which in turn moved in circles about the earth.
(a) scrutinizers; believe
(b) observers; agree
(c) scrutinizers; suggest
(d) observers; concede

35. Every human being, after the first few days of his life, is a product of two factors: on the one hand, there is his _____ endowment; and on the other hand, there is the effect of environment, including _____.
(a) constitutional; weather
(b) congenital; education
(c) persona!; climate
(d) genetic; pedagogy

36. Exhaustion of natural resources, destruction of individual initiative by governments, control over man's minds by central _____ of education and propaganda are some of the major evils which appear to be on the increase as a result of the impact of science upon minds suited by _____ to an earlier kind of world.
(a) tenets; fixation
(b) aspects; inhibitions
(c) institutions; inhibitions
(d) organs; tradition

37. But _____ are now regularly written not just for tools, but well-established practices, organisations and institutions, not all of which seem to be _____ away.
(a) reports, withering (b) stories, trading
(c) books, dying (d) obituaries, fading

38. The Darwin who _____ is most remarkable for the way in which he _____ the attributes of the world class thinker and head of the household.
(a) comes, figures
(b) arises, adds
(c) emerges, combines
(d) appeared, combines

39. Since her face was free of _____ there was no way to _____ if she appreciated what had happened.
(a) make-up, realise
(b) expression, ascertain
(c) emotion, diagnose
(d) scars, understand

40. In this context, the _____ of the British labour movement is particularly _____.
(a) affiar, weird
(b) activity, moving
(c) experience, significant
(d) atmosphere, gloomy

41. Indian intellectuals may boast, if they are so inclined, of being _____ to the most elitist among the intellectual _____ of the world.
(a) subordinate, traditions
(b) heirs, cliques
(c) ancestors, societies
(d) heir, traditions

Directions for Q. 42 to 47.

Fill the gaps in the passages below with the most appropriate word from the options given for each gap. The right words are the ones used by the author. Be guided by the author's overall style and meaning when you choose the answers.

Von Nuemann and Morgenstern assume a decision framework in which all options are thoroughly considered, each option being independent of others, with a numerical value derived for the utility of each possible outcome (these outcomes reflecting, in turn, all possible combinations of choices). The decision is then made to maximize the expected utility.

(42)_____, such a model reflects major simplifications of the way decisions are made in the real world. Humans are not able to process information as quickly and effectively as the model assumes; they tend not to think (43) _____ as easily as the model calls for; they often deal with a particular option without really assessing its (44) _____, and when they do assess alternatives, they may be extremely nebulous about their criteria of evaluation.

42. (a) Regrettably (b) Firstly
(c) Obviously (d) Apparently
43. (a) quantitatively (b) systematically
(c) scientifically (d) analytically
44. (a) implications (b) disadvantages
(c) utility (d) alternatives

In a large company, (45) _____ people is about as common as using a gun or a switch-blade to (46) _____ an argument. As a result, most managers have little or no experience of firing people, and they find it emotionally traumatic; as result, they often delay the act interminably, much as an unhappy spouse will prolong a bad marriage. And when the firing is done, it's often done clumsily, with far worse side effects than are necessary.

Do the world-class softward organizations have a different way of firing people ? No, but they do the deed swiftly, humanely, and professionally.

The key point here is to view the fired employees as a "failed product" and to ask how the process (47) _____ such a phenomenon in the first place.

45. (a) dismissing (b) punishing
(c) firing (d) admonishing
46. (a) resolve (b) thwart
(c) defeat (d) close
47. (a) derived (b) engineered
(c) produced (d) allowed

Directions for Q. 48 to 57 :

Fill up the blanks, numbered [48], [49]....up to [57], in the two passages below with the most appropriate word from the options given for each blank.

"Between the year 1946 and the year 1955, I did not file any income tax returns." With that [48] statement, Ramesh embarked on an account of his encounter with the Income Tax Department. "I originally owed Rs. 20,000 in unpaid taxes. With [49] and [50], the 20,000 became 60,000. The Income Tax Department then went into action, and I learned first hand just how much power the Tax Department wields. Royalties and trust funds can be [51]; automobiles may be [52], and auctioned off. Nothing belongs to the [53] until the case is settled."

48. (a) devious (b) blunt
(c) tactful (d) pretentious
49. (a) interest (b) taxes
(c) principal (d) returns
50. (a) sanctions (b) refunds
(c) fees (d) fines
51. (a) closed (b) detached
(c) attached (d) impounded
52. (a) smashed (b) seized
(c) dismantled (d) frozen
53. (a) purchaser (b) victim
(c) investor (d) offender

At that time the White House was as serene as a resort hotel out of season. The corridors were [54]. In the various offices, [55] gray men in waistcoats talked to one another in low-pitched voices. The only colour, or choler, curiously enough, was provided by President Eisenhower himself. Apparently, his [56] was easily set off; he scowled when he [57] the corridors.

54. (a) striking (b) hollow
(c) empty (d) white
55. (a) quiet (b) faded
(c) loud (d) stentorian
56. (a) laughter (b) curiosity
(c) humour (d) temper
57. (a) paced (b) strolled
(c) stormed (d) prowled.

EXERCISE - II

(QUESTIONS FROM PREVIOUS GATE EXAMS)

MCQ TYPE QUESTIONS

2015

1. Choose the most appropriate word from the options given below to complete the following sentence. The official answered _____ that the complaints of the citizen would be looked into.
(a) respectably (b) respectfully
(c) reputably (d) respectively

2. Choose the most appropriate word from the options given below to complete the following sentence.
The principal presented the chief guest with a _____, as token of appreciation.
(a) momento (b) memento
(c) momentum (d) moment

3. Choose the appropriate word/phrase, out of the four options given below, to complete the following sentence :
Frogs _____.
(a) Croak (b) Roar
(c) Hiss (d) Patter

4. Extreme focus on syllabus and studying for tests has become such a dominant concern of Indian students that they close their minds to anything _____ to the requirements of the exam
(a) related
(b) extraneous
(c) outside
(d) useful

5. The Tamil version of _____ John Abraham-starrer Madras café _____ cleared by the Censor Board with no cuts last week but the film's distributors _____ no takers among the exhibitors for a release in Tamil Nadu _____ this Friday.
(a) Mr., was, found, on
(b) a, was, found, at
(c) the, was, found, on
(d) a, being, find at

6. Choose the appropriate word/phase, out of the four options given below, to complete the following sentence:
Apparent lifelessness _____ dormant life.
(a) harbours (b) lead to
(c) supports (d) affects

7. Choose the correct verb to fill in the blank below :
Let us _____.
(a) Introvert (b) Alternate
(c) Atheist (d) Altruist

8. Choose the most appropriate word from the options given below to complete the following sentence :
If the Athlete had wanted to come first in the race, he _____ several hours everyday
(a) should practice
(b) should have practiced
(c) practiced
(d) should be practicing

9. Choose the appropriate word/phrase, out of the four options given below, to complete the following sentence:
Dhoni, as well as the other team members of Indian team, _____ present on the occasion.
(a) were (b) was
(c) has (d) have

10. We _____ our friend's birthday and we _____ how to make it up to him.
(a) completely forgot — don't just known
(b) forget completely — don't just know
(c) completely forget —just don't know
(d) forgot completely —just don't know

11. Which of the following options is the closest in meaning to the sentence below?
She enjoyed herself immensely at the party.
(a) She had a terrible time at the party
(b) She had a horrible time at the party
(c) She had a terrific time at the party
(d) She had a terrifying time at the party

12. Didn't you buy _____ when you went shopping?
(a) any paper (b) much paper
(c) no paper (d) a few paper

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13. Choose the most appropriate word from the options given below to complete the following sentence:
A person suffering from Alzheimer's disease _____ short-term memory loss.
(a) experienced
(b) has experienced
(c) is experiencing
(d) experiences

14. Choose the most appropriate word from the options given below to complete the following sentence:
_____ is the key to their happiness; they are satisfied with what they have.
(a) Contentment
(b) Ambition
(c) Perseverance
(d) Hunger

15. Choose the most appropriate word from the options given below to complete the following sentence:
One of his biggest _____ was his ability to forgive.
(a) vice (b) virtues
(c) choices (d) strength

16. After the discussion, Tom said to me, 'Please revert!' He expects me to _____.
(a) retract (b) get back to him
(c) move in reverse (d) retreat

17. The value of one U.S. Dollar is 65 Indian Rupees today, compared to 60 last year. The Indian Rupee has _____.
(a) Depressed (b) Depreciated
(c) Appreciated (d) Stabilized

18. Choose the most appropriate word from the options given below to complete the following sentence:
Communication and interpersonal skills are _____ important in their own ways.
(a) each (b) both
(c) all (d) either

19. Choose the most appropriate pair of words from the options given below to complete the following sentence:
She could not _____ the thought of _____ the election to her bitter rival.
(a) bear, loosing
(b) bare, loosing
(c) bear, losing
(d) bare, losing

20. Choose the most appropriate phrase from the options given below to complete the following sentence:
The aircraft _____ take off as soon as its flight plan was filed.
(a) is allowed to (b) will be allowed to
(c) was allowed to (d) has been allowed to

21. Choose the most appropriate word from the options given below to complete the following sentence:
Many ancient cultures attributed disease to supernatural causes. However, modern science has largely helped _____ such notions.
(a) impel (b) dispel
(c) propel (d) repel

22. If she _____ how to calibrate the instrument, she _____ done the experiment.
(a) knows, will have
(b) knew, had
(c) had known, could have
(d) should have known, would have

23. Who _____ was coming to see us this evening ?
(a) you said
(b) did you say
(c) did you say that
(d) had you said

24. Choose the most appropriate word from the options given below to complete the following sentence:
He could not understand the judges awarding her the first prize, because he thought that her performance was quite _____.
(a) Superb
(b) Medium
(c) Mediocre
(d) Exhilarating

2013

25. Friendship, no matter how _____ it is, has its limitations.
(a) cordial (b) intimate
(c) secret (d) pleasant

26. Were you a bird, you_____ in the sky.
(a) would fly (b) shall fly
(c) should fly (d) shall have flown

27. Complete the sentence:
Dare _____ mistakes.
(a) commit (b) to commit
(c) committed (d) committing

28. The Headmaster _____ to speak to you.
Which of the following options is incorrect to complete the above sentence?
(a) is wanting (b) wants
(c) want (d) was wanting

2012

29. Choose the most appropriate alternative from the options given below to complete the following sentence :
Despite several _____ the mission succeeded in its attempt to resolve the conflict.
(a) attempts (b) setbacks
(c) meetings (d) delegations

30. Choose the most appropriate alternative from the options given below to complete the following sentence :
If the tired soldier wanted to lie down, he ___ the mattress out on the balcony.
(a) should take
(b) shall take
(c) should have taken
(d) will have taken

31. Choose the most appropriate word from the options given below to complete the following sentence :
Given the seriousness of the situation that he had to face, his ___ was impressive.
(a) beggary (b) nomenclature
(c) jealousy (d) nonchalance

2011

32. Choose the most appropriate word from the options given below to complete the following sentence:
If you are trying to make a strong impression on your audience, you cannot do so by being understated, tentative or _____ .
(a) hyperbolic
(b) restrained
(c) argumentative
(d) indifferent

33. Choose the most appropriate word(s) from the options given below to complete the following sentence.
I contemplated _____ Singapore for my vacation but decided against it.
(a) to visit (b) having to visit
(c) visiting (d) for a visit

34. Choose the most appropriate word from the options given below to complete the following sentence:

Under ethical guidelines recently adopted by the Indian Medical Association, human genes are to be manipulated only to correct diseases for which_____treatments are unsatisfactory.

(*a*) similar (*b*) most

(*c*) uncommon (*d*) available

35. Choose the most appropriate word from the options given below to complete the following sentence:

It was her view that the country's problems had been_____by foreign technocrats, so that to invite them to come back would be counter-productive.

(*a*) identified (*b*) ascertained

(*c*) exacerbated (*d*) analysed

2010

36. Choose the most appropriate word from the options given below to complete the following sentence:

If we manage to _____ our natural resources, we would leave a better planet for our children.

(*a*) uphold

(*b*) restrain

(*c*) cherish

(*d*) conserve

37. Choose the most appropriate word from the options given below to complete the following sentence:

His rather casual remarks on politics _____his lack of seriousness about the subject.

(*a*) masked

(*b*) belied

(*c*) betrayed

(*d*) suppressed

ANSWERS

EXERCISE – I

MCQ Type Questions

1. (1)	**2.** (*b*)	**3.** (*b*)	**4.** (*d*)	**5.** (*b*)	**6.** (*c*)	**7.** (*d*)	**8.** (*d*)	**9.** (*c*)	**10.** (*c*)
11. (*a*)	**12.** (*d*)	**13.** (*a*)	**14.** (*d*)	**15.** (*a*)	**16.** (*d*)	**17.** (*b*)	**18.** (*a*)	**19.** (*d*)	**20.** (*c*)
21. (3)	**22.** (*c*)	**23.** (*d*)	**24.** (*b*)	**25.** (*b*)	**26.** (*d*)	**27.** (*a*)	**28.** (*c*)	**29.** (*d*)	**30.** (*c*)
31. (*c*)	**32.** (*d*)	**33.** (*c*)	**34.** (*d*)	**35.** (*d*)	**36.** (*d*)	**37.** (*d*)	**38.** (*c*)	**39.** (*b*)	**40.** (*c*)
41. (*d*)	**42.** (*d*)	**43.** (*d*)	**44.** (*d*)	**45.** (*c*)	**46.** (*a*)	**47.** (*d*)	**48.** (*b*)	**49.** (*a*)	**50.** (*d*)
51. (*c*)	**52.** (*b*)	**53.** (*d*)	**54.** (*c*)	**55.** (*a*)	**56.** (*d*)	**57.** (*d*)			

EXERCISE – II

MCQ Type Questions

1. (*b*)	**2.** (*b*)	**3.** (*a*)	**4.** (*b*)	**5.** (*c*)	**6.** (*a*)	**7.** (*b*)	**8.** (*b*)	**9.** (*b*)	**10.** (*c*)
11. (*c*)	**12.** (*a*)	**13.** (*d*)	**14.** (*a*)	**15.** (*b*)	**16.** (*b*)	**17.** (*b*)	**18.** (*b*)	**19.** (*c*)	**20.** (*c*)
21. (*b*)	**22.** (*c*)	**23.** (*b*)	**24.** (*c*)	**25.** (*b*)	**26.** (*a*)	**27.** (*b*)	**28.** (*c*)	**29.** (*b*)	**30.** (*a*)
31. (*d*)	**32.** (*b*)	**33.** (*c*)	**34.** (*d*)	**35.** (*c*)	**36.** (*a*)	**37.** (*c*)			

EXPLANATIONS

EXERCISE – I

MCQ TYPE QUESTIONS

1. They in choice (*a*) means 'some of our ruling politicians' (the subject). Also grammatically 'they have a right to plunder .....' is correct wheras'..... right to the punder .....' is incorrect.

2. 'Bizarre' means 'weird'. Hence, choice (*a*) is ruled out. "Its haunting images have endured' fits syntactically. choice (*c*) can also be ruled out as 'munificence' means generosity and this is not intended in the sentence. Also, the word 'prints' in the given sentence gets with the word 'images' of choice (*b*).

3. 'They' in the part after the comma must refer to somebody or some-thing. Hence, only choice (*b*) will fit as 'they' in that case refers to 'things'.

4. The right word to precede the proposition 'on' is imposed. Tax is not collected on ..... people. Tax is not placed on people, Tax is not charged on people. Hence, choices (*a*), (*b*) and (*c*) are not the answers. Tax is imposed ..... on ..... .

5. Choice (*d*) is ruled out as it does not logically fit. Choice (*a*) is also ruled out because of the expression 'tears in me'.

6. The key words are 'plunging' and 'investor confidence'. If markets are plunging, investor confidence is low. This aspect is brought out best by choice (*c*).

7. (*a*) 'Discovery' refers to something that existed previously and came to knowledge later. Home is not a discovery.

(*b*) Home is not a certain. It is a set up which comes into being as a result of a certain composition of various factors.

(*c*) Home is not a new invention of anyone at any stage of development.

So choice (*d*) is the best answer. Home is an institution just like 'marriage. Home is a reality only when its members accept certain principles.

8. (*a*) A man's toil may help to make out wealth, but it is not the ultimate aim.

(*b*) A man's toil may secure a lot of things for his children or friends . That too is not something great.

(*c*) A man's toil may help him to overcome many difficulties, but that does not give him much pleasure.

So, choice (*d*) is the best answer. What one becomes by one's toil is greater than what one gets.

9. (*a*) 'Reopens the first chapter' means begins from the very start.

(*b*) 'Writes an epilogue' means ends the book.

(*c*) 'Opens new books' means creates new problems.

The above three are only partly correct. An act of justice pardons for ever an act of vengeance creates new problems. It is a saying. "An act of justice closes the book on a misdeed; an act of vengeance writes one of its own."

10. Here, the trail is the important thing and hence the excitement of the trail is what you will be looking for.

11. (*b*) The expression must be "it is of cardinal importance".

(*c*) There is no urgency in understanding, "It should be understood" is enough.

(*d*) Instead of "it cannot be,' the expression should be it must be'.

So choice (*a*) is the best, It is important that the two things are not to be interfered with.

13. (*b*) It is not mentioned with what do wines bear semblance.

(*c*) The 'similarity' is only in terms of production. The similarity in production does not achieve commercial profit.

(*d*) For the same commercial profit wines need not be produced in the same of similar manner.

So, choice (*a*) is the best. It does not mention that the wines of similar profit are similarly produced or have semblance with one another.

14. (*a*) A sociological analysis does not stop at the outer limits of the subjects.

(*b*) No one can indicate how far into the subject can a sociological analysis go.

(*c*) 'Relative distance' refers to the distance of something else which is not mentioned.

So, choice (*d*) is the best. This indicates the probable distance to which the sociological analysis can go to.

30. Choices (*a*) and (*d*) are ruled out because 'suggest' does not go with 'to+ verb' form. Choice (2) cannot fit in the blank because of the error in parallelism. The appropriate structure is "Many people suggest.....and others advocate .....".

31. The tense in choices (*a*) and (*b*) is not consistent. Moreover, choice (*a*) uses both 'essential' and 'must'. thus rendering either one of these words redundant. In choice (*d*) 'things' the plural form cannot take it'.

32. Choice (*d*) fits into the blank in choice (*a*) the usage of the indefinite article is erroneous. Choices (*b*) and (*c*) are erroneous due to the incorrect ordering of words.

33. 'Disingenuous' means 'insincere' and this best fits the context as the phrase 'ethnic cleansing' is marked by apostrophes indicating a misrepresentation. Also, the word 'victims' is best in contrast to 'perpetrators'. Hence, option (*c*) is correct.

36. 'Navigators, calendar makers' are best clubbed under the category of 'observers' rather than 'scrutinizers', as the latter term is more used to refer to specialists in a particular field. 'Students' can also fit to a certain extent and this leaves us with options (*b*), (*d*). 'Concede' has more to do with defeat or revision which is more apt in this context than 'agree' or 'conclude'. Hence, option (*c*) is correct.

42. Going by options, 'regrettably' doesn't fit as there is nothing to regret. 'Firstly' is no choice and nothing 'secondly' is mentioned. Paragraph has nothing aggresive tone so 'obviously' does not suit. 'Apparently' is only and most suitable option.

43. All options are suitable for the space but to choose one, the best, is required here 'Analytically' appears to be best. It expresses to analyse something using statistically proven methods to reach a decision.

44. It is a very obvious choice determined from the paragraph giving it an access.

45. 'Firing' comes as best suited option as paragraph explains it as a subject of discussion.

46. An argument is always 'resolved', it is never thwart, defeat or closed.

47. Going by options makes choice easier. Only "how the process allowed....." suits logically.

48. 'Blunt' means 'very direct' with respect to a remark. 'Devious' in choice (*d*) means 'dishonest' and is incorrect as the statement is a frank one. For the same reason 'tactful' in choice (*c*) is also not appropriate.

49. 'Interest' is the most appropriate work because the original amout he owed to the department accumulated to the said figure. 'Taxes is redundant while. 'principal' usually refers to a loan (in this case it isn't). 'Returns' is what one gets rather than pays.

50. 'Sanctions is used for an official permission or order that limits trade and is rarely used for an amout. 'Fees' are paid for a service and 'refunds' are what one gets back. 'Fines is the best choice because of the penalty component.

51. 'Attach' is the word used for officially taking something away. 'Impound' means 'confiscate' and is generally used in the context of illegal goods or contraband. Other choices are in no way comparable.

52. 'Smashed in choice (*a*) and 'dismantled' in choice (*c*) are far from being logical. 'Frozen is a wrong choice. 'Frozen is used in the context of making bank accounts inoperable etc. Automobiles are seized.

53. Clearly the topic is about a person who has violated incom tax regulations and he should be called an 'offender' of the law.

55. Choices (*c*) and (*d*) have similar meaning and are misnomers in the given context, as the keywords 'lowpitched voices' require a word that is synonymous with 'quiet' or 'calm'. The usage of the adjective 'faded' in the sentence to describe men is untenable.

56. The given sentence depicts the President as a very angry person as suggested by the keyword 'scowled' Choices (*a*), (*b*) and (*c*) are either positive or neutral and cannot fit in the sentence.

57. 'Paced' in choice (*a*) is not consistent with the mood of the President as depicted in the first part of the sentence. It means 'to walk fast'. 'Strolled' in (*b*) means 'Walk casually' and hence is discarded. 'Stormed' in choice (*c*) refers to 'attack a place suddenly' or 'say something in a loud, angry way'. But here, neither of them fits in. 'Prowled' refers to moving in a restless manner and hence fits.

EXERCISE – II

MCQ TYPE QUESTIONS

2. The principal presented the chief guest with a memento, as token of appreciation.

3. Frogs make 'croak' sound.

4. Extraneous -irrelevant or unrelated to the subject being dealt with.

5. John-Abraham starrer Madras Cafe talks about the movie not the person, so Mr. is ruled out.
'Find no takers' is not the correct phrase. At this Friday is incorrect.
So, option (*c*) is correct.

6. Apparent: looks like
dormant: hidden
Harbour: give shelter
Effect (verb): results in

8. For condition regarding something which already happened, should have practiced is the correct choice.

13. The correct word to fill in the blank is experiences. This is because the sentence is in present tense and the sense is in present indefinite.

14. Contentment is the right word which is appropriate to the meaning of the sentence. Contentment is the sign of true happiness as it gives a sense of fulfillment.

29. The word despite tells us that mission succeeded even though there were problems. Hence the correct answer is Setbacks.

32. Tone of the sentence clearly indicates a word that is similar to understated is needed for the blank. Alternatively, word should be antonym of strong (fail to make strong impression). Therefore, best choice is restrained which means controlled / reserved / timid.

33. Contemplate is a transitive verb and hence is followed by a gerund Hence the correct usage of contemplate is verb + ing form.

34. The context seeks to take a deviation only when existing / present / current / alternative treatments are unsatisfactory. So word for the blank should be a close synonym of existing / present / current / alternative. Available is the closest of all.

35. Clues: foreign technocrats did something negatively to the problems – so it is counter-productive to invite them. All other options are non-negative. Best choice is exacerbated which means aggravated or worsened.

37. Betrayed, **means** 'showed' or revealed. ■■

3 CHAPTER

Synonyms

INTRODUCTION

A Synonym may be defined as a word that has the same meaning or the same general meaning as a particular word in the same Language or is in some applications a more or less satisfactory application of it. Thus the synonym of a word can be an exact equivalent in meaning or should atleast convey its meaning in one way or the other or should show in what sense it can best be used.

SOME IMPORTANT WORDS WITH THEIR SYNONYMS

A

abandon : desert, for sake, jilt, leave, evacuate, quilt, vacate, abdicate, renounce.
abduct : kidnap, run away with, seize, snatch.
abet : aid, assist, back, condone, help, promote, sanction, support, sustain, uphold.
abeyance : adjournment,deferral, discontinuation, inactivity, intermission, postponement, recess.
abide : accept, bear, brook, endure, dwell, linger, live, lodge, reside, sojourn, persist.
ability : adeptness, aptitude, capability, capacity, competence, dexterity, endowment, expertise.
ablaze : afire, aflame, alight, blazing, burning, fiery, flaming, aglow, brilliant, flashing, gleaming, illuminated, radiant, sparkling.
abstain : avoid, decline, deny, desist, renounce, refrain, withhold, forgo.
abundance : affluence, ampleness, bounty, copiousness, exuberance, pro-fusion, fortune, wealth.
abysmal : bottomless, boundless, complete, deep, extreme, incalculable, profound, vast.
accede : accept, admit, agree, comply, concede, concur, consent, endorse, assume, attain.
access : admission, admittance, approach, avenue, course, entering, outburst, paroxysm.
acclaim : applaud, approve, celebrate, eulogize, exalt, hail, praise, salute, clapping.
accompany : attend, chaperon, conduct, convoy, squire, usher, supplement.
accord : agree, assent, concur, conform, harmonize, bestow, concede, endow, render, vouchsafe.
accuse : allege, blame, cite, denounce, impench, incriminate, indict, recriminate.
accustom : acclimatize, acquaint, adapt, descipline, familiarize, habituate, inure.
acquaint : advice, announce, apprise, disclose, enlighten, notify, reveal, tell.
acrid : caustic, harsh, irritating, pungent, stinging, acrimonious, mordant, nasty, sarcastic, trenchant, vitriolic.
acrimony : bitterness, harshness, churlishness, peevishness, rancour, spleen, tartness, virulence.
adamant : determined, firm, immovable, inexorable, intransigent, obdurate, resolute, stubborn, unrelenting, flinty, steely, stony.
adept : adroit, dexterous, masterful, proficient, versed, expert, genius.
adequate : capable, commensurate, competent, requisite, tolerable, sufficient.
adhere : cleave, cling, faster, obey, observe, respect.
adieu : conge, goodbye, valediction.
adjourne : defer, delay, discontinue, prorogue.
admirable : commendable, excellent, exquisite, wonderful, worthy.
admirer : bean, lover, wooer, devotee, enthusiast, votary.
adrift : drifting, unanchored, aimless, amiss, astray.
adroit : clever, cunning, deft, dexterous, ingeneous, nimble, proficient, quick-witted, able.
audacious : courageous, daring, dauntless, intrepid, rash.
austere : exacting, forbidding, grave, grim, harsh, rigorous, severe, solemn, sterr, stringent.
averse : antipathetic, disinclined, hostile, inimical, loath, reluctant, unwilling.
avid : ardent, eager, fanatical, ferrent, passionate, zealous, avaricious,

B

baptism : immersion, purification, sprinkling, debut.

bastion : citadel, defence, fortress, mainstay.
blaspheme : abuse, anathematize, curse, desecrate, execrate, profane, revile, swear.
breach : break, aperture, charm, cleft, fissure, infraction, infringement, estrangement.
broaden : augment, develop, fatten, spread, stretch.
brood : agonize, dwell, mope, ponder, repine, ruminate.

C

catastrophe: affliction, adversity, cataclysm, devastation, disaster, fiasco, culmination.
chaos : anarchy, bedlam, confusion, disorganization, entropy, tumult, lawlessness.
chastity : celibacy, continence, innocence, modesty, purity.
clan : band, brotherhood, fraternity, gens.
clinch : assure, conclude, confirm, determine, secure.
colloquial : conversational, demotic, everyday, familiar, idiomatic, informal, vernacular.
commotion : ado, agitation, bustle, disorder, disturbance, excitement.
complicate : confuse, entangle, involve, muddle.
contour : curve, figure, lines, profile, relief.
conviction : assurance, certainty, confidence, ferrour.
creditable : admirable, commendable, deserving, estimable, honourable, loudable.
crest : apex, zenith, pinnacle, summit, ridge.
culminate : climax, close, conclude, finish.
cynical : sarcastic, scornful, sneering, distrustful.

D

daunt : dismay, alarm, appal, frighten, intimidate, scare, subdue, terrify.
deed : achievement, act, action, exploit, feat.
deflate : collapse, confract, empty, exhaust, shrink, chasten.
deliberate : cogitate, consult, debate, mediate, ponder, reflect.
diligent : active, attentive, busy, careful, constant.
dire : alarming, appalling, awful, catastrophic, cruel, disastrous.
discreet : careful, cautious, circumspect, diplomatic, judicious, politic, prudent.
dissent : decline, differ, disagree, protest, refuse, disagreement, discord.
ditch : channel, drain, furrow, trench, discord.
dodge : dant, duck, shift, sidestep, avoid, deceive.
dubious : doubtful, hesitant, uncertain, unconvinced, undecided, unsure, ambiguous
dwell : abide, establish, inhabit, settle, sojourn.

E

edgy : anxious, irritable, nervous, touchy.
elusive : difficult to catch, shifty, slippery, tricky, baffling, fleeting, puzzling, subtle.
emblem : badge, gest, device, figure, image, representation.
enchant : beguile, bewitch, captivate, charm, delight, enamour, enrapture, fascinate, hypnotize.
enrage : aggrarate, anger incense, incite, inflame, irritate, provoke.
exalt : advance, dignify, elevate, honour, promote, acclaim, applaud, extol.
exhilarating: breathtaking, cheering, exalting, exciting, vigorating, stimulating.
expedite : accelerate, advance, assist, dispatch, facilitate, hasten, precipitate.

F

fame : celebrity, eminence, glory, honour, prominence, renown, repute, stardom.
famine : scarcity, dearth, destitution, hunger, starvation.
feeble : weak, delicate, etiolated, exhausted, infirm, languid, puny, shilpit.
fierce : barbarous, brutal, cruel, dan-gerous, ferocious, murderous, passionate, savage.
fishy : doubtful, dubious, funny, improbable, suspect, suspicious, deadpan, dull.
flawless : faultless, impeccable, perfect, unblemished.
foresee : anticipate, divine, envisage, forebode, portend, predict, prophesy, presage.
fortress : castle, citadel, fastness, redoubt, stronghold.
fraternity : brotherhood, association, company, comradship.
fretful : captious, complaining, fractions, irritable, peevish, splenetic, touchy.

G

gaiety : animation, blitheness, cheerfulness, elation, exhilaration, glee, humour.
gaudy : bright, brilliant, flashy, florid, garish, glaring, meretricious, showy, vulgar.
gestation : development, evolution, incubation, pregnancy, repening.
gracious : affable, amiable, beneficent, benevolent, cheritable, chivalrous, cordial.
grimace : face, frown, mouth, scowl, sneer.
grubby : dinty, filthy, frowzy, manky, shabby.

H

haggle : bargain, barter, beat down, dicker, higgle, dispute, quarrel, wrangle.
haphazard : accidental, arbitrary, chance, fluky, random, aimless, careless.

hatred : abomination, animosity, animus, antagonism, antipathy, enmity.
hectic : animated, boisterous, chaotic, frantic, furious, turbulent, wild.
hilarious : amusing, comical, convivial, exhilarated, humorous, jolly, jovial, joyful, joyous.
holocaust : annihilation, carnage, destruction, devastation, massacre.
hurricane : cyclone, gale, storm, tempest, tornado, twister.

I

idle : dead, empty, gathering, dust, inactive, jobless, redundant.
immense : enormous, extensive, giant, gigantic, huge, illimitable,
: interminable, jumbo.
implicated : incriminated, involved, suspected, under suspicion.
impudence : assurance, audacity, boldness, bumptiousness, insolence.
incense : anger, enrage, excite, aroma, balm, fragrance, redolence, scent.
indent : order, request, dint, nick, notch.
inept : awkward, bumbling, bungling, clumsy, gauche, inexpert, maladroit, absurd.
insolent : abusive, bold, contemptuous, impertinent, impudent, insulting, rude, saucy.
instigate : actuate, encourage, forment, impel, incite, influence, provoke, persuade.

J, K, L, M

jaunty : airy, breezy, buoyant, carefree, dapper, gay, lively, perky, showy, smart.
jurisdiction : authority, command, control, influence, prerogative, sway.
keepsake : emblem, favour, memento, relic, reminder, souvenir, symbol, token.
latch : bar, bolt, catch, fastening, lock, sneck.
link : component, constituent, division, association, attachment.
meander : ramble, snake, straving, stroll, wander, bend, curve, loop, twist.
melancholy : dejection, depression, despondency, gloom.

N, O, P, Q

nostalgic : homesick, longing, regretful, sentimental, wistful.
obnoxious : abhorrent, abominable, detestable, disagreeable, disgusting, hateable, horrid.
onerous : burdensome, crushing, demanding, difficult, exciting, exhausting, exigent.
paw : grab, handle, roughly, manhandle, maul, molest.
percolate : drain, drip, exude, filter, leache, ooze, penetrate, permeate, pervade.
piety : devotion, dutifulness, faith, grace, piousness, religion, sanctify, veneration.
pudding : dessert, sweet.
quandary : bewilderment, stick, delicate, situation, difficult, dilemma, embarrassment.
quiver : agitate, oscillate, palpitate, pulsate, shudder, tremble, vibrate.

R, S, T

rapt : absorbed, carried away, enthralled, entranced, fascinated, gripped, intent.
reveal : announce, betray, broadcast, communicate, disclose, divulge, impart, proclaim.
rowdy : bristerous, disordely, loutish, noisy, unruly, uproarious, brawler.
sage : acute, canny, intelligent, judicious, polite, prudent, sagacious.
scorn : contempt, despite, disdain, disparagement, slight, sneer, curl, condemn.
shake : fluctuate, bump, joggle, jounce, quake, shiver, shudder, totter, tremble.
sinner : offender, reprobate, trespasser.
snigger : giggle, laugh, smirk, sneer, titter.
stimulus : encouragement, incentive, incitement, inducement, provocation.
tepid : slightly warm, luke warm.
timid : afraid, bashful, cowardly, diffident.

U, V, W, Y, Z

unmask : disclose, discover, expose, reveal, unveil.
urbane : civil, civilized, cosmopolitan, courteous, cultivated, debonair.
vehement : ardent, eager, earnest, emphatic, enthusiastic, fervent, fervid, fierce, forcible.
weired : bizarre, creepy, cerie, freakish, ghostly, mysterious, queer, spooky.
writhe : contont, distant, jerk, struggle, thrash, wiggle, twist, wriggle.
yearn : ache, covet, crave, desire, hanker, hunger, itch, languish.
zenith : acme, apex, apogee, climax, height, meridian, peak, pinnacle, summit, top.

QUESTIONS

Directions: Each of the questions below consists of a word followed by four lettered words or phrases. Choose the lettered word or phrase that is most nearly same in meaning to the word.

1. Abase
(a) incur (b) tax
(c) estimate (d) humiliate

2. Aberration
(a) deviation (b) abhorrence
(c) dislike (d) absence

3. Abet
(a) conceive (b) wager
(c) encourage (d) evade

4. Aabeyance
(a) obedience (b) discussion
(c) excitement (d) suspended action

5. Abjure
(a) discuss (b) renounce
(c) run off secretly (d) perjure

6. Ablution
(a) censure (b) forgiveness
(c) mutiny (d) washing

7. Abnegation
(a) blackness (b) self-denial
(c) selfishness (d) cause

8. Aborigine
(a) first design
(b) absolution
(c) finale
(d) primitive inhabitant

9. Abortive
(a) unsuccessful
(b) consuming
(c) financing
(d) familiar

10. Abstinence
(a) restrained (b) vulgar display
(c) deportment (d) reluctance

11. Abstruse
(a) profound (b) irrespective
(c) suspended (d) protesting

12. Abut
(a) stimulate (b) grasp
(c) oppose (d) adjoin

13. Abysmal
(a) bottomless (b) eternal
(c) meteric (d) diabolic

14. Accede
(a) fail (b) compromise
(c) correct (d) consent

15. Acclivity
(a) index
(b) report
(c) upslope of a hill
(d) negotiator

16. Astute
(a) sheer (b) noisy
(c) astral (d) clever

17. Atrocity
(a) endurance (b) fortitude
(c) session (d) heinous act

18. Atrophy
(a) capture (b) waste away
(c) govern (d) award prize

19. Attenuate
(a) appear (b) be absent
(c) weaken (d) testify

20. Atypical
(a) superfluous (b) fortitude
(c) unusual (d) clashing

21. Audacity
(a) boldness (b) asperity
(c) strength (d) stature

22. Augment
(a) make noble (b) anoint
(c) increase (d) harvest

23. Auxiliary
(a) righteous (b) prospective
(c) assistant (d) archaic

24. Avarice
(a) easiness (b) greed
(c) statement (d) invoice

25. Avert
(a) entertain (b) transform
(c) turn away (d) lead toward

26. Awry
(a) recommended
(b) commiserating
(c) started
(d) crooked

27. Baleful
(a) doubtful (b) virtual
(c) deadly (d) conventional

28. Balmy
(a) venturesome (b) dedicated
(c) mild (d) fanatic

29. Banal
(a) philosophical (b) trite
(c) dramatic (d) heedless

30. Baneful
(a) intellectual (b) thankful
(c) decisive (d) poisonous

31. Boisterous
(a) conflicting (b) noisy
(c) testimonial (d) grateful

32. Bombastic
(a) sensitive (b) pompous
(c) rapid (d) sufficient

33. Bouillon
(a) insight
(b) chowder
(c) gold
(d) clear soup

34. Brackish
(a) careful (b) salty
(c) chosen (d) tough

35. Braggadocio
(a) weaponry (b) boasting
(c) skirmish (d) encounter

36. Brazen
(a) shameless (b) quick
(c) modest (d) pleasant

37. Brindled
(a) equine (b) pathetic
(c) hasty (d) spotted

38. Brochure
(a) opening (b) pamphlet
(c) censor (d) bureau

39. bucolic
(a) diseased (b) replusive
(c) rustic (d) twinkling

40. Buxom
(a) voluminous (b) indecisive
(c) convincing (d) plump

41. Cache
(a) lock
(b) hiding place
(c) tide
(d) automobile

42. Cacophony
(a) discord
(b) dance
(c) applause
(d) type of telephone

43. Callow
(a) youthful (b) holy
(c) mild (d) colored

44. Candid
(a) vague (b) outspoken
(c) experienced (d) anxious

45. Chaste
(a) loyal (b) timid
(c) curt (d) pure

46. Chide
(a) unite (b) fear
(c) record (d) scold

47. Chimerical
(a) developing (b) brief
(c) distant (d) fantastic

48. Choleric
(a) musical (b) episodic
(c) hot headed (d) global

49. Churlish
(a) marine (b) economical
(c) impolite (d) compact

50. Ciliated
(a) foolish (b) swift
(c) early (d) hairy

51. Circuitous
(a) indirect (b) complete
(c) obvious (d) aware

52. City
(a) galvanize (b) visualize
(c) locate (d) quote

53. Clandestine
(a) abortive (b) secret
(c) tangible (d) doomed

54. Claustrophobia
(a) lack of confidence
(b) fear of spiders
(c) love of books
(d) fear of closed places

55. Cleft
(a) split (b) waterfall
(c) assembly (d) adherence

56. Cliche
(a) increase (b) vehicle
(c) morale (d) platitude

57. Coerce
(a) recover (b) begin
(c) force (d) license

58. Cognizance
(a) policy (b) knowledge
(c) advance (d) omission

59. Cohere
(a) hold together
(b) occur simultaneously
(c) recollect
(d) materialize

60. Conjecture
(a) magic (b) guess
(c) position (d) form

61. Connoisseur
(a) gourmand (b) lover of art
(c) humidor (d) delinquent

62. Consanguinity
(a) kinship (b) friendship
(c) bloodletting (d) relief

63. Consensus
(a) general agreement (b) project
(c) insignificance (d) sheaf

64. Construe
(a) explain (b) promote
(c) reserve (d) erect

65. Contaminate
(a) arrest (b) prepare
(c) pollute (d) beam

66. Contentious
(a) squealing (b) surprising
(c) quarrelsome (d) smug

67. Continence
(a) humanity (b) research
(c) embryology (d) self-restraint

68. Contraband
(a) purpose (b) rogue
(c) rascality (d) difficulty

69. Contrite
(a) smart
(b) penitent
(c) restful
(d) recognized

70. Controvert
(a) turnover (b) contradict
(c) mind (d) expolain

71. Convene
(a) propose (b) restore
(c) question (d) gather

72. Conversant
(a) ignorant
(b) speaking
(c) incorporated
(d) familiar

73. Copious
(a) plentiful (b) cheating
(c) dishonorable (d) adventurous

74. Corpulent
(a) regenerate (b) obese
(c) different (d) hungry

75. Decimate
(a) kill (b) disgrace
(c) search (d) collide

76. Declivite
(a) trap (b) quadrangle
(c) quarter (d) downward slope

77. Decollete
(a) flavored (b) demure
(c) flowery (d) low-necked

78. Decorous
(a) momentary (b) emotional
(c) suppressed (d) proper

79. Decrepitude
(a) feebleness (b) disease
(c) coolness (d) melee

80. Default
(a) failure to act
(b) tendency to err
(c) desire to remedy
(d) debt

81. Defection
(a) determination (b) desertion
(c) invitation (d) affection

82. Defile
(a) maincure (b) ride
(c) pollute (d) assemble

83. Degraded
(a) surpprised (b) lowered
(c) ascended (d) learned

84. Deleterious
(a) delaying
(b) experimental
(c) harmful
(d) graduating

85. Deluge
(a) confusion (b) deception
(c) flood (d) mountain

86. Denigrate
(a) refuse (b) blacken
(c) terrify (d) admit

87. Denouement
(a) action (b) scenery
(c) resort (d) solution

88. Depravity
(a) wickedness (b) sadness
(c) heaviness (d) tidiness

89. Deranged
(a) insane (b) systematic
(c) neighborly (d) alphabetical

90. Dwindle
(a) blow (b) inhabit
(c) spin (d) lessen

91. Ecstasy
(a) joy (b) speed
(c) treasure (d) warmth

92. Edify
(a) mystify (b) suffice
(c) improve (d) erect

93. Efface
(a) countenance
(b) encourage
(c) recognize
(d) rub out

94. Effigy
(a) requisition (b) organ
(c) charge (d) dummy

95. Egregious
(a) pious (b) shocking
(c) anxious (d) sociable

96. Egress
(a) entrence (b) bird
(c) exit (d) double

97. Elated
(a) debased
(b) respectful
(c) drooping
(d) gay

98. Elusive
(a) deadly (b) eloping
(c) evasive (d) simple

99. Emaciated
(a) garrulous (b) primeval
(c) vigorous (d) thin

100. Emancipate
(a) set free (b) take back
(c) make worse (d) embolden

101. Embellish
(a) doff (b) don
(c) abscond (d) adon

102. Embroil
(a) cherish (b) overtheat
(c) entangle (d) assure

103. Emendation
(a) correction
(b) interpretation
(c) exhumation
(d) inquiry

104. Eminent
(a) purposeful (b) high
(c) delectable (d) currious

105. Flored
(a) ruddy (b) rusty
(c) ruined (d) patient

106. Foil
(a) bury (b) frustrate
(c) shield (d) desire

107. Foment
(a) spoil (b) instigate
(c) interrogate (d) spray

108. Foolhardy
(a) strong (b) unwise
(c) brave (d) futile

109. Foppish
(a) scanty (b) radical
(c) orthodox (d) dandyish

110. Foray
(a) exursion (b) contest
(c) ranger (d) intuition

111. Formidable
(a) dangerous
(b) outlandish
(c) grandiloquent
(d) impenetrable

112. Foster
(a) accelerate (b) fondle
(c) become infected (d) raise

113. Franchise
(a) subway (b) discount
(c) license (d) reason

114. Fritter
(a) sour (b) chafe
(c) dissipate (d) cancel

115. Frugality
(a) foolishness (b) extremity
(c) indifference (d) economy

116. Fulminate
(a) fulfill (b) contemplate
(c) talk nonsense (d) protest

117. Furor
(a) excitement (b) worry
(c) flux (d) anteroom

118. Furtive
(a) underhanded (b) coy
(c) brilliant (d) quick

119. Gadfly
(a) humorist
(b) nuisance
(c) scholar
(d) bum

120. Galleon
(a) liquid measure (b) ship
(c) armada (d) company

121. Garisth
(a) sordid (b) flashy
(c) prominent (d) lusty

122. Garner
(a) prevent (b) assist
(c) collect (d) compute

123. Garnish
(a) paint (b) gamer
(c) adorn (d) abuse

124. Garrulity
(a) credulity
(b) senility
(c) loquaciousness
(d) speciousness

125. Garrulous
(a) arid (b) hasty
(c) sociable (d) talkative

126. Gauche
(a) rigid (b) swift
(c) awkward (d) taciturn

127. Gaudy
(a) holy (b) showy
(c) sentimental (d) mild

128. Gaunt
(a) victorious (b) tiny
(c) stylish (d) haggard

129. Genuflect
(a) falsify (b) trick
(c) project (d) bend the knee

130. Germane
(a) bacteriological
(b) middle european
(c) prominent
(d) relevant

131. Germtinal
(a) creative (b) excused
(c) sterilized (d) primitive

132. Gist
(a) chaff (b) summary
(c) expostulation (d) expiation

133. Glib
(a) slippery (b) fashionable
(c) antiquated (d) articulate

134. Gnome
(a) fury (b) giant
(c) dwarf (d) native

135. Inclement
(a) unfavorable (b) abandoned
(c) kindly (d) selfish

136. Incompatible
(a) capable
(b) reasonable
(c) faulty
(d) alienated

137. Inconsequential
(a) disorderly (b) insignificant
(c) subsequent (d) insufficient

138. Incontinent
(a) insular (b) complaisant
(c) crass (d) wanton

139. Incorrigible
(a) narrow (b) straight
(c) inconceivable (d) unreformable

140. Incriminate
(a) exacerbate (b) involve
(c) intimidate (d) lacerate

141. Inculcate
(a) exculpate (b) educate
(c) exonerate (d) prepare

142. Indigent
(a) lazy (b) pusillanimous
(c) penurious (d) affluent

143. Indigniy
(a) pomposity (b) bombast
(c) obeisance (d) insult

144. Indolence
(a) sloth (b) poverty
(c) latitude (d) aptitude

145. Indubitably
(a) flagrantly (b) doubtfully
(c) carefully (d) cortainly

146. Inebriety
(a) revelation (b) drunkenness
(c) felony (d) starvation

147. Inept
(a) outward (b) spiritual
(c) foolish (d) clumsy

148. Infallible
(a) final
(b) unbelievable
(c) perfect
(d) inaccurate

149. Infirmity
(a) disability (b) age
(c) inoculation (d) hospital

150. Ingenuous
(a) clever (b) stimulating
(c) naive (d) worried

151. Inimical
(a) antagonistic
(b) anonymous
(c) fanciful
(d) accurate

152. Innocuous
(a) not capable (b) not dangerous
(c) not eager (d) not frank

153. Insinuate
(a) resist (b) suggest
(c) report (d) rectify

154. Insipid
(a) witty (b) flat
(c) wily (d) talkative

155. Integrate
(a) tolerate (b) unite
(c) flow (d) copy

156. Inter
(a) bury (b) amuse
(c) relate (d) frequent

157. Interdict
(a) acclain (b) dispute
(c) prohibit (d) decide

158. Intermittent
(a) heavy (b) fleet
(c) occasional (d) fearless

159. Intractable
(a) culpable (b) flexible
(c) unruly (d) efficient

160. Intransigence
(a) lack of training
(b) stubbornness
(c) novelty
(d) timidity

161. Intrepid
(a) cold (b) hot
(c) understood (d) courageous

162. Intrinsic
(a) extrinsic (b) abnormal
(c) above (d) basic

163. Inundate
(a) abuse (b) deny
(c) swallow (d) flood

164. Inveigh
(a) speak violently
(b) orate
(c) disturb
(d) apply

165. Lethargy
(a) dynamism (b) orate
(c) disturb (d) apply

166. Loquacious
(a) verbose (b) tacturn
(c) rational (d) alluring

167. Loath
(a) tired (b) unwilling
(c) sickly (d) spirited

168. Mawkish
(a) sentimental (b) true
(c) certain (d) devious

169. Mediocre
(a) average (b) bitter
(c) medieval (d) industrial

170. Melee
(a) heat (b) brawl
(c) attempt (d) weapon

171. Mellifluous
(a) porous (b) honeycombed
(c) strong (d) smooth

172. Menial
(a) intellectual (b) clairvoyant
(c) servile (d) arrogant

173. Mentor
(a) guide (b) genius
(c) talker (d) philosopher

174. Mesmerize
(a) remember (b) hypnotize
(c) delay (d) bore

175. Meticulous
(a) steadfast (b) recent
(c) quaint (d) painstaking

176. Miasma
(a) dream (b) noxious fumes
(c) scenario (d) quantity

177. Militant
(a) combative (b) dramatic
(c) religious (d) quaintity

178. Minon
(a) monster (b) quorum
(c) majority (d) dependent

179. Mirage
(a) dessert (b) illusion
(c) water (d) mirror

180. Misanthrope
(a) benefactor
(b) philanderer
(c) man-hater
(d) aesthete

181. Mischance
(a) gamble (b) ordinance
(c) aperture (d) anecdote

182. Misdemeanor
(a) felony (b) peccadillo
(c) indignity (d) flat

183. Modish
(a) sentimental (b) stylish
(c) vacillating (d) contrary

184. Mollify
(a) avenge (b) attenuate
(c) attibute (d) appease

185. Monetary
(a) boring (b) fascinating
(c) fiscal (d) stationary

186. Moratorium
(a) burial (b) gathering
(c) delay (d) refusal

187. Mordant
(a) dying (b) trenchant
(c) fabricating (d) controlling

188. Moribund
(a) dying (b) trenchant
(c) fabricating (d) controlling

189. Motley
(a) active (b) disguised
(c) variegated (d) somber

190. Muggy
(a) attacking (b) fascinating
(c) humid (d) characteristic

191. Mulct
(a) swindle (b) hold
(c) record (d) pring

192. Multilingual
(a) variegated (b) polyglot
(c) multilateral (d) polyandrous

193. Mundane
(a) global (b) futile
(c) spatial (d) worldly

194. Munificent
(a) grandiose (b) puny
(c) philanthropic (d) poor

195. Musty
(a) flat (b) necessary
(c) indifferent (d) nonchalant

196. Myopic
(a) visionary (b) nearsighted
(c) moral (d) glassy

197. Nascent
(a) incipient (b) ignorant
(c) loyal (d) treacherous

198. Phlegmatic
(a) calm (b) cryptic
(c) practical (d) salivary

199. Physiognomy
(a) posture (b) head
(c) physique (d) face

200. Piebald
(a) motley (b) coltish
(c) hairless (d) thoroughbred

201. Pillage
(a) hoard (b) plunder
(c) versify (d) denigrate

202. Pinion
(a) express (b) report
(c) reveal (d) restrain

203. Pinnacle
(a) foothills (b) card game
(c) pass (d) peak

204. Pious
(a) historic (b) devout
(c) multiple (d) fortunate

205. Pique
(a) pyramid (b) revolt
(c) resentment (d) struggle

206. Placate
(a) determine (b) transmit
(c) pacify (d) allow

207. Plagiarism
(a) theft of funds (b) theft of ideas
(c) belief in god (d) arson

208. Plaintive
(a) moumful (b) senseless
(c) persistent (d) rural

209. Platitude
(a) fatness (b) bravery
(c) dimension (d) trite remark

210. Polemic
(a) blackness (b) lighting
(c) magnetism (d) controversy

211. Precarious
(a) priceless (b) premature
(c) primitive (d) hazardous

212. Refractory
(a) articulate (b) sinkable
(c) vaunted (d) unmanageable

213. Regal
(a) oppressive (b) royal
(c) major (d) basic

214. Reiterate
(a) gainsay (b) revive
(c) revenge (d) repeat

215. Relish
(a) desire (b) nibble
(c) savor (d) violating

216. Remiss
(a) lax (b) lost
(c) foolish (d) violating

217. Remonstrate
(a) display (b) restate
(c) protest (d) resign

218. Repartee
(a) witty retort (b) willful departure
(c) spectator (d) monologue

219. Repllent
(a) propulsive (b) unattractive
(c) porous (d) stiff

220. Rpercussion
(a) reaction (b) restitution
(c) resistance (d) magnificence

221. Replenish
(a) polish (b) repeat
(c) reinstate (d) refill

222. Replica
(a) museum piece (b) famous site
(c) battle emblem (d) facsimile

223. Reprisal
(a) reevaluation (b) assessment
(c) loss (d) retaliation

224. Reprove
(a) prevail (b) rebuke
(c) ascertain (d) prove false

225. Repudiate
(a) besmirch (b) appropriate
(c) annoy (d) reject

226. Repugnance
(a) belligerence (b) tanacity
(c) renewal (d) loathing

227. Tarry
(a) polish (b) restrain
(c) surpass (d) linger

228. Tautology
(a) memory (b) repetition
(c) tension (d) smile

229. Tawdry
(a) orderly (b) meretricious
(c) reclaimed (d) filtered

230. Temerity
(a) timidity (b) resourcefulness
(c) boldness (d) tremulousness

231. Temporal
(a) priestly (b) scholarly
(c) secular (d) sleeply

232. Tenacious
(a) fast running (b) intertional
(c) obnoxious (d) holding fast

233. Tenacity
(a) splendor (b) perseverance
(c) tendency (d) ingratitide

234. Tendentious
(a) biased (b) likely
(c) absurd (d) festive

235. Tentative
(a) prevalent (b) portable
(c) mocking (d) wry

236. Tenuous
(a) vital (b) thin
(c) careful (d) dangerous

237. Tepid
(a) boiling (b) luckewarm
(c) freezing (d) gaseous

EXERCISE – II
(QUESTIONS FROM PREVIOUS GATE EXAMS)
MCQ TYPE QUESTIONS

2015

1. The word similar in meaning to 'dreary' is
(a) cheerful (b) dreamy
(c) hard (d) dismal

2. Choose the word most similar in meaning to the given word :
Educe
(a) Exert (b) Educate
(c) Extract (d) Extend

3. Choose the most suitable one word substitute for the following expression
Connotation of a road or way
(a) Pertinacious (b) Viaticum
(c) Clandestine (d) Ravenous

4. Choose the word most similar in meaning to the given word:
Awkward
(a) Inept
(b) Graceful
(c) Suitable
(d) Dreadful

5. A generic term that includes various items of clothing such as a skirt, a pair of trousers and a shirt as
(a) fabric (b) textile
(c) fibre (d) apparel

6. Which of the following combinations is incorrect?
(a) Acquiescence – Submission
(b) Wheedle – Roundabout
(c) Flippancy – Lightness
(d) Profligate – Extravagant

2014

7. A student is required to demonstrate a high level of comprehension of the subject, especially in the social sciences.
The word closest in meaning to comprehension is
(a) understanding (b) meaning
(c) concentration (d) stability

8. While receiving the award, the scientist said, "I feel vindicated". Which of the following is closest in meaning to the word 'vindicated'?
(a) punished (b) substantiated
(c) appreciated (d) chastened

9. Choose the word that is opposite in meaning to the word "coherent".
(a) sticky (b) well-connected
(c) rambling (d) friendly

10. Match the columns:

Column-1	Column-2
1. eradicate	P. misrepresent
2. distort	Q. soak completely
3. saturate	R. use
4. utilize	S. destroy utterly

(a) 1 : S, 2 : P, 3 : Q, 4 : R
(b) 1 : P, 2 : Q, 3 : R, 4 : S
(c) 1 : Q, 2 : R, 3 : S, 4 : P
(d) 1 : S, 2 : P, 3 : R, 4 : Q

2013

11. Which of the following options is the closest in meaning to the word given below:
Primeval
(a) Modern (b) Historic
(c) Primitive (d) Antique

12. Which one of the following options is the closest in meaning to the word given below?
Nadir
(a) Highest (b) Lowest
(c) Medium (d) Integration

13. They were requested not to **quarrel** with others. Which one of the following options is the closest in meaning to the word **quarrel**?
(a) make out (b) call out
(c) dig out (d) fall out

2012

14. Which one of the following options is the closest in meaning to the word given below?
Mitigate
(a) Diminish (b) Divulge
(c) Dedicate (d) Denote

15. Which one of the following options is the closest in meaning to the word given below ?
Latitude
(a) Eligibility (b) Freedom
(c) Coercion (d) Meticulousness

2011

16. Which of the following options is the closest in the meaning to the word below :
Inexplicable
(a) Incomprehensible (b) Indelible
(c) Inextricable (d) Infallible

2010

17. Which of the following options is the closest in meaning to the word below:
Circuitous
(a) cyclic (b) indirect
(c) confusing (d) crooked

EXERCISE – I

MCQ Type Questions

1. (d)	2. (a)	3. (c)	4. (d)	5. (b)	6. (d)	7. (b)	8. (d)	9. (a)	10. (a)
11. (a)	12. (d)	13. (a)	14. (d)	15. (d)	16. (d)	17. (d)	18. (b)	19. (c)	20. (e)
21. (a)	22. (c)	23. (c)	24. (b)	25. (c)	26. (d)	27. (c)	28. (c)	29. (b)	30. (d)
31. (b)	32. (b)	33. (d)	34. (b)	35. (b)	36. (a)	37. (d)	38. (b)	39. (c)	40. (d)
41. (b)	42. (a)	43. (a)	44. (c)	45. (d)	46. (d)	47. (d)	48. (c)	49. (c)	50. (d)
51. (a)	52. (d)	53. (b)	54. (d)	55. (a)	56. (d)	57. (c)	58. (b)	59. (a)	60. (b)
61. (b)	62. (a)	63. (a)	64. (a)	65. (c)	66. (c)	67. (d)	68. (d)	69. (b)	70. (b)
71. (d)	72. (d)	73. (a)	74. (b)	75. (a)	76. (d)	77. (d)	78. (d)	79. (a)	80. (a)
81. (b)	82. (c)	83. (b)	84. (c)	85. (c)	86. (b)	87. (d)	88. (a)	89. (a)	90. (d)
91. (a)	92. (c)	93. (d)	94. (d)	95. (b)	96. (c)	97. (d)	98. (c)	99. (d)	100. (a)
101. (d)	102. (c)	103. (a)	104. (b)	105. (a)	106. (b)	107. (b)	108. (b)	109. (d)	110. (a)
111. (a)	112. (c)	113. (c)	114. (c)	115. (d)	116. (d)	117. (a)	118. (a)	119. (b)	120. (b)
121. (b)	122. (c)	123. (c)	124. (c)	125. (d)	126. (c)	127. (b)	128. (d)	129. (d)	130. (d)
131. (a)	132. (b)	133. (d)	134. (c)	135. (a)	136. (d)	137. (b)	138. (d)	139. (d)	140. (b)
141. (b)	142. (c)	143. (d)	144. (a)	145. (d)	146. (b)	147. (d)	148. (c)	149. (a)	150. (c)
151. (a)	152. (b)	153. (b)	154. (b)	155. (a)	156. (a)	157. (c)	158. (c)	159. (c)	160. (b)
161. (d)	162. (d)	163. (d)	164. (a)	165. (b)	166. (a)	167. (b)	168. (d)	169. (a)	170. (b)
171. (d)	172. (c)	173. (a)	174. (b)	175. (d)	176. (b)	177. (a)	178. (d)	179. (b)	180. (c)
181. (d)	182. (b)	183. (b)	184. (d)	185. (c)	186. (c)	187. (b)	188. (a)	189. (c)	190. (c)
191. (a)	192. (b)	193. (d)	194. (c)	195. (a)	196. (b)	197. (a)	198. (a)	199. (d)	200. (a)
201. (b)	202. (d)	203. (d)	204. (b)	205. (c)	206. (c)	207. (b)	208. (a)	209. (d)	210. (d)
211. (d)	212. (d)	213. (b)	214. (d)	215. (c)	216. (a)	217. (c)	218. (a)	219. (b)	220. (a)
221. (d)	222. (d)	223. (d)	224. (b)	225. (d)	226. (d)	227. (d)	228. (b)	229. (b)	230. (c)
231. (c)	232. (d)	233. (b)	234. (a)	235. (d)	236. (b)	237. (b)			

EXERCISE – II

MCQ Type Questions

1. (d)	2. (c)	3. (b)	4. (a)	5. (d)	6. (b)	7. (a)	8. (b)	9. (c)	10. (a)
11. (c)	12. (b)	13. (b)	14. (a)	15. (b)	16. (a)	17. (b)			

EXPLANATIONS

EXERCISE – II

MCQ TYPE QUESTIONS

1. *dreary* - depressingly dull and bleak or repetitive.

2. The word similar in meaning to Educe is Extract.

12. **Given:** A word "Nadir"

To find: Closest meaning word to 'Nadir' in the given options.

Analysis: The dictionary meaning of 'Nadir' is 'Lowest' which is potion (*b*)

Source: Google

Given: A sequence 44, 42, 40,.....

To find: Maximum sum of the sequence

Analysis: To find the maximum sum, we have to ignore all the negative terms (they will only decrease the sum). Hence, we terminate the sequence at 0.

Now, the sequence is 44, 42,.....,2, 0.

This sequence is an Arithmetic progression with common difference –2. Now, the problem reduces to finding the sum of this A.P.

To find the sum of A.P., we have to find the number of terms in A.P.

Let n be the number of terms

a be the first term

l be the last term

d be the common difference.

$n = ?$

$a = 44, \quad l = 0, \quad d = -2$

Now, $l = a + (n - 1)d$

14. 'Mitigate' means to make something less harmful, serious, intense etc. Hence correct answer is 'Diminis'

16. Inexplicable means not explicable; that cannot be explained, understood, or accounted for. So the best synonym here is incomprehensible.

17. **Circuitous** means round about or not direct

So circuitous : indirect.

■■

CHAPTER **4**

Antonyms

The basic idea of a antonym question is to find an opposite for a word. Antonym items are first and foremost a test of vocabulary.

If you have no idea of the meaning of the given word, there's not much you can do. When you don't know the meaning of an antonym, don't waste a lot of time trying to fiture it out. In other words, once you recognize that you are out of ammunition, just do a random guess, confirm it, and move on the next item.

Each antonym starts with a capitalized word, which is followed by four words of phrases in lowercase letters. The task is to find the word or phrase that is most nearly opposite in meaning to the bold word.

Plan to Answer Antonym Questions

1. Define the capitalized word.
2. Think of a meaning that is opposite to this word.
3. Read all the answer choices. Eliminate those that do not relate to the meaning you thought of. If only one choice remains, mord it and go on.
4. If more than one choice remains, go back and refine your thinking about the bold word.
5. Pick the answer choice that is most nearly opposite to the bold word.

Tedious

(*a*) unlimited (*b*) confined

(*c*) enthralling (*d*) appetizing

(*i*) The capitalized word means "boring" or "tiresome"

(*ii*) The opposite of "boring" would be something like *exciting or interesting*.

(*iii*) Looking at all the answer choices, you can immediately eliminate (*a*), and (*b*) as having nothing to do with excitement. But that leaves (*c*) *enthralling and* (*d*) *apperizing*.

(*iv*) Of the two remaining choices, *enthralling, which means "captivating"* or fascinating," is more nearly opposite in meaning to *tedious than appetizing*, which means "savory," or "delicious".

(*v*) Mark (c), enthralling, as the correct choice.

Note :

(*i*) **Makes sure to read all the answers :** To answer antonym questions you'll sometimes have to distinguish fine shades of meaning. That's why you have to consider very answer choice.

(*ii*) **Verb or Noun?** If the part of speech of the capitalized word is ambiguous, check the answer choices.

6. Sometimes the test writer selects a word you are likely to be familiar with, but sets up the question to test a meaning you do not ordinarily associate with the word. Here is an example:

e.g. precipitous

(*a*) pleasantly sweet

(*b*) overly ambitious

(*c*) agreeably situated

(*d*) gently sloping

The best choice is (*d*). We most often use the word precipitous to mean rash or foothardy, but its central meaning is related to precipice, a sharp drop-off.

7. **Eleminate non-answers**

The simple way to eleminate answer choices is to toss out any words that don't have opposite meanings.

8. **Concentrate on answer choices**

Words can have different meanings based upon their part of speech. If you are uncertain about the part of speech of the bold word, just check the naswer choices. If they're all verbs, for example, so is the bold word.

9. **Play around with the part of speech of a word**

You can alter the part of speech of the bold word and answer choices in your own mind. Sometimes an antonym will use a word you know but as a part of speech that is unfamiliar to you.

***e.g.* sublimity**

(*a*) erosion (*b*) baseness

(*c*) conciseness (*d*) insistence

The best choice is (*b*). You may know the word sublimity better as the adjective sublime, meaning "lofty, high, or noble". So you may find it easire to think about the antonym by changing *sublimity* to the more familiar form *sublime*". As you think about each answer choice, you would then change it in your mind to an adjective *Baseness*, therefore, would become *base*; and *base* is an opposite of *sublime*.

10. **Use word connotations to eliminate answer choices**

Even if you don't know the exact meaning of a word, you may have a vague recollection of the context in which you first encountered it. So you may know whether the word has positive overtones or negative ones. This recollection may be sufficient to get a correct answer.

e.g. raffish
(a) grotesque (b) delinquent
(c) uncaring (d) noble

The correct choice is (d). Let's assume that you do not know that is means "low, vulgar, the base. 'Let's further assume, however, that you have a vague knowledge of the word. You've seen it used to describe a character who is dishonest and not trustworthy. So even though you don't know the exact meaning of the word, you know that it has negative overtones. Sine you an looking for the opposite of a word with negative overtones, you would eliminate every answer choice with negative overtones. As it turns out, this strategy workd perfectly with this antonym; only one word is left. *Noble* is the only word with positive overtones.

11. Unlock word meanings by taking the word apart

Even when you encounter a word for the first time, you may be able to ascertain its meaning from its parts.

e.g. congnoscitive
(a) courageous (b) expensive
(c) unconscious (d) redundant

The best answer is (c). This is very unusual word, but you can probably figure out its meaning by looking at its root cog-. This is the same root found in words such as *cognition and recognize*, and it has to do with knowledge. So we infer that *cognoscitive* has something to do with awareness, and (c) looks like a good opposite. A word of caution, however : Don't spend too much time trying to decipher the meaning of a word.

EXERCISE – I

MCQ TYPE QUESTIONS

Directions : *Each question below consists of a word printed in capital lettes, followed by four lettered words of phrases. Choosed the lettered word or phrase that is most nearly opposite in meaning to the word in bold letters.*

1. Abrogate
(a) transgress (b) signify
(c) allevate (d) ratify

2. Admonition
(a) premonit ion (b) hallucination
(c) escape (d) commendation

3. Adulation
(a) youth (b) purity
(c) brightness (d) criticism

4. Advocate
(a) define (b) oppose
(c) remove (d) inspect

5. Affable
(a) rude (b) ruddy
(c) needy (d) useless

6. Affecfted
(a) weary (b) unfriendly
(c) divine (d) unfeigned

7. Affluence
(a) poverty (b) fear
(c) persuasion (d) consideration

8. Agility
(a) awkwardness (b) solidity
(c) temper (d) harmony

9. Alacrity
(a) slowness (b) plenty
(c) fifth (d) courtesy

10. Alleviate
(a) endure (b) worsen
(c) enlighten (d) maneuver

11. Allure
(a) hinder (b) repel
(c) ignore (d) leave

12. Aloof
(a) triangular (b) gregarious
(c) comparable (d) honorable

13. Amalgamate
(a) equip (b) separate
(c) generate (d) materialize

14. Ambiguous
(a) salvageable (b) corresponding
(c) responsible (d) clear

15. Amble
(a) befriend (b) hasten
(c) steal (d) browse

16. Ambulatory
(a) convalescent (b) valedictory
(c) bedridden (d) emergency

17. Ameliorate
(a) make slow (b) make sure
(c) make young (d) make worse

18. Amicable
(a) penetrating (b) compensating
(c) unfriendly (d) zigzag

19. Amoral
(a) unusal (b) unfriendly
(c) ethical (d) suave

20. Amorphous
(a) nauseous (b) obscene
(c) providential (d) definite

21. Amplify
(a) distract (b) infer
(c) publicize (d) decrease

22. Analogous
(a) not comparable (b) not capable
(c) not culpable (d) not corporeal

23. Anthematize
(a) locate (b) deceive
(c) regulate (d) bless

24. Anemic
(a) pailid (b) cruel
(c) red-blooded (d) ventilating

25. Animated
(a) worthy (b) dull
(c) humorous (d) lengthy

26. Animus
(a) pterodactyl (b) bastion
(c) giraffe (d) favor

27. Anomaly
(a) desperation (b) requisition
(c) registry (d) regularity

28. Anonymous
(a) desperate (b) signed
(c) defined (d) expert

29. Antediluvian
(a) transported (b) subtle
(c) isolated (d) modern

30. Antipathy
(a) profundity (b) objection
(c) willingness (d) foundness

31. Antithesis
(a) velocity (b) maxim
(c) similarity (d) acceleration

32. Aphasia
(a) volubility (b) necessity
(c) pain (d) crack

33. Baroque
(a) polished (b) constant
(c) transformed (d) simple

34. Beatific
(a) glorius (b) dreadful
(c) theatrical (d) crooked

35. Belittle
(a) disobey (b) forget
(c) magnify (d) extol

36. Bellicose
(a) peaceful (b) naval
(c) amusing (d) piecemeal

37. Benign
(a) tenfold (b) peaceful
(c) blessed (d) malignant

38. Benison
(a) curse (b) bachelor
(c) wedding (d) orgy

39. Berate
(a) grant (b) praise
(c) refer (d) purchase

40. Bestial
(a) animated (b) noble
(c) zoological (d) clear

41. Bigotry
(a) arrogance (b) approval
(c) mourning (d) tolerance

42. Bizarre
(a) roomy (b) veiled
(c) subduced (d) normal

43. Blanch
(a) bleach (b) scatter
(c) darken (d) analyze

44. Bland
(a) caustic (b) meager
(c) soft (d) uncooked

45. Blase
(a) fiery (b) clever
(c) intriguing (d) ardent

46. Bleak
(a) pale (b) sudden
(c) dry (d) cheerful

47. Blithe
(a) spiritual (b) profuse
(c) cheerless (d) hybrid

48. Capacious
(a) warlike (b) cordial
(c) curious (d) not spacious

49. Capricious
(a) satisfied (b) insured
(c) photographic (d) steadfast

50. Captious
(a) tolerant (b) capable
(c) frivolous (d) winning

51. Carnal
(a) impressive (b) minute
(c) spiritual (d) actual

52. Carnivorous
(a) gloomy (b) tangential
(c) productive (d) vegetarian

53. Carping
(a) rapid (b) uncritical
(c) unintellectual (d) illegal

54. Castigation
(a) commendation (b) patience
(c) generosity (d) understatement

55. Categorical
(a) negative (b) ironic
(c) impartial (d) qualified

56. Catholic
(a) religious (b) pacific
(c) narrow (d) weighty

57. Celerity
(a) assurance (b) state
(c) acerbity (d) delay

58. Celibate
(a) investing (b) married
(c) retired (d) commodious

59. Censure
(a) process (b) enclose
(c) interest (d) praise

60. Centrifugal
(a) centripetal (b) ephemeral
(c) lasting (d) barometric

61. Cessation
(a) premium (b) gravity
(c) beginning (d) composition

62. Chaffing
(a) achieving (b) serious
(c) caitalistic (d) sneezing

63. Coy
(a) weak (b) airy
(c) brazen (d) old

64. Cozen
(a) amuse (b) treat honestly
(c) prate (d) shackle

65. Craven
(a) desirous (b) direct
(c) bold (d) civilized

66. Crux
(a) affliction (b) spark
(c) events (d) trivial point

67. Cryptic
(a) tomblike (b) futile
(c) famous (d) candid

68. Cupidity
(a) anxiety (b) tragedy
(c) generosity (d) entertainment

69. Curtail
(a) mutter (b) lengthen
(c) express (d) burden

70. Cynical
(a) trusting (b) effortless
(c) conclusive (d) gallant

71. Dank
(a) dry (b) guiltless
(c) warm (d) babbling

72. Dapper
(a) unintelligent (b) untidy
(c) uncertain (d) ungrateful

73. Dauntless
(a) stolid (b) cowardly
(c) irrelevant (d) peculiar

74. Dearth
(a) life (b) abundance
(c) brightness (d) terror

75. Debacle
(a) progress (b) refusal
(c) masque (d) cowardice

76. Debilitate
(a) bedevil (b) repress
(c) strengthen (d) animate

77. Debonair
(a) awkward (b) windy
(c) balmy (d) strong

78. Derogatory
(a) roguish (b) immediate
(c) openionated (d) praising

79. Desecrate
(a) desist (b) integrate
(c) confuse (d) consecrate

80. Despicable
(a) streeing (b) worthy of esteem
(c) inevitable (d) featureless

81. Destitute
(a) affluent (b) dazzling
(c) stationary (d) characteristic

82. Devoid
(a) latent (b) eschewed
(c) full of (d) suspecting

83. Devout
(a) quiet (b) dual
(c) impious (d) straightforward

84. Diabolical
(a) mischievous (b) lavish
(c) seraphic (d) redolent

85. Diatribe
(a) mass (b) range
(c) eulogy (d) elegy

86. Diffidence
(a) sharapness (b) boldness
(c) malcontent (d) dialogue

87. Dilate
(a) procrastinate (b) contract
(c) conclude (d) participate

88. Dilatory
(a) narrowing (b) prompt
(c) enlarging (d) portentous

89. Diminution
(a) expectation (b) context
(c) validity (d) appreciation

90. Din
(a) lightness (b) safety
(c) silence (d) hunger

91. Disabuse
(a) crash (b) violate
(c) renege (d) deceive

92. Disconsolate
(a) unprejudiced (b) thankful
(c) threatrical (d) joyous

93. Enervate
(a) strengthen (b) sputter
(c) arrange (d) scrutinize

94. Enhance
(a) degrade (b) doubt
(c) scuff (d) gasp

95. Ennui
(a) hate (b) excitement
(c) seriousness (d) humility

96. Enunciate
(a) pray (b) request
(c) deliver (d) mumble

97. Ephemeral
(a) sensuous (b) passing
(c) popular (d) eternal

98. Equable
(a) flat (b) decisive
(c) stormy (d) dishonest

99. Equanimity
(a) agitation (b) stirring
(c) volume (d) identity

100. Equilibrium
(a) imbalance (b) peace
(c) inequity (d) directness

101. Equitable
(a) able to leave (b) able to learn
(c) unfair (d) preferable

102. Equivocal
(a) mistaken (b) quaint
(c) azure (d) clear

103. Erratic
(a) unromantic (b) free
(c) popular (d) steady

104. Erroneous
(a) accurate (b) dignified
(c) curious (d) abrupt

105. Erudite
(a) professorial (b) stately
(c) short (d) ignorant

106. Ethereal
(a) long-lasting (b) earthy
(c) ill (d) critical

107. Eulogistic
(a) pretty (b) critical
(c) brief (d) stern

108. Euphonious
(a) strident (b) lethargic
(c) literary (d) significant

109. Evasive
(a) frank (b) correct
(c) empty (d) fertile

110. Exasperate
(a) confide (b) formalize
(c) placate (d) betray

111. Excoriate
(a) scandalize (b) encourage
(c) avoid (d) praise

112. Exculpate
(a) blame (b) prevail
(c) acquire (d) ravish

113. Execrable
(a) innumerable (b) philosophic
(c) physical (d) excellent

114. Excrate
(a) decipher (b) sadden
(c) integrate (d) admit

115. Exhume
(a) decipher (b) sadden
(c) integrate (d) inter

116. Exodus
(a) neglect (b) consent
(c) entry (d) gain

117. Exonerate
(a) forge (b) accuse
(c) record (d) doctor

118. Exorbitant
(a) moderate (b) partisan
(c) military (d) barbaric

119. Extemporaneous
(a) rehearsed (b) hybrid
(c) humilating (d) satetesmanlike

120. Extraneous
(a) modern (b) decisive
(c) essential (d) effective

121. Extrinsic
(a) reputable (b) inherent
(c) swift (d) ambitious

122. Extrovert
(a) clown (b) hero
(c) ectomorph (d) introvert

123. Grandiose
(a) false (b) ideal
(c) proud (d) simple

124. Gratuitous
(a) warranted (b) frank
(c) ingenuous (d) frugal

125. Gregarious
(a) antisocial (b) anticipatory
(c) glorious (d) horrendous

126. Grisly
(a) suggestive (b) doubtful
(c) untidy (d) pleasant

127. Gullible
(a) incredulous (b) fickle
(c) tantamount (d) easy

128. Gusto
(a) noise (b) panic
(c) atmosphere (d) distaste

129. Ghusty
(a) clam (b) noisy
(c) fragrant (d) routine

130. Hackneyed
(a) carried (b) original
(c) banned (d) timely

131. Haggard
(a) shrewish (b) inspired
(c) plump (d) maidenly

132. Halcyon
(a) wasteful (b) prior
(c) subsequent (d) martial

133. Haphazard
(a) safe (b) indifferent
(c) deliberate (d) tense

134. Hapless
(a) cheerful (b) consistent
(c) fortunate (d) considerate

135. Heed
(a) ignore (b) hope
(c) overtake (d) nurture

136. Heretic
(a) sophist (b) believer
(c) interpreter (d) pacifist

137. Heterogeneous
(a) orthodox (b) pagan
(c) unlikely (d) similar

138. Hibernal
(a) musical (b) summerlike
(c) local (d) seasonal

139. Hilarity
(a) gloom (b) heartiness
(c) weakness (d) casualty

140. Hirsute
(a) scaly (b) bald
(c) erudite (d) quiet

141. Hortatory
(a) inquiring (b) denying
(c) killing (d) dissuading

142. Hovr
(a) commence (b) soothe
(c) leave (d) transform

143. Hunbbub
(a) calm (b) fury
(c) capital (d) axle

144. Hummock
(a) ummusical (b) scorn
(c) wakefulness (d) vale

145. Husbandry
(a) sportsmanship (b) dishonesty
(c) wastefulness (d) friction

146. Hybrid
(a) productive (b) special
(c) purebred (d) oafish

147. Hyperbole
(a) velocity (b) climax
(c) curve (d) understatement

148. Hypercritical
(a) tolerant (b) false
(c) extreme (d) inarticulate

149. Hypocritical
(a) sincere (b) narrow-minded
(c) shameful (d) amiable

150. Hypothetial
(a) rational (b) fantastic
(c) wizened (d) axiomatic

151. Ignoble
(a) produced by fire (b) worthy
(c) given to questioning (d) huge

152. Illusive
(a) not deceptive (b) not certain
(c) not obvious (d) not choherent

153. Irksome
(a) interesting (b) lazy
(c) tireless (d) devious

154. Irrelevant
(a) lacking piety (b) fragile
(c) congruent (d) pertinent

155. Irreparable
(a) legible (b) correctable
(c) proverbial (d) concise

156. Irtreverent
(a) related (b) mischievous
(c) respective (d) pious

157. Jaded
(a) upright (b) stimulated
(c) aspiring (d) applied

158. Jaundiced
(a) whitened (b) inflamed
(c) quickened (d) unbiased

159. Jaunty
(a) youthful (b) ruddy
(c) strong (d) sedate

160. Jeopardy
(a) patience (b) courage
(c) safety (d) willingness

161. Jettison
(a) salvage (b) submerge
(c) descend (d) decelerate

162. Jocular
(a) arterial (b) bloodless
(c) verbose (d) serious

163. Judicious
(a) punitive (b) unwise
(c) criminal (d) licit

164. Kindle
(a) dislike (b) quench
(c) gather (d) sparkle

165. Lachrymose
(a) cheering (b) smooth
(c) passionate (d) curt

166. Lackadasical
(a) monthly (b) possessing time
(c) ambitious (d) pusillanimous

167. Laconic
(a) milky (b) verbose
(c) wicked (d) flagrant

168. Lampoon
(a) darken (b) praise
(c) abandon (d) sail

169. Languor
(a) vitality (b) length
(c) embarrassment (d) wine

170. Latent
(a) trim (b) forbidding
(c) execrable (d) obvious

171. Lavish
(a) hostile (b) unwashed
(c) timely (d) frugal

172. Laudatory
(a) dritying (b) disclaiming
(c) defamatory (d) inflammatory

173. Lax
(a) salty (b) strick
(c) shrill (d) boring

174. Lechery
(a) trust (b) compulsion
(c) zeal (d) purity

175. Lethargic
(a) convalescent (b) beautiful
(c) enervating (d) invigorating

176. Levite
(a) bridge (b) dam
(c) praise (d) solemnity

177. Lilliputian
(a) destructive (b) proper
(c) gigantic (d) elegant

178. Limpid
(a) erect (b) turbid
(c) tangential (d) timid

179. Lithe
(a) stiff (b) limpid
(c) facetious (d) insipid

180. Livid
(a) alive (b) mundane
(c) positive (d) undiscolored

181. Loath
(a) loose (b) evident
(c) deliberate (d) eager

182. Loquacious
(a) taciturn (b) sentimental
(c) soporific (d) soothing

183. Moist
(a) crisp (b) hard
(c) dry (d) parched

184. Mask
(a) hit (b) expose
(c) benign (d) pompous

185. Moisture
(a) dryness (b) delegate
(c) dampness (d) fdefect

186. Magnify
(a) produce (b) support
(c) reduce (d) destroy

187. Mendacious
(a) broken (b) efficacious
(c) truthful (d) destructive

188. Manumit
(a) throw (b) lock
(c) promise (d) enslave

189. Nebulous
(a) starry (b) clear
(c) cold (d) fundamental

190. Nefarious
(a) various (b) lacking
(c) benign (d) pompous

191. Negation
(a) postulation (b) hypothecation
(c) affirmation (d) violation

192. Neophyte
(a) veteran (b) satellite
(c) desperado (d) handwriting

193. Niggardly
(a) protected (b) biased
(c) prodigal (d) bankrupt

194. Nocturnal
(a) harsh (b) marauding
(c) patrolling (d) daily

195. Noisome
(a) quiet (b) dismayed
(c) fragrant (d) sleepy

196. Notorious
(a) fashionable (b) renowned
(c) inactive (d) interepid

197. Obdurate
(a) yielding (b) fleeting
(c) finite (d) fascinating

198. Obese
(a) skillful (b) cadaverous
(c) clever (d) unpredictable

199. Objective
(a) indecisive (b) apathetic
(c) markedly inferior (d) emotionally involved

200. Obligatory
(a) demanding (b) optional
(c) facile (d) friendly

201. Obloquy
(a) praise (b) rectangle
(c) circle (d) dialogue

202. Obsequious
(a) successful (b) democratic
(c) supercilious (d) ambitious

203. Obsession
(a) whim (b) loss
(c) phobia (d) delusion

204. Obsolete
(a) heated (b) desolate
(c) renovated (d) frightful

205. Obstreperous
(a) turbid (b) quiet
(c) remote (d) lucid

206. Obtuse
(a) sheer (b) transparent
(c) tranquil (d) shrewd

207. Odious
(a) fragrant (b) redolent
(c) fetid (d) delightful

208. Odium
(a) noise (b) liking
(c) dominant (d) hasty

209. Omnipotent
(a) weak (b) democratic
(c) despotic (d) passionate

210. Omniscient
(a) sophisticated (b) ignorant
(c) essential (d) trivial

211. Opiat
(a) distress (b) sleep
(c) stimulant (d) laziness

212. Opportune
(a) occasional (b) fragrant
(c) fragile (d) awkward

213. Opportunist
(a) man of destiny (b) man of principle
(c) changeling (d) adversary

214. Opporbrium
(a) delineation (b) aptitude
(c) majesty (d) praise

215. Optimum
(a) pessimistic (b) knowledgeable
(c) worst (d) minimum

216. Opulence
(a) pessimism (b) patrionsm
(c) potency (d) poverty

217. Ostentatious
(a) inactive (b) unassuming
(c) impolite (d) illicit

218. Overweening
(a) humble (b) impotent
(c) avid (d) acrimonious

219. Pejorative
(a) positive (b) legal
(c) determining (d) delighting

220. Pellucid
(a) logistical (b) philandering
(c) incomperhensible (d) vagrant

221. Penchant
(a) distance (b) imminence
(c) dislike (d) attitude

222. Penurious
(a) imprisoned (b) captivated
(c) generous (d) vacant

223. Perfunctory
(a) official (b) through
(c) insipid (d) vicarious

224. Perigee
(a) eclipse (b) planet
(c) apogee (d) refugee

225. Peripatetic
(a) wordly (b) stationary
(c) disarming (d) seeking

226. Permeable
(a) perishable (b) effective
(c) plodding (d) impenetrable

227. Pernicious
(a) practical (b) comparative
(c) harmless (d) tangible

228. Perpetual
(a) momentary (b) standard
(c) serious (d) industrial

229. Perspicuity
(a) grace (b) feature
(c) review (d) vagueness

230. Pert
(a) polite (b) perishable
(c) moral (d) deliberate

231. Pertinacious
(a) vengeful (b) consumptive
(c) superficial (d) skilled

232. Pertient
(a) understood (b) living
(c) discontented (d) irrelevant

233. Petulant
(a) angry (b) moral
(c) declining (d) uncomplaining

234. Precipitate
(a) dull (b) anticipatory
(c) cautious (d) considerate

235. Prefatory
(a) outstanding (b) magnificent
(c) conclusive (d) intelligent

236. Prelude
(a) intermezzo (b) diva
(c) aria (d) aftermath

237. Presumption
(a) assertion (b) activation
(c) motivation (d) humility

238. Pretentious
(a) ominous (b) calm
(c) unassuming (d) futile

239. Prim
(a) informal (b) prior
(c) exterior (d) private

240. Pristine
(a) cultivated (b) condemned
(c) irreligious (d) cautious

241. Probity
(a) aregert (b) assumption
(c) corruptibility (d) extent

242. Prodigal
(a) large (b) thrifty
(c) consistent (d) compatible

243. Prodigious
(a) infinitesimal (b) indignant
(c) indifferent (d) indisposed

244. Profane
(a) sanctify (b) desecrate
(c) define (d) manifest

245. Prolific
(a) unworkable (b) backward
(c) barren (d) controversial

246. Prolix
(a) stupid (b) indifferent
(c) redundant (d) pithy

247. Prophylactic
(a) causing growth (b) causing disease
(c) antagonistic (d) brushing

248. Propinquity
(a) remoteness (b) uniqueness
(c) health (d) virtue

249. Propitious
(a) rich (b) induced
(c) promoted (d) unfavorable

250. Prosaic
(a) pacified (b) reprieved
(c) pensive (d) imaginative

251. Protean
(a) amateur (b) catholic
(c) unchanging (d) rapid

252. Protract
(a) make circular (b) shorten
(c) further (d) rectrace

253. Provident
(a) unholy (b) rash
(c) miserable (d) disabled

254. Provincial
(a) wealthy (b) crass
(c) literary (d) sophisticated

255. Psychotic
(a) dangerous (b) clairvoyant
(c) criminal (d) sane

256. Puerile
(a) fragrant (b) adult
(c) lonely (d) feminine

257. Pugnacious
(a) pacific (b) feline
(c) mature (d) angular

258. Puissant
(a) pouring (b) fashionable
(c) articulate (d) weak

259. Pulchritude
(a) ugliness (b) notoriety
(c) bestiality (d) masculinity

260. Punctilious
(a) happy (b) active
(c) vivid (d) careless

261. Punitive
(a) large (b) humorous
(c) rewarding (d) restive

262. Pusillanimous
(a) poverty-stricken (b) chained
(c) posthumous (d) courageous

263. Putative
(a) colonial (b) quarrelsome
(c) undisputed (d) powerful

264. Resilient
(a) pungent (b) foolish
(c) worthy (d) unyielding

265. Restive
(a) buoyant (b) placid
(c) remorseful (d) resistant

266. Retentive
(a) forgetful (b) accepting
(c) repetitive (d) avoiding

267. Reticence
(a) fatigue (b) fashion
(c) treachery (d) loquaciousness

268. Retrograde
(a) progressing (b) inclining
(c) evaluating (d) concentrating

269. Revere
(a) advance (b) dishonor
(c) age (d) precede

270. Rife
(a) direct (b) scant
(c) peaceful (d) grim

271. Robust
(a) weak (b) violent
(c) vicious (d) villainous

272. Rotundity
(a) promenade (b) nave
(c) grotesqueness (d) slimness

273. Rubble
(a) artificial facade (b) unbroken stone
(c) pale complexion (d) strong defense

274. Ruddy
(a) robust (b) witty
(c) wan (d) exotic

275. Rudimentary
(a) pale (b) polite
(c) asinine (d) developed

276. Rueful
(a) trite (b) content
(c) capable (d) capital

277. Rustic
(a) urban (b) slow
(c) corroded (d) mercenary

278. Ruthless
(a) merciful (b) majestic
(c) mighty (d) militant

279. Sadistic
(a) happy (b) quaint
(c) kindhearted (d) vacant

280. Sagacious
(a) foolish (b) bitter
(c) voracious (d) veracious

281. Sallow
(a) salacious (b) ruddy
(c) colorless (d) permitted

282. Salubrious
(a) salty (b) bloody
(c) miasmic (d) maudlin

283. Salvage
(a) remove (b) outfit
(c) burn (d) lose

284. Sanctimonious
(a) hypothetical (b) paltry
(c) mercenary (d) pious

285. Sanguine
(a) choleric (b) sickening
(c) warranted (d) pessimistic

286. Satiety
(a) emptiness (b) warmth
(c) erectness (d) straightness

287. Scanty
(a) collected (b) remote
(c) invisible (d) plentiful

288. Scurrilous
(a) savage (b) scabby
(c) decent (d) volatile

289. Secular
(a) vivid (b) clerical
(c) punitive (d) positive

290. Sedentary
(a) vicarious (b) loyal
(c) accidental (d) active

291. Sedulous
(a) indolent (b) guileless
(c) vindictive (d) upright

292. Senlity
(a) verility (b) loquaciousness
(c) forgetfulness (d) youth

293. Sententious
(a) paragraphed (b) positive
(c) posthumous (d) wordy

294. Tractable
(a) unmanageable (b) irreligious
(c) mortal (d) incapable

295. Traduce
(a) exhume (b) increase
(c) purchase (d) extol

296. Tranquillity
(a) lack of sleep (b) lack of calm
(c) emptiness (d) renewal

297. Transient
(a) carried (b) close
(c) permanent (d) removed

298. Tremulous
(a) steady (b) obese
(c) young (d) healthy

299. Trenchant
(a) lacking bite (b) imperious
(c) inessential (d) unafraid

300. Trepidation
(a) slowness (b) amputation
(c) fearlessness (d) adroitness

301. Trite
(a) correct (b) original
(c) distinguished (d) premature

302. Truculent
(a) juicy (b) overflowing
(c) peaceful (d) determined

303. Tumult
(a) scarcity (b) defeat
(c) cooleness (d) secrenity

304. Turbid
(a) clear (b) improbable
(c) invariable (d) honest

305. Turbulence
(a) reaction (b) approach
(c) impropriety (d) calm

306. Turgid
(a) rancid (b) shrunken
(c) cool (d) explosive

307. Turpitude
(a) amplitude (b) heat
(c) wealth (d) virtue

308. Tyro
(a) infant (b) rubber
(c) personnel (d) expert

309. Unearth
(a) conceal (b) gnaw
(c) clean (d) fling

310. Unfeigned
(a) pretended (b) fashionable
(c) wary (d) switched

311. Ungainly
(a) ignorant
(b) graceful
(c) detailed
(d) dancing

312. Unimpeachable
(a) fruitful
(b) rampaging
(c) faulty
(d) pensive

313. Unkempt
(a) bombed (b) washed
(c) neat (d) tawdry

314. Unruly
(a) chatting (b) obedient
(c) definite (d) lined

315. Unseemly
(a) effortless (b) proper
(c) conducive (d) pointed

316. Unsullied
(a) tranished (b) countless
(c) soggy (d) papered

317. Untenable
(a) supportable (b) tender
(c) sheepish (d) tremulous

318. Unwitting
(a) clever (b) intense
(c) sensitive (d) intentional

319. Vacillation
(a) remorse (b) relief
(c) respect (d) steadfastness

320. Valedictory
(a) sad (b) collegiate
(c) derivative (d) salutatory

321. Valor
(a) admonition
(b) injustice
(c) cowardice
(d) generosity

322. Vanguard
(a) regiment
(b) rear
(c) echelon
(d) protection

323. Vaunted
(a) unvanquished
(b) fell
(c) belittled
(d) believed

EXERCISE – II

(QUESTIONS FROM PREVIOUS GATE EXAMS)

MCQ TYPE QUESTIONS

2015

1. Which word is not a synonym for the word *vernacular*?
(a) regional (b) indigeneous
(c) indigent (d) colloquial

2011

2. Choose the word from the options given below that is most nearly opposite in meaning to the given word:
Amalgamate
(a) merge
(b) split
(c) collect
(d) separate

3. Choose the word from the options given below that is most nearly **opposite** in meaning to the given word :
Frequency
(a) periodicity (b) rarity
(c) gradualness (d) persistency

MCQ Type Questions

1. (d)	2. (b)	3. (d)	4. (b)	5. (a)	6. (d)	7. (a)	8. (a)	9. (a)	10. (b)
11. (b)	12. (b)	13. (b)	14. (d)	15. (b)	16. (c)	17. (d)	18. (c)	19. (c)	20. (d)
21. (d)	22. (a)	23. (d)	24. (c)	25. (b)	26. (d)	27. (b)	28. (b)	29. (d)	30. (d)
31. (c)	32. (a)	33. (d)	34. (b)	35. (d)	36. (a)	37. (d)	38. (a)	39. (b)	40. (b)
41. (d)	42. (d)	43. (c)	44. (a)	45. (d)	46. (d)	47. (c)	48. (d)	49. (d)	50. (a)
51. (c)	52. (d)	53. (b)	54. (a)	55. (d)	56. (c)	57. (b)	58. (a)	59. (b)	60. (a)
61. (c)	62. (b)	63. (c)	64. (b)	65. (c)	66. (d)	67. (d)	68. (c)	69. (b)	70. (a)
71. (a)	72. (b)	73. (b)	74. (b)	75. (a)	76. (c)	77. (a)	78. (d)	79. (d)	80. (b)
81. (a)	82. (c)	83. (c)	84. (d)	85. (c)	86. (b)	87. (b)	88. (b)	89. (d)	90. (c)
91. (d)	92. (d)	93. (a)	94. (a)	95. (b)	96. (d)	97. (d)	98. (c)	99. (a)	100. (a)
101. (c)	102. (b)	103. (d)	104. (a)	105. (d)	106. (b)	107. (b)	108. (a)	109. (a)	110. (c)
111. (d)	112. (a)	113. (d)	114. (b)	115. (d)	116. (c)	117. (b)	118. (a)	119. (b)	120. (c)
121. (b)	122. (d)	123. (d)	124. (a)	125. (a)	126. (d)	127. (a)	128. (d)	129. (d)	130. (b)
131. (c)	132. (d)	133. (c)	134. (c)	135. (a)	136. (b)	137. (d)	138. (b)	139. (a)	140. (b)
141. (d)	142. (c)	143. (a)	144. (d)	145. (c)	146. (c)	147. (d)	148. (a)	149. (a)	150. (d)
151. (b)	152. (a)	153. (a)	154. (d)	155. (b)	156. (d)	157. (b)	158. (d)	159. (d)	160. (c)
161. (a)	162. (d)	163. (a)	164. (b)	165. (a)	166. (c)	167. (d)	168. (b)	169. (a)	170. (d)
171. (d)	172. (c)	173. (b)	174. (d)	175. (d)	176. (d)	177. (c)	178. (b)	179. (a)	180. (d)
181. (d)	182. (a)	183. (c)	184. (b)	185. (a)	186. (c)	187. (c)	188. (d)	189. (b)	190. (c)
191. (c)	192. (a)	193. (c)	194. (d)	195. (c)	196. (b)	197. (d)	198. (d)	199. (d)	200. (b)
201. (a)	202. (c)	203. (a)	204. (c)	205. (b)	206. (d)	207. (d)	208. (b)	209. (a)	210. (a)
211. (c)	212. (d)	213. (b)	214. (d)	215. (c)	216. (d)	217. (b)	218. (a)	219. (a)	220. (c)
221. (c)	222. (c)	223. (b)	224. (c)	225. (b)	226. (d)	227. (c)	228. (a)	229. (d)	230. (a)
231. (c)	232. (d)	233. (d)	234. (c)	235. (c)	236. (d)	237. (d)	238. (c)	239. (a)	240. (a)
241. (c)	242. (b)	243. (a)	244. (a)	245. (c)	246. (d)	247. (b)	248. (a)	249. (d)	250. (d)
251. (c)	252. (b)	253. (b)	254. (d)	255. (d)	256. (b)	257. (a)	258. (d)	259. (a)	260. (b)
261. (c)	262. (d)	263. (c)	264. (b)	265. (a)	266. (d)	267. (a)	268. (b)	269. (b)	270. (b)
271. (a)	272. (d)	273. (b)	274. (c)	275. (d)	276. (b)	277. (a)	278. (a)	279. (c)	280. (a)
281. (b)	282. (c)	283. (d)	284. (d)	285. (d)	286. (d)	287. (d)	288. (c)	289. (b)	290. (d)
291. (a)	292. (d)	293. (d)	294. (a)	295. (d)	296. (b)	297. (c)	298. (a)	299. (a)	300. (c)
301. (b)	302. (c)	303. (d)	304. (a)	305. (d)	306. (b)	307. (d)	308. (d)	309. (a)	310. (a)
311. (b)	312. (c)	313. (d)	314. (b)	315. (b)	316. (a)	317. (a)	318. (d)	319. (d)	320. (d)
321. (c)	322. (b)	323. (d)							

EXERCISE – II

MCQ Type Questions

1. (c) 2. (b) 3. (b)

EXPLANATIONS

EXERCISE – II

MCQ TYPE QUESTIONS

1. *Vernacular* - expressed or written in the native language of a place

   Indigent - deficient in what is requisite

2. Amalgamate means combine or unite to form one organization or structure. So the best option here is split. Separate on the other hand, although a close synonym, it is too general to be the best antonym in the given question while Merge is the synonym; Collect is not related.

3. Best antonym is rarity which means shortage or scarcity.

5 CHAPTER

Reasoning Ability

Reasoning is the process of drawing conclusions from facts. These conclusions must follow inevitably from the facts from which they are drawn. Reasoning is not concerned with some conclusion that has a good chance of being true when facts are true. Indeed, reasoning as used here refers to logical reasoning, not of commonsense reasoning or probabilistic reasoning. The only conclusions that are acceptable are those that follow logically from the supplied facts.

TYPES OF REASONING

Reasoning can be classified in two major parts :

- Verbal Reasoning
- Non-verbal Reasoning

VERBAL REASONING

Verbal reasoning tests words, letters and numbers, and require logical reasoning and a reasonable knowledge of the English Language. It is also necessary to be familiar with simple manipulations with figures, like addition, subtractions division and multiplication. The problems of number in *Test of Reasoning* test how logical we are i.e., how well we reason and think while carrying out simple arithmetic manipulations.

CLASSIFICATION OF VERBAL REASONING

I. SERIES COMPLETION

In verbal series, words, letters or digits are given in a specific sequence/order and we have to find out next word, letter or digit to complete the given series.

Numbers or alphabetical letters, are generally called *terms of the series*. These terms follow a certain pattern throughout. In the questions we have to identify last one or two terms to continue the series or to find a missing term in between given terms to continue the sequence followed in the question. There is no set pattern and each question may follow a different pattern or sequential arrangement of letters or digits, which have to detect using common sense and reasoning ability.

TYPES OF SERIES COMPLETION QUESTIONS

Mainly following four types of question are asked.

1. **Alphabet series**

   In this series, given alphabets follow a particular sequence or order. We have to detect the pattern from the given alphabets and find missing alphabet or the next alphabet to continue the pattern.

- There are no set rules.
- There can be omission of alphabets in an order.
- Alphabets may also be omitted in an increasing/decreasing order, which may be direct increase or decrease.
- There can also be alternate order.
- There may also be alternate sequences
- There may be several other patterns in the letter series.
- To tackle letter series questions, vary position of the alphabet and its position number in both forward and backward sequences.
- To continue the series after Z, we again begin with A. In other words, the sequence is kept in a circular order.
- In solving these questions pattern of the alphabet series should be noted.

Some Skipping Patterns.

(*i*) ***Regular Order :*** Number of alphabets skipped remains the same.

Example. A, D, G, J, ...?

Ans. M

(*ii*) ***Increasing Order :*** Each time the number of alphabets skipped increases in a given pattern.

Example. A C F J O ?

Ans. U

Here, each time number of letters skipped increases by one.

(*iii*) ***Decreasing Order :*** Each time the number of letters skipped decreases in a given pattern.

Example. A G L P S ?

Ans. U

Here number of letters skipped decreases by one each time, *i.e.*, first 5, then 4, then 3, and so on.

(*iv*) ***Interlinked Series :*** In this two or more different series are attached together. These different series follow their own different rules.

Example. A D F J M R ?

Ans. V

Here, there are two interlinked series.

2. **Letter Series**

   This type of questions usually consist of a series of small letters which follow a certain pattern.

However, some letters are missing from the series. Then these missing letters are given in a proper sequence as one of the alternatives.

Example. aaa... b...a...

(*a*) baa (*b*) abb (*c*) bab (*d*) aab (*e*) bbb

Ans. (*a*)

Method :

- First blank space should be filled in by 'b', so that we have two a's followed by two b's.
- Second blank space should be filled in either by 'a', so that we have four a's followed by two b's, or by 'b' so that we have three a's followed by three b's.
- Last space must be filled in by 'a'.
- *Thus, we have two possible answers :* 'baa' and 'bba'.

But, only 'baa' appears in the alternatives. So, the answer is (*a*).

Note : In case, we had both the possible answers in the alternatives, we would have chosen the one that forms a more prominent pattern, which is aabb/ aaabbbaa. Thus, our answer would have been "*bba*".

3. Number series

In the number series, some numbers are arranged in a particular sequence. All the numbers form a series and change in a certain order. Sometimes, one or more numbers are wrongly put in the number series. We are required to observe the trend in which numbers change in the series and find out which number/numbers misfit into the series that number/numbers is *odd number* of the series.

Pure Series.

In this type of number series, the number itself obeys certain order so that the character of the series can be found out.

Number may be Perfect square; Perfect cube; Prime and Combination

Difference Series.

In this type of number series, change in order for the difference between each consecutive number of the series is found out.

Ratio Series.

In this type of number series, change in order for the ratios between each consecutive number of the series is found out.

4. Mixed Series

In this type of number series, numbers obeying various orders of two or more different types of series are arranged alternately in a single number series.

Rules:

1. Difference between consecutive numbers is same.
2. Differences between consecutive numbers are in Arithmetic Progression (A.P.).
3. Difference between consecutive numbers is a perfect square.
4. Differences between consecutive numbers are multiples of a number.
5. Differences between consecutive numbers are prime numbers.
6. Difference between consecutive numbers is a perfect cube.
7. Difference between consecutive numbers are in Geometric Progression (G.P.)
8. Ratio between each consecutive number is the same.
9. Ratio between each consecutive numbers is in Arithmetic Progression (A.P.)
10. Ratio between consecutive number is perfect square number.
11. Ratio between consecutive number is the multiple of a number.
12. Ratio between consecutive numbers is a prime number.
13. Ratio between consecutive numbers is a perfect cube number.
14. Ratios between consecutive number are in Geometric Progression (G.P.)

5. Correspondence Series

This type of series consists of three sequences with three different elements (usually capital letters, digits and small letters). On the basis of the similarity in positions in the three sequences, a capital letter is found to correspond with a unique digit and a unique small letter, whenever it occurs. We are required to trace out this correspondence and accordingly choose the elements to be filled in at the desired places.

II. ANALOGY

'*Analogy*' means 'correspondence'.

In questions based on analogy, a particular relationship is given and another similar relationship has to be identified from the given alternatives.

Analogy means similarity or some kind of relationship. Analogy test, therefore tests the ability to see a relationship between two words and to recognize a similar relationship between two other words. It is a test of verbal reasoning ability rather than vocabulary. The key to analogy success is being able to express the relationship between the pair of words.

KINDS OF RELATIONSHIPS WITH EXAMPLES

1. Instrument and Measurement

- Thermometer : Temperature (thermometer is an instrument used to measure temperature).
- Barometer : Pressure
- Anemometer : Wind vane
- Odometer : Speed
- Scale : Length
- Balance : Mass
- Sphygmomanometer: Blood Pressure
- Rain Gauge : Rain
- Hygrometer : Humidity
- Ammeter : Current
- Screw Gauge : Thickness
- Seismograph : Earthquakes
- Taseometer : Strains

2. Quantity and Unit

- Mass : Kilogram (kilogram is the unit of mass)
- Length : Metre
- Force : Newton
- Energy : Joule
- Resistance : Ohm
- Volume : Litre
- Angle : Radians
- Time : Seconds
- Potential : Volt
- Work : Joule
- Current : Ampere
- Luminosity : Candela
- Pressure : Pascal
- Area : Hectare
- Temperature : Degrees
- Power : Watt
- Conductivity : Mho
- Magnetic field : Oersted/ Tesla

3. Individual and Group

- Soldiers : Army (group of soldiers is called an army).
- Cattle : Herd
- Flowers : Bouquet
- Grapes : Bunch
- Singer : Chorus
- Artist : Troupe
- Fish : Shoal
- Sheep : Flock
- Riders : Cavalcade
- Bees : Swarm
- Man : Crowd
- Sailors : Crew
- Nomads : Horde

4. Animal and Young one

- Cow : Calf (calf is the young one of cow)
- Horse : Pony
- Cat : Kitten
- Sheep : Lamb
- Butterfly : Caterpillar
- Insect : Larva
- Dog : Puppy
- Cock/Hen : Chicken
- Lion : Cub
- Duck : Duckling
- Man : Child

5. Male and Female

- Horse : Mare (mare is the female horse)
- Dog : Bitch
- Stag : Doe
- Son : Daughter
- Lion : Lioness
- Sorcerer : Sorceress
- Drone . Bee
- Gentleman : Lady
- Nephew : Niece
- Tiger : Tigress

6. Individual and Class

- Monkey : Mammal (monkey belongs to the class of mammal).
- Man : Mammal
- Ostrich : Bird
- Snake : Reptile
- Butterfly : Insect
- Whale : Mammal
- Rat : Rodent
- Lizard : Reptile

7. Individual and Dwelling Place

- Dog : Kennel (dog lives in a kennel)
- Bee : Apiary
- Cattle : Shed
- Lion : Den
- Poultry : Farm
- Monk : Monastery
- Fish : Aquarium
- Birds : Aviary
- Horse : Stable

8. Study and Topic

- Ornithology : Birds (ornithology is the study of birds).
- Seismology : Earthquakes
- Botany : Plants
- Onomatology : Names
- Ethnology : Human Races
- Ontology : Reality
- Herpetology : Amphibians
- Pathology : Diseases
- Astrology : Future
- Anthropology : Man
- Palaeography : Writings
- Ichthyology : Fishes
- Semantics : Language
- Nephrology : Kidney
- Concology : Shells
- Haematology : Blood
- Craniology : Skull
- Mycology : Fungi
- Entomology : Insects
- Zoology : Animals
- Occultism : Supernatural

9. Worker and Tool

- Warrior : Sword (sword is the tool used by a warrior).
- Carpenter : Saw
- Woodcutter : Axe
- Tailor : Needle
- Labourer : Spade
- Soldier : Gun
- Sculptor : Chisel
- Chef : Knife
- Doctor : Stethoscope
- Farmer : Plough
- Author : Pen

10. Tool and Action

- Needle : Sew (needle is used for sewing).
- Knife : Cut
- Sword : Slaughter
- Mattock : Dig
- Filter : Purify
- Steering : Drive
- Pen : Write
- Spanner : Grip
- Spoon : Feed
- Microscope : Magnify
- Gun : Shoot
- Shovel : Scoop
- Chisel : Carve
- Oar : Row
- Axe : Grind
- Auger : Bore
- Spade : Dig

11. Worker and Working Place

- Chef : Kitchen (chef works in a kitchen)

- Farmer : Field
- Warrior : Battlefield
- Engineer : Site
- Sailor : Ship
- Pilot : Cockpit
- Beautician : Parlour
- Artist : Theatre
- Actor : Stage
- Mechanic : Garage
- Lawyer : Court
- Scientist : Laboratory
- Teacher : School
- Doctor : Hospital
- Clerk : Office
- Servant : House
- Driver : Cabin
- Grocer : Shop
- Painter : Gallery
- Waiter · Restaurant
- Worker : Factory
- Umpire : Pitch
- Gambler : Casino

12. Worker and Product

- Mason : Wall (mason builds a wall)
- Farmer : Crop
- Teacher : Education
- Hunter : Prey
- Chef : Food
- Carpenter : Furniture
- Judge : Justice
- Author : Book
- Choreographer : Ballet
- Goldsmith : Ornaments
- Producer : Film
- Butcher : Meat
- Architect : Design
- Cobbler : Shoes
- Tailor : Clothes
- Poet : Poem
- Dramatist : Play

13. Product and Raw Material

- Prism : Glass (prism is made of glass).
- Butter : Milk
- Cloth : Fibre
- Wall : Brick
- Road : Asphalt
- Furniture : Wood
- Book : Paper
- Shoes : Leather
- Sack : Jute
- Pullover : Wool
- Omelette : Egg
- Metal : Ore
- Jewellery : Gold
- Rubber : Latex
- Linen : Flax
- Jaggery : Sugarcane
- Oil : Seed
- Wine : Grapes
- Paper : Pulp
- Fabric : Yarn

14. Part and Whole Relationship

- Pen : Nib (nib is a part of a pen).
- Pencil : Lead
- House : Kitchen
- Fan : Blade
- Class : Student
- Room : Window
- Aeroplane : Cockpit
- Book : Chapter

15. Word and Intensity

- Anger : Rage (rage is of higher intensity than Anger)
- Wish : Desire
- Kindle : Burn
- Sink : Drown
- Quarrel : War
- Error : Blunder
- Famous : Renowned
- Unhappy : Sad
- Refuse : Deny

16. Word and Synonym

- Abode : Dwelling (abode means almost the same as Dwelling)
- Blend : Mix
- Ban : Prohibition
- Assign : Allot
- Vacant : Empty
- Abduct : Kidnap
- Dearth : Scarcity
- Dissipate : Squander
- Sedate : Calm
- Brim : Edge
- House : Home
- Solicit : Request
- Presage : Predict
- Haughty : Proud
- Flaw : Defect
- Fierce : Violent
- Fallacy : Illusion
- Substitute : Replace
- Mend : Repair
- Alight : Descend
- Presume : Assume

17. Word and Antonym

- Attack : Defend (defend means opposite of Attack)
- Advance : Retreat
- Cruel : Kind
- Best : Worst
- Fresh : Stale
- Ignore : Notice
- Initial : Final
- Condense : Expand
- Chaos : Peace
- Create : Destroy
- Gradual : Abrupt
- Sink : Float
- Robust : Weak
- Gentle : Harsh
- Deep : Shallow
- Cordial : Hostile
- Affirm : Deny
- Mourn : Rejoice
- Lethargy : Alertness

18. Part

Gill : Fin (Gill and fin are two parts of a fish)

19. Type

Sword : Weapon(Sword is a type of weapon)

20. Tool & Worker

Paint Brush : Artist (Paintbrush is a tool to artist)

21. Degree

Warm : Hot (Warm is a greater degree of hot)

Fond : Doting (Fond is less extreme than doting)

22. Sign of

Grimace : Pain(Grimace is a sign of pain)

Brush : Discomfiture (A blush signifies discomfiture)

23. A place for

Miner : Quarry (A miner works in a quarry)

24. Sequence relationship

Foreward : Appendix (In a book a foreword precedes the Appendix)

25. Worker and Work

Poet : Sonnet (A sonnet is written by a poet)

Mason : Wall (A mason builds a wall)

26. Cause and Effect

Soporific : Sleepiness (Soporific causes sleepiness)

COMMON TYPES OF ANALOGY QUESTIONS

1. Completing Analogous Pair

In this type of questions, given two words are, which related to each other in some way. Another word is also given and we required to find out the relationship between first two words and choose the word from the given alternatives, which bears the same relationship to the third word, as the first two bear.

Example. Doctor : Nurse : : ? : Follower

(*a*) Employer (*b*) Leader (*c*) Worker
(*d*) Manager (*e*) Union

Ans. (*b*)

Just as a nurse follows the Doctor's instructions, so also a follower works as directed by the leader.

2. Choosing Analogous Pair

In this type of questions, a pair of words is given, followed by four pairs of words as alternatives and we are required to choose the pair in which words bear the same relationship to each other as the words of the given pair bear.

Example. Sink : Float : : ?

(*a*) Brim : Edge (*b*) Mend : Repair
(*c*) Gentle : Harsh (*d*) Flow : defect
(*e*) Solicit : Request

Ans. (*b*)

Just as sink is antonym of float, so also gentle is opposite of harsh.

3. Choosing Similar word

In this type of questions, a group of three words is given, followed by four other words as alternatives. The candidate is required to choose the alternative, which is similar to the given three words.

Example. Sitar : Guitar : Tanpura

(*a*) Trumpet (*b*) Violin (*c*) Harmonium
(*d*) Mridanga (*e*) Fluet

Ans. (*b*)

Sitar, Guitar and Tanpura are all string instruments. Violin is also a string instrument.

4. Detecting Analogies

In this type of questions, we are required to trace out hidden analogy or common characteristic among the given words or to choose the word which possesses the same characteristic as the given word.

Example. Judo : Karate : Taekwando

(*a*) They are names of martial arts.
(*b*) They can be performed by obese persons.
(*c*) They are performed on stage.
(*d*) They are important items of Asian Games.
(*e*) They have origin in Kerala

Ans. (*a*)

Judo, Karate and Taekwando are martial arts and (*a*) is most suitable description for all the three.

5. Three word Analogy

In this type of questions, a group of three interrelated words is given and we are required to trace out the relationship among these three words and choose another group with similar analogy, from among the alternatives provided.

Example. Pen : Pencil : Ink

(*a*) Orange : Banana : Juice
(*b*) Table : Chair : Wood
(*c*) Cow : Milk : Curd
(*d*) Fish : Shark : Water
(*e*) Car : Engine : Circle

Ans. (*a*)

Pen contains ink and pencil belongs to the same category as pen i.e. stationery. Similarly, orange contains juice and banana belongs to the same category as orange, *i.e.* fruits.

6. Number Analogy

It includes deals two types of questions :

(*i*) Choosing a similarly related pair as the given number pair on the basis of relation between the numbers in each pair.

Example. 582 : 194 :: 258 : ?

(*a*) 82 (*b*) 86 (*c*) 92 (*d*) 58

Ans. (*b*)

Just as 582 is related to 194 as 582 is 3 times 194, in the same way 258 is 3 times the 86.

(*ii*) Choosing a number similar to a group of numbers on the basis of certain common properties that they possess.

Example. Which number belongs to given set of numbers?

2, 3, 5, 7, 11, 13, 17, 19, 23.....

(*a*) 15 (*b*) 9 (*c*) 21 (*d*) 29 (*e*) 27

Ans. (*d*)

Given set of numbers belong to the prime numbers. In the given option, only prime number is 29.

7. Alphabet Analogy

In this type of questions, two groups of alphabets related to each other in same way, are given and we are required to find out this relationship and choose a group of alphabets which is related in the same way to a third group provided in the question.

Example. B C D : P Q R : : X Y Z : ?

(*a*) FGH (*b*) LMN (*c*) RST (*d*) TUV (*e*) STU

Ans. (*b*)

Each letter of the first group is moved fourteen steps forward to obtain the corresponding letter of the second group. A similar relationship will exist between the third and fourth groups.

HOW TO ANSWER ANALOGY QUESTIONS

1. Figure out how the capitalized words are related.
2. Create a sentence that expresses that connection.
3. Test the choices with your sentence and eliminate the ones that don't work.
4. If we are left with more than one answer—or no answer at all—go back and refine your sentence.
5. Choose the best answer. If none of the choices fits exactly, choose the one that works best.

PLAN TO SOLVE ANALOGY QUESTIONS

1. Make your Sentence More Precise

The analogies get more difficult as we work our way through each group. Use the common categories as a starting point, but be prepared to refine the relationship by making your sentence more precise.

Example. Grain : Silo ::

(*a*) Pilot : plane (*b*) judge : courtroom

(*c*) water : reservoir (*d*) clock : time

If we apply the "place where" idea without thinking, here is what happens.

A silo is a place where we would find grain.

(*a*) A *plane* is a place where you would find a *pilot*.

(*b*) A *courtroom* is a place where you would find a *judge*.

(*c*) A *reservoir* is a place where you would find *water*.

(*d*) A *clock* is a place where you would find *time*.

We can eliminate (d), but that still leaves you with four possible answers. Now is express the relationship between silo and grain more precisely?

A silo is a place where grain is stored.

(*a*) A plane is a place where a pilot is stored.

(*b*) A courtroom is a place where a judge is stored.

(*c*) A reservoir is a place where water is stored.

(*d*) A highway is a place where automobiles are stored.

2. Analogies Work only if there's a clear cut connection

An analogy depends upon a necessary connection between pairs of words based on the meaning of the words. This clear connection must exist for both the original capitalized word pair and the correct answer choice, i.e. we can eliminate any answer choice for which we cannot describe a necessary relationship between the words. Use the "clear connection" test to rule out answer choices even when we don't know one of the capitalized words.

Example. Letter : Alphabet : :

(*a*) note : scale

(*b*) ocean : merchandise

(*c*) expert : automobile

(*d*) victory : farmland

We dont't know what the capitalized words are in this analogy, but you can still eliminate choices (*b*), (*c*) and (*d*). There is no clear-cut connection between the words of those choices. The actual analogy above is LETTER : ALPHABET :: note : scale.

3. Some Analogies Work Better by Working Backward

Sometimes the capitalized words fall easily into a sentence that expresses their relationship–and sometimes they don't. If we are having trouble making up a sentence that relates the two words, be prepared to shift gears. Try reversing the order of the original word pair.

Example. *Ice : Glacier ::*

(*a*) train : trestle (*b*) sand : dune

(*c*) path : forest (*d*) feather : bird

If we can't come up with a sentence relating ICE to GLACIER, try relating GLACIER to ICE :

A *glacier* is made of *ice*.

If we reverse the order of the capitalized words, we must also reverse the order of the words in each answer choice.

(*a*) A *trestle* is made up of a *train*.

(*b*) A *dune* is made up of *sand*.

(*c*) A *forest* is made up of a *path*.

(*d*) A *bird* is made up of a *feather*.

Clearly (*b*) exhibits the same relationship as the original pair.

4. Check from the Answer Choices

Many words have different meanings depending upon whether they are used as nouns or verbs or adjectives. If we are not sure how one of the capitalized words is being used, just check the answer choices. In analogies, all the answer choices will have the same grammatical structure as the capitalized words pair. That means if the answer choices are noun : noun, the capitalized pair will be NOUN : NOUN. On the other hand, if the answer choices are adjective : noun, then the capitalized pair will be ADJECTIVE : NOUN.

If we have eliminated all the answer choices with words that we know or sort of know, and we are left with a couple of choices containing words that we have never seen before, just guess and move on.

III. CLASSIFICATION

'Classification' means arrangement of given items on the basis of some common character. In this test, a group of certain items are given, out of which some are similar to in some manner and one is different from the rest. We are required to choose this one item which does not fit into the given group.

TYPES OF CLASSIFICATION QUESTIONS

1. Choosing Odd word

In this type of questions, four (or five) words are given out of which one is dissimilar to the others and students have to find that odd one from the given set.

Example. *Choose the word which is least like the other words in the group.*

(*a*) chair (*b*) cupboard (*c*) Table

(*d*) paper weight (*e*) Bed

Ans. (*d*)

Here, all except paper weight are furnitures.

2. Choosing Odd pair of words

In this type of questions, certain pairs of words are given out of which words in all the pairs except one, bear a certain common relationship. We are required to decipher this relationship and choose the pair in which words are differently related, as the answer.

Example. *Choose the odd pair of words.*

(*a*) Loom : Cloth (*b*) Table : Drawer

(*c*) Book : Page (*d*) Car : Wheel

Ans. (*a*)

In all other pairs, second is part of the first.

3. Choosing Odd numeral

In this type of questions, certain numbers are given, out of which all except one are alike in some manner while one is different and this number is to be chosen as the answer.

Example. *Choose the number which is different from others in the group.*

(*a*) 49 (*b*) 64

(*c*) 121 (*d*) 144

(*e*) 156

Ans. (*e*)

Each of the numbers except 156 is complete square.

4. Choosing the Odd numeral pair/group

In this type of questions, certain pairs/groups of numbers are given out of which all except one are similar in some manner while one is different. The numbers in these similar pairs may have the same property or may be related to each other according to the same rule. We are is required to choose the odd pair/group.

Example. *Choose the numeral pair/group which is different from others.*

(*a*) 83 – 75 (*b*) 58 – 50 (*c*) 49 – 42

(*d*) 25 – 17 (*e*) 131–123

Ans. (*c*)

In each of the pairs, first number is eight more than the second.

5. Choosing the Odd letter group

In this type of questions, usually four groups of letters are given. Three of them are similar to each other in some manner while one is different and we have to the answer.

Example. *Choose the group of letters which is different from others.*

(*a*) SU (*b*) PN (*c*) IK (*d*) BD

Ans. (*b*)

All other are two alternate letters

IV. CODING-DECODING

A CODE is a '*system of signals*'. Therefore, coding is a method of transmitting a message between the sender and the receiver without a third person knowing it.

Coding and Decoding Test is set up to judge the candidate's ability to decipher the rule that codes a particular word/ message and break the code to decipher the message.

TYPES OF CODING DECODING QUESTIONS

1. Letter Coding

In these questions, real alphabets in a word are replaced by certain other alphabets according to a specific rule to form its code and we are required to detect the common rule and answer the questions accordingly.

Example. If TAP is coded as SZO, then how is FREEZE coded?

(*a*) EQDFYG (*b*) ESDFYF (*c*) GQFDYF

(*d*) EQDDYD (*e*) EPDEZD

Ans. (*d*)

Each letter in the word TAP is moved one step backward to obtain the corresponding letter of the code.

	S	Z O
–1	↑	↑ ↑
	T	A P

Thus, in FREEZE, F will be coded as E, R as Q, E as D and Z as Y. So, the code becomes EQDDYD.

2. Number Coding

In these questions, either numerical code values are assigned to a word or alphabetical code letters are assigned to the numbers and we are is required to analyse the code as per the directions.

Example. *If CHINA is written as 38126 and NEPAL is 25769, how is PLAINE is coded ?*

Solution : Clearly Alphabets are coded as shown:

C	H	I	N	A	E	P	L
3	8	1	2	6	5	7	9

Hence, PLAINE is coded as 796125.

Example. In a certain code '37' means 'which class' and '583' means 'caste and class'. What is the code for 'caste' ?

(*a*) 3 (*b*) 7 (*c*) 8

(*d*) Either 5 or 8

Ans. (*d*)

Given, which class' ≡ '37' ...(*i*)

and'caste and class' ≡ '583' ...(*ii*)

From (*i*) and (*ii*), the common word 'class' is coded as '3'.

From (*ii*), 'caste' is coded as '5' or '8'.

3. Substitution

In this type of questions, some particular objects are assigned code names. Then a question is asked that is to be answered in the code language.

Example. If cook is called *butler*, butler is called manager, manager is called teacher, teacher is called clerk and clerk is called principal, who will teach in a class ?

(a) Cook (b) Butler (c) Manager
(d) Teacher (e) Clerk

Ans. (e)

A 'teacher' teaches in a class and as given 'teacher' is called 'clerk'. So, a 'clerk' will teach in the class.

4. Mixed letter Coding

In this type of questions, three or four complete messages are given in the coded language and the code for a particular word is asked. To analyse such codes, any two messages bearing a common word are picked up. The common code word will mean that word. Proceeding similarly by picking up all possible combinations of two, the entire message can be analysed.

Example. If 'tee see pee' means 'Drink fruit juice'; 'see kee lee' means 'Juice is sweet' and 'fee ree mee' means. "He is intelligent', which word in that language means 'sweet' ?

(a) see (b) kee (c) lee
(d) pee (e) None of these

Ans. (b)

In first and second statements, common word is 'Juice' and common code word is 'see'. So, 'see' means 'Juice'. In second and third statements, common word is 'is' and common code is lee'. So, 'lee' means 'is'. Thus, in second statement, remaining word 'sweet' is coded as 'kee'.

5. Mixed number Coding

In this type of questions, a few groups of numbers each coding a certain short message, are given. Through a comparison of the given coded messages, taking two at a time, the candidate is required to find the number code for each word and then formulate the code for the message given.

Example. In a certain code, '786' means 'study very hard', '958' means 'hard work pays' and '645' means 'study and work'. Which of the following is the code for 'very' ?

(a) 8 (b) 6 (c) 7 (d) 5

Ans. (c)

In first and second statements, common word is *'hard'* and common code digit is '8'. So, '8' means 'hard'.

In the first and third statements, common word is 'study' and the common code digit is '6'. So, '6' means 'study'.

Thus, in the first statement, '7' means 'very'.

6. Deciphering individual letter codes by analysis

In this type of questions, certain sample words are given along with their codes. The candidate is required to decipher individual codes for different letters by comparing, taking two words at a time, and then answer the given questions accordingly.

V. BLOOD RELATIONS

In these tests, the success of a candidate depends upon the knowledge of the blood relations.

Following Blood relations help solve these Questions :

Mother's or father's son	— Brother
Mother's or father's daughter	— Sister
Mother's or father's brother	— Uncle
Mother's or father's sister	— Aunt
Mother's or father's father	— Grand father
Mother's or father's mother	— Grand mother
Son's wife	— Daughter-in-Law
Daughter's husband	— Son-in-Law
Husband's or wife's sister	— Sister-in-Law
Husband's or wife's brother	— Brother-in-Law
Brother's son	— Nephew
Brother's daughter	— Niece
Uncle or aunt's son or daughter	— Cousin
Sister's husband	— Brother-in-Law
Brother's wife	— Sister-in-Law
Grandson's or Grand daughter's daughter	— Great grand daughter

Types of Blood Relation Questions

(i) Deciphering Jumbled up descriptions

In this type of questions, a round-about description is given in the form of certain small relationships and direct relationship between the persons concerned is to be deciphered.

Example. Pointing to a photograph, a man said, *"I have no brother or sister but that man's father is my father's son."* Whose photograph was it ?

(a) His own (b) His son's
(c) His father's (d) His nephew's

Ans. (a)

Since man has no brother, his father's son is he himself.

(ii) Relation Puzzle

In this type, mutual blood relations or other informations of more than two persons are mentioned and information about any two is mentioned.

Example. A and B are brothers. C and D are sisters. A's son is D's brother. How is B related to C.

(a) Father (b) Brother (c) Grandfather
(d) Uncle (e) None of these

Ans. (d)

B is the brother of A; A's son is D's brother. This means D is the daughter of A. Since C and D are sisters, C is also the daughter of A. So, B is the uncle of C.

(iii) Coded Relations

In such questions, relationships are represented by certain codes or symbols such as +, −, ×, ÷, *, Δ etc. Then relationships between certain persons, given in the form of these codes, are to be analysed.

Example. If A + B means A is the sister of B; A − B means A is the brother of B; A × B means A is the daughter of B, which of the following shows the relation that E is the maternal uncle of D?

(a) D + F × E (b) D − F × E
(c) D × F + E (d) D × F × E
(e) None of these.

Ans. (c)

E is the maternal uncle of D means D is the daughter of the sister (say F) of E i.e. D × F + E.

VI. PUZZLE TEST

This type of questions put in the form of puzzles involving certain number or items, be it persons or things and we are required to analyse the given information, condense it in a suitable form and answer the questions asked.

TYPE OF PUZZLE TESTS

1. Classification type Questions

This type consists of questions in which certain items belonging to different groups or possessing different qualities are given along with some clues with the help of which the candidate is required to group and analyse the given items and answer the questions accordingly.

Example. *Read the following information carefully and answer the questions that follow:*

(i) Five friends P, Q, R, S and T travelled to five different cities of Chennai, Calcutta, Delhi, Bangalore and Hyderabad by five different modes of transport of Bus, Train, Aeroplane, Car and Boat from Mumbai.

(ii) The person who travelled to Delhi did not travel by boat.

(iii) R went to Bangalore by car and Q went to Calcutta by aeroplane.

(iv) S travelled by boat whereas T travelled by train.

(v) Mumbai is not connected by bus to Delhi and Chennai.

1. Which of the following combinations of person and mode is not correct ?

(a) P — Bus (b) Q — Aeroplane
(c) R — Car (d) S — Boat
(e) T — Aeroplane

2. Which of the following combinations is true for S ?

(a) Delhi — Bus (b) Chennai — Bus
(c) Chennai — Boat (d) Data inadequate
(e) None of these

3. Which of the following combinations of place and mode is not correct ?

(a) Delhi — Bus (b) Calcutta — Aeroplane
(c) Bangalore — Car (d) Chennai — Boat
(e) Hyderabad — Bus

4. The person travelling to Delhi went by which of the following modes ?

(a) Bus (b) Train
(c) Aeroplane (d) Car
(e) Boat

5. Who among the following travelled to Delhi ?

(a) R (b) S
(c) T (d) Data inadequate
(e) None of these

Ans. *Given information can be analysed as follows :*

Mode of Transport.

R travels by Car, Q by Aeroplane, S by Boat and T by Train.

Now, only P remains. So, P travels by Bus.

Place of Travel.

R goes to Bangalore, Q to Calcutta.

Now, bus transport is not available for Delhi or Chennai. So, P who travels by Bus goes to Hyderabad.

S travels by boat and hence, S did not go to Delhi. So, S goes to Chennai.

Now, only T remains. So, T goes to Delhi.

Person	Place	Mode
P	Hyderabad	Bus
Q	Calcutta	Aeroplane
R	Bangalore	Car
S	Chennai	Boat
T	Delhi	Train

1. (e). Incorrect combination is T — Aeroplane
2. (c). Correct combination for S is Chennai — Boat
3. (a). Incorrect combination is Delhi — Bus
4. (b). T travelled to Delhi by Train
5. (c). T travelled to Delhi

2. Seating / Placing Arrangements

In this type of questions, some clues regarding seating or placing sequence (linear or circular) of some persons or items is given. The candidate is required to form the proper sequence using these clues and answer the questions accordingly.

Example. Four girls are sitting on a bench to be photographed. Shikha is to the left of Reena. Manju is to the right of Reena. Rita is between Reena and Manju. Who would be second from the left in the photograph ?

(a) Reena (b) Shikha (c) Manju
(d) Rita (e) Either Shikha or Reena

Ans. (a)

Shikha is to the left of Reena and Manju is to her right.
Rita is between Reena and Manju.
Hence order is : Shikha, Reena, Rita, Manju.
In the photograph, Reena will be second from left.

3. Comparison Type Questions

In this type of questions, clues are given regarding comparisons among a set of persons or things with respect to one or more qualities and we are required to analyse the whole information, form a proper ascending/descending sequence and then answer the given questions accordingly.

Example. *Read the information given below and answer the questions that follow :*

(*i*) There is a group of five girls.
(*ii*) Kamini is second in height but younger than Rupa.
(*iii*) Pooja is taller than Monika but younger in age.
(*iv*) Rupa and Monika are of the same age but Rupa is tallest between them,
(*v*) Neelam is taller than Pooja and elder to Rupa.

1. If they are arranged in the ascending order of height, who will be in third position ?
(*a*) Monika
(*b*) Rupa
(*c*) Monika or Rupa
(*d*) Data inadequate
(*e*) None of these

2. If they are arranged in the descending order of their ages, who will be in fourth position ?
(*a*) Monika or Rupa
(*b*) Kamini
(*c*) Monika
(*d*) Data inadequate
(*e*) None of these

3. To answer the question "who is the youngest person in the group", which of the given statements is superfluous ?
(*a*) Only (*i*) (*b*) Only (*ii*) (*c*) Only (*v*)
(*d*) Either (*i*) or (*iv*) (*e*) None of these

Ans. First find sequence of heights
By (*iii*), we have M < P
By (*v*), we have P < N.
Now, Rupa is tallest and Kamini is second in height, hence sequence of heights is :
M < P < N < K < R.
Now, find age sequence :
By (*ii*), we have K < R.
By (*ii*), we have P < M.
By (*iv*), we have R = M.
By (*v*), we have R < N.
Hence sequence of ages is:
P < K <R = M < N or K < P < R = M < N

1. (*e*). In increasing order of height, Neelam is in third position.
2. (*e*). In descending order of ages, Neelam will be in fourth position (because Monika and Rupa both lie at third position).
3. (*a*). Only statement (*i*) is not necessary.

4. Sequential Order of Things

In this type of questions, some clues are given regarding order of occurrence of certain eventsand we are required to analyse the given information, frame the right sequence and then answer the questions accordingly.

Example. *Read the following information carefully and answer the questions given below it :*

(*i*) Eight doctors P, Q, R, S, T, U, V and W visit a charitable dispensary every day except on a holiday i.e. Monday.
(*ii*) Each doctor visits for one hour from Tuesday to Sunday except Saturday. The timings are 9 a.m. to 1 p.m. and 2 p.m. to 6 p.m.; 1 p.m. to 2 p.m. is lunch break.
(*iii*) On Saturday, it is open only in the morning i.e. 9 a.m. to 1 p.m. and each doctor visits for only half an hour.
(*iv*) No other doctor visits the dispensary before doctor Q and after doctor U.
(*v*) Doctor-W comes immediately after lunch break and is followed by R.
(*vi*) S comes in the same order as P in the afternoon session.

1. Doctor P visits in between which of the following pairs of doctors ?
(*a*) S and V (*b*) U and W (*c*) R and W
(*d*) R and U (*e*) None of these

2. At what time the visit of doctor R is over on Sunday?
(*a*) 1 p.m. (*b*) 3 p.m. (*c*) 4p.m.
(*d*) 5p.m. (*e*) None of these

3. At what time the visit of Doctor T would be over on Saturday?
(*a*) 10 a.m.
(*b*) 11 a.m.
(*c*) Either 10 a.m. or 11 a.m.
(*d*) Data inadequate
(*e*) None of these

4. If the lunch break and subsequent visiting hours are reduced by 15 minutes, at what time Doctor U is expected to attend the dispensary ?
(*a*) 3.15 p.m. (*b*) 4 p.m. (*c*) 4.15 p.m.
(*d*) 4.45 p.m. (*e*) None of these

Ans. First form sequence of visit using (*iv*), (*v*) and (*vi*).
From (*iv*),Q visits first and U visits last.
From (*v*),W visits first after break and is followed by R.
From (*vi*), P visits after break.
Hence sequence of visit after break becomes W R P U
Also, S has the same position in morning session as P in afternoon session. Thus, sequence of visit before break is Q, T/V, S, V/T.

1. (*d*). P visits between R and U.
2. (*c*). Time of visit of W is 2 p.m. to 3 p.m., Time of visit of R is 3 p.m. to 4 p.m. So, the visit of doctor R is over at 4 p.m.
3. (*c*). T visits either second or fourth. So, the time of visit on Saturday will be either 9.30 a.m. or 10.30 a.m. Thus, T's visit will be over at either 10 a.m. or 11 a.m.
4. (*b*). As mentioned, lunch break will be over and doctor W will visit at 1.45 p.m., doctor R will visit at 2.30 p.m., doctor P will visit at 3.15 p.m. and U will visit at 4 p.m.

5. Selection based on given conditions

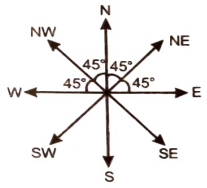
In this type of questions, a few essential criteria for selection of a group of items are given and we have to keep these conditions in mind and make the required selection as per the directions given in each question.

6. Jumbled problems

In this type of questions, some mixed clues regarding three or more qualities of given items or persons is given and we are required to analyse this mixed information with respect to different qualities and classify the items accordingly.

VII. DIRECTION SENSE

In this type of test, the questions consist of a sort of direction puzzle. A successive follow-up of directions is formulated and we are required to ascertain final direction or the distance between two points. The test is meant to judge ability to trace and follow correctly and sense the direction correctly.



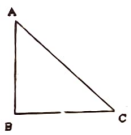
Above figure shows four main directions : North N, South S, East E, West W
Four cardinal directions : North East NE, North West NW, South East SE, South West SW

Pythagoras theorem

The theorem is applicable only in right angle triangles, in which from theorem one angle is of 90°

$$AC^2 = AB^2 + BC^2$$

where, AC is Hypotense
AB is Perpendicular (Height)
BC is Base



Shortcut method

1. If 2 opposite turns are coming together, they will cancel each other
Example. left, right and right, left.
2. If similar turns are coming together they will opposite the earlier direction.
Example. if a person is going in North and taking two left turns it means now person is moving in opposite to North i.e. South.
3. Start from either left side or right side

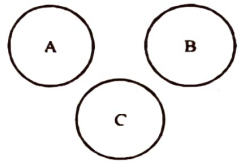
VIII. LOGICAL VENN DIAGRAMS

This type of questions aim at analysing ability to relate a certain given group of items and illustrate it diagrammatically.

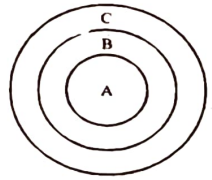
1. LOGICAL DIAGRAMS.

In this type a few types of Venn diagrams with their implications made clear. Let a group of three items are given.

1. If items evidently belong to three different groups, then Venn diagram representing it would be asC shown alongside.
Example. Doctors, Engineers, Lawyers
These three items bear no relationship to each other. So, they are represented by 3 disjoint figures as shown:



2. If one item belongs to the class of the second and second belongs to the class of third, then representation is in the form of three concentric circles, as shown below:



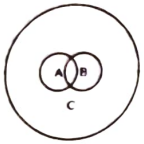
Example. Seconds, Minutes, Hours
Seconds are a part of minutes and minutes are a part of hours. So, Venn diagram would be as shown in the figure with circle A representing Seconds, circle B representing Minutes and circle C representing Hours.

3. If two separate items belong to the class of the third, they are represented by two disjoint circles inside a bigger circle as shown below

Example. Table, Chair, Furniture

Table and chair are separate items but both are items of furniture. So, they would be represented as in the figure with circle A representing Table, circle B representing Chair and circle C representing Furniture.

4. If two separate things having something common belongs to the class of third, they are represented by two joint figures inside one bigger figure is shown below.

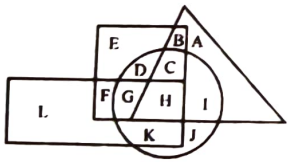


In this type of questions, generally a Venn diagram is given.

2. VENN DIAGRAMS

Each geometrical figure in the diagram represents a certain class. The candidate is required to study and analyse the figure carefully and then answer certain questions regarding the given data.

Example. In the following diagram, square represents girls, the circle represents tall persons, the triangle is for tennis players and the rectangle stands for the swimmers.



On the basis of the above diagram, answer the following questions :

1. Which letter represents tall girls who are swimmers but don't play tennis ?

(*a*) C (*b*) D (*c*) G (*d*) H (*e*) B

Ans. (*c*)

Tall girls, who are swimmers are represented by the region common to the square, circle and the rectangle i.e., G and H. But, according to the given conditions, the girls shouldn't be tennis players. So, required region should not be a part of the triangle i.e., H should be excluded. Thus, region representing the persons satisfying given conditions is G.

2. Which letter represents girls who are swimmers, play tennis but are not tall?

(*a*) B (*b*) E (*c*) H (*d*) F

(*e*) None of these

Ans. (*e*)

Girls who are swimmers and play tennis are represented by the region common to the square, triangle and rectangle i.e., H. But, it is given that the girls shouldn't be tall. So, required region should not be a part of the circle. Since H is a part of the circle. So none of these is correct

IX. ALPHABETICAL QUIBBLE

In this type of questions, we are given alphabets from A to Z. The position of a letter is given in the form of a puzzle and we are required to find this letter. However, sometimes a random letter series is given and we are required to find out how many times a letter satisfying the conditions specified in the question occurs.

Example. *Following question is based on the following alphabet series.*

ABCDEFGHIJKLMNOPQRSTUVWXYZ

Which letter is exactly midway between H and S in the given alphabet ?

(*a*) L (*b*) M (*c*) N (*d*) O

(*e*) No such letter

Ans. (*e*)

From given series we get that M & N are in midway between H and S. So, exactly midway is no such letter.

X. NUMBER, RANKING & TIME SEQUENCE

1. Number Test

In this type of questions, generally you are given a long series of numbers and we are required to find out how many times a number satisfying the conditions, specified in the question, occurs.

Example. How many 5's are there in the following sequence which are immediately followed by 3 but not immediately preceded by 7 ?

8953253855687335775365335738

(*a*) One (*b*) Two (*c*) Three

(*d*) Four (*e*) More than four

Ans. (*c*)

A number which comes after a given number is said to follow it while the one which comes before the given number precedes it. Thus, numbers satisfying the given conditions, can be shown as follows. Hence there are three such numbers.

2. Ranking Test

In this type of test, generally ranks of a person both from top and bottom are mentioned and total number of persons is asked. However, sometimes this question is put in the form of a puzzle of interchanging seats by two persons.

Example. Anil ranked ninth from the top and thirty eighth from the bottom in a class. How many students are there in the class?

(*a*) 45 (*b*) 46 (*c*) 47

(*d*) 48 (*e*) 49

Ans. (*b*)

Whole class consists of :

(*i*) 8 students who have a rank higher than Anil

(*ii*) Anil; and

(*iii*) 37 students who have rank lower than Anil i.e.,(8 + 1 + 37) = 46 students.

3. Time Sequence Test

Example. Raman remembers that his brother's birthday is after fifteenth but before eighteenth of February whereas his sister Kamla remembers that her brother's birthday is after sixteenth but before nineteenth of February. On which day in February is Raman's brother's birthday ?

(*a*) 16th (*b*) 17th (*c*) 18th

(*d*) 19th (*e*) None of these

Ans. (*b*)

According to Raman, brother's birthday is on one of the days among 16th and 17th February

According to Kamla, the brother's birthday is on one of the days among 17th and 18th February. Hence Raman's brother's birthday is on the day common to both the above groups i.e., 17th February.

XI. MATHEMATICAL OPERATIONS

This type questions are based on simple mathematical operations.

Here, four fundamental operations :

Addition, subtraction, multiplication and division

and statements such as

'less than', 'greater than', 'equal to', 'not equal to', etc.

are represented by symbols, different from the usual ones.

The questions involving these operations are set using artificial symbols. The candidate has to substitute the real signs and solve the questions accordingly, to get the answer.

TYPE OF PROBLEMS.

1. Solving by Substitution

In this type of questions, we are provided with substitutes for various mathematical symbols, followed by a question involving calculation of an expression or choosing the correct/ incorrect equation. We are required to put in the real signs in the given equation and then solve the questions as required.

While solving a mathematical expression, proceed according to the rule BODMAS

i.e., Brackets, Of, Division, Multiplication, Addition, Subtraction.

2. Deriving appropriate Conclusions

In such questions, the usual algebraic signs and mathematical symbols may be given different meanings or may even be replaced by new symbols. For instance (÷) may denote addition, (×) may denote subtraction (<) may denote division and so on. The simplest way to solve such an equation is to rewirite the problem replacing the given symbols with the signs they are supposed to stand for. For instance, if :

+ means –	÷ means +
'V' means <	– means >
× means =	> means ×
&, < means ÷	

Then equation, 65 < 5 ÷ 2.5 — 15 will be rewritten as 65 ÷ 5 + 2.5 > 15, and problem may be solved.

In yet another type of questions included under this system, even the digits (instead of the symbols)

0, 1, 2, ... 8, 9, may be replaced by alphabets and the candidate may then be asked to perform calculations and solve the given problem. For instance, codes may be used as (a = 1, b = 2, c = 3 etc.) or in reverse (i.e., z = 1, y = 2, x = 3 and so on).

Following question employs such a code.

Q. If y × w = s and u — v = z, what is r/x ?

(*a*) y (*b*) z (*c*) s (*d*) x

Solution : If we count backwards from z, we get y as the 2nd letter, w as the 4th letter, and s as the 8th letter. In this order u is the 6th, v is the 5th and z the 1st letter of the code.

So, $2 \times 4 = 8$ and $6 - 5 = 1$

Hence, r = 9 and x = 3.

So $\frac{r}{x} = \frac{9}{3} = 3$. So the answer is (*d*) x.

In certain cases, arithmetical operations may not be defined. Consider the following question.

Q. If 8 + 12 = 2, and 7 + 13 = 3, then what is the value of 10 + 18 ?

(*a*) 9 (*b*) 6 (*c*) 4 (*d*) 7

Solution : $\frac{12-8}{2} = 2$, and $\frac{13-7}{2} = 3$. So, code employed here is special.

In x + y, '+' stands for $\frac{y-x}{2}$

$\therefore \quad 10 + 18 = \frac{18-10}{2} = 4$, so the answer is (c).

XII. LOGICAL SEQUENCE OF WORDS

In this type of questions, a group of words is given. The candidate is required to arrange these words in a meaningful order such as the sequence of occurrence of events, sequence from a part to the whole, sequence of increasing/decreasing size, value, intensity etc., and then choose the correct sequence accordingly.

Examples.

1. *Arrange the following in a meaningful sequence :*

1. Consultation
2. Illness
3. Doctor
4. Treatment
5. Recovery

(*a*) 2, 3, 1, 4, 5 (*b*) 2, 3, 4, 1, 5
(*c*) 4, 3, 1, 2, 5 (*d*) 5, 1, 4, 3, 2
(*e*) 2, 3, 4, 5, 1

Ans (*a*)

Illness occurs first. One then goes to the doctor and after consultation with him, undergoes treatment to finally attain recovery. Thus, the correct order is 2, 3, 1, 4, 5.

2. *Arrange the following in a logical order :*

1. Euphoria
2. Happiness
3. Ambivalence
4. Ecstasy
5. Pleasure

(*a*) 4, 1, 3, 2, 5 (*b*) 3, 2, 5, 1, 4
(*c*) 2, 1, 3, 4, 5 (*d*) 1, 4, 2, 5, 3
(*e*) 5, 4, 3, 2, 1

Ans. (*b*)

All the given words stand for 'Joy', but the intensity increases in the order — Ambivalence, Happiness, Pleasure, Euphoria, Ecstasy. Thus, the correct order is 3, 2, 5, 1, 4.

XIII. DECISION MAKING

It with questions in which we have to decide the course of action to be taken upon a candidate who has applied for a vacancy or allotment or membership to an institution, keeping in mind the essential requisites and the data given for the candidate.

TYPE I

In such type of test, a vacancy is declared. The necessary qualifications required in the candidates coming up to fill the vacancy are provided and merits of the candidates mentioned. The decision about each candidate has to be made from amongst the five choices named (*a*), (*b*), (*c*), (*d*), (*e*) which state the courses of action to be taken as per the candidate's potentials.

Example. *Read the following information to answer the given questions*

Following are the criteria for selecting & marketing officer by a company. The candidate must :

(1) be a graduate with at least 50% marks.

(2) have secured at least 40% marks in the written test.

(3) not be less than 24 years and more than 29 years as on 10th October, 2007.

(4) should have work experience of at least two years as an officer. However, if a candidate:

(5) fulfils all other criteria except at (4) above but has a diploma in Marketing Management, his/her case is to be referred to General Manager, Marketing.

(6) fulfils all other criteria except at (3) above but has worked as Marketing Officer at least for three years, his/her case is to be referred to Director, Marketing.

Based on the above criteria and information given in each of the following questions, we have to take the decision in regard to each case. We are not to assume anything. These cases are given as on 10th October, 2007.

Mark answer (*a*) if candidate is to be appointed;

Mark answer (*b*) if the candidate is not to be appointed;

Mark answer (*c*) if the data given are not sufficient to take decision;

Mark answer (*d*) if to be referred to General Manager - Marketing; and

Mark answer (*e*) if to be referred to Director – Marketing.

1. Amit, born on 5th June, 1983, has done his post-graduation in Marketing Management with first class. He has secured 50% marks in the written test. He has been working in an organisation as a Marketing Officer for the last four years.

2. Rohit has been working in an organisation as Officer for the last ten years. His date of birth is 17th February, 1964. He has secured 60% marks in the degree examination and 40% marks in the written test.

3. Manju is a first class graduate and has done a diploma in Marketing Management. She has secured 50% marks in the written test. She was 23 years old as on 5th September, 2006.

4. Nitin was born on 25th August, 1985. He has secured 60% and 50% marks in graduation and in the written test, respectively. He has been working in an organisation as Officer for the last four years.

5 Suman is a graduate with first class and has secured 60% marks in the written test. She has been working as an Officer for the last three years. She was born on 20th May, 1982.

Ans.

1. (*a*). All conditions of eligibility are satisfied.

2. (*e*). Condition (3) is violated but it can be waived by condition (6).

3. (*d*). The candidate satisfies all conditions except (4). But he fulfils condition (5) so that (4) is waived.

4. (*e*). The candidate satisfies condition (2) instead of condition (3).

5. (*a*). All conditions of eligibility are satisfied.

TYPE II. ASSERTION AND REASON

This test is meant to judge the candidate's knowledge and his ability to reason out correctly. In this test, two statements referred to as the Assertion (A) and Reason (R) respectively are provided. Five alternative comments on these are given and the correct one is to be chosen.

Example. *For the Assertions (A) and Reasons (R) below, choose the correct alternative from the following:*

(*a*) Both A and R are true and R is the correct explanation of A.

(*b*) Both A and R are true but R is NOT the correct explanation of A.

(*c*) A is true but R is false.

(*d*) A is false but R is true.

(*e*) Both A and R are false.

1. **Assertion (A) :** Moon cannot be used as a satellite for communication.

Reason (R) : Moon does not move in the equatorial plane of the earth.

Ans. (*a*)

Since R explains A.

2. **Assertion (A) :** Salt is added to cook food at higher altitudes.

Reason (R): Temperature is lower at higher altitudes.

Ans. (*b*)

Since both statements A and R are separately true but R does not explain A.

TYPE III. SITUATION REACTION TEST

In this test, certain situation is described and the candidate is required to choose the most suitable reaction to the given situation from amongst the alternatives provided. This test judges reasoning power of the candidate and his ability to act correctly and promptly to a situation that may arise in emergency.

Example. While travelling in your car, certain persons stop you on the way asking you to take an injured child to the hospital. You would :

(*a*) ask them to leave your way and then drive away.

(*b*) ask them to first call the police.

(*c*) immediately take the child to hospital.

(*d*) get out of the car and ask some other person to help them.

(*e*) None of these

Ans. (*c*)

Situation described above demands that the person should immediately render the help asked for and take the child to the hospital.

Examples

1. While playing cricket in the school, suddenly when you hit the ball, it strikes your classmate on the forehead and blood starts oozing out. You would :

(*a*) run away from the field.

(*b*) start fighting with the boy why he came in the way.

(*c*) blame somebody else for the accident.

(*d*) take the boy to the first aid room.

(*e*) None of these

Ans. (*d*)

In the above situation, urgent need is to provide first aid to the boy so that the bleeding may stop.

2. You are visiting a place for the first time and are travelling in a bus. Suddenly you realise that the driver is taking the bus to a lonely place with no right intentions. You would :

(*a*) with the help of some other passengers, try to baffle the driver and take over the bus.

(*b*) sit and wait to face the repercussions.

(*c*) jump out of the running bus.

(*d*) console the worried passengers.

Ans. (*a*)

When a wrong doing is expected, immediate action to prevent it is the need.

3. You have gone to enjoy a Diwali Mela organised by a club. Suddenly you come across a lost child crying desperately.

You would :

(*a*) neglect and walk away.

(*b*) ask the child to find his parents.

(*c*) ask him to stop crying and wait patiently for his parents.

(*d*) contact with the club authorities and make an announcement for the parents.

Ans. (*d*)

Immediate need is to find the child's parents and for this, best way is to announce the child's name and appearance so that his parents might know where the child is.

EXERCISE – I

MCQ TYPE QUESTIONS

1. Given two numbers x and y, define

Addition	A(x, y) = x + y
Subtraction	S (x, y) = x – y
Multiplication	M (x, y) = x, y
Division	D (x, y) = x/y

S[M(D(A(a, b), 2), D(A(a, b), 2)), M(D(S(a, b), 2), D(S(a, b),2))] is equal to

(a) $(a - b)^2$ (b) ab

(c) $(a + b)^2$ (d) None of these

Directions Q. 2 – 5 :

Kamal Babu came home just after judging a beauty contest where there were four semi-finalists, Ms. Andhra Pradesh, Ms. Uttar Pradesh, Ms. Maharashtra and Ms. West Bengal. His wife was very keen on knowing who the winner was and Kamal Babu replied immediately that it was the one wearing the yellow saree. When his wife asked for more details, he gave the following information:

* The four girls were wearing sarees of different colours (yellow, red, green, white) and the runner-up was wearing green.
* The four girls were sitting in a row, and Ms. West Bengal was not sitting at either end.
* There was only one runner-up and she was sitting next to Ms. Maharashtra.
* The girls wearing yellow and white sarees occupied the seats at either end.
* Ms. West Bengal was nither the winner nor the runner-up.
* Ms. Maharashtra was wearing white.
* The winner and the runner-up were not sitting next to each other.
* The girl wearing the green saree was not Ms. Andhra Pradesh.

Answer the following questions based on the above information:

2. What was the colour of the saree that Ms. Andhra Pradesh was wearing?

(a) White (b) Yellow

(c) Red (d) Cannot be determined

3. Between which two was Ms.West Bengal sitting?

(a) Ms. Andhra Pradesh and Ms. Maharashtra

(b) Ms. Andhra Pradesh and Ms. Uttar Pradesh

(c) Ms. Uttar Pradesh and Maharashtra

(d) Cannot be determmined

4. Who was wearing the red saree?

(a) Ms. Andhra Pradesh

(b) Ms. Uttar Pardesh

(c) Ms. West Bengal

(d) Cannot be determined

5. What was the colour of the saree that Ms. Uttar Pradesh was wearing?

(a) White (b) Green

(c) Red (d) Yellow

Directions Q. 6 – 7 :

A, B, C, D, E and F are a group of friends from a club. There are two housewives, one lecturer, one architect, one accountant and one lawyer in the group. There are two married couples in the group. The lawyer is married to D who is a housewife. No lady in the group is either an architect or an accountant C. The accountant, is married to F who is a lecturer. A is married to D and E is not a housewife.

6. What is E ?

(a) Lawyer (b) Architect

(c) Lecturer (d) Accountant

7. How many members of the group are males?

(a) 2 (b) 3

(c) 4 (d) none of these

Directions Q. 8 – 9 :

Seven university cricket players are to be honoured at a special luncheon. the players will be seated on the dais along one side of a single rectangular table.

A and G have to leave for the lunch early and must be seated at the extreme right end of the table which is closest to the exit.

B will receive the Man of the Match award and must be in the center chair.

C and D, who are bitter rivals for the position of wicket keeper, dislike each other and should be seated as far apart as possible.

E and F are best friends and want to sit together.

8. Which of the following may not be seated at either end of the table ?

(a) C (b) D

(c) G (d) F

9. Which of the following pairs may not be seated together ?

(a) E & A (b) B & D

(c) C & F (d) G & D

Directions Q. 10 –13 :

A study was conducted to ascertain the relative importance that employees in five different countries assigned to five different traits in their Chief Executive Officers. The traits were compassion (C), decisiveness (D), negotiation skills (N), public visibility (P), and vision (V). The level of dissimilarity between two countries is the maximum difference in the ranks allotted by the two countries to any of the five traits. The following table indicates the rank order of the five traits for each country.

RANK	Country				
	INDIA	CHINA	JAPAN	MALAYSIA	THAILAND
1	C	N	D	V	V
2	P	C	N	D	C
3	N	P	C	P	N
4	V	D	V	C	P
5	D	V	P	N	D

10. Which of the following pairs of countries are most dissimilar?

(a) China and Japan

(b) India and China

(c) Malaysia and Japan

(d) Thailand and Japan

11. Which of the following countries is least dissimilar to India?

(a) China (b) Japan

(c) Malaysia (d) Thailand

12. Which amongst the following countries is most dissimilar to India?

(a) China (b) Japan

(c) Malaysia (d) Thailand

13. Three of the following four pairs of countries have identical levels of dissimilarity. Which pair is the odd one out?

(a) Malaysia and China

(b) China and Thailand

(c) Thailand and Japan

(d) Japan and Malaysia

Directions Q. 14 – 17 :

Four families decided to attend the marriage ceremony of one of their colleagues. One family has no kids, while others have at least one kid each. Each family with kids has at least one kid attending the marriage.

Given below is some information about the families, and who reached when to attend the marriage.

The family with 2 kids came just before the family with no kids.

Shanthi who does not have any kids reached just before Sridevi's family.

Sunil and his wife reached last with their only kid.

Anil is not the husband of Joya.

Anil and Raj are fathers.

Sridevi's and Anita's daughters go to the same school.

Joya came before Shanthi and met Anita when she reached the venue.

Raman stays the farthest from the venue.

Raj said his son could not come because of his exams.

14. Which woman arrived third ?

(a) Shanthi (b) Sridevi

(c) Anita (d) Joya

15. Name the correct pair of husband and wife ?

(a) Raj and Shanthi (b) Sunil and Sridevi

(c) Anil and Sridevi (d) Raj and Anita

16. Of the following pairs, whose daughters go to the same school ?

(a) Anil and Raman (b) Sunil and Raman

(c) Sunil and Anil (d) Raj and Anil

17. Whose family is known to have more than one kid for certain ?

(a) Raman's (b) Raj's (c) Anil's (d) Sunil's

Directions Q. 18 – 20 :

Answer the following questions based on the statements given below:

(i) There are three houses on each side of the road.

(ii) These six houses are labeled as P, Q, R, S, T and U.

(iii) The houses are of different colours, namely, Red, Blue, Green, Orange, Yellow and White.

(iv) The houses are of different heights.

(v) T, the tallest house, is exactly opposite to the Red coloured house.

(vi) The shortest house is exactly opposite to the Green coloured house.

(vii) U, the Orange coloured house, is located between P and S.

(viii) R, the Yellow coloured house, is exactly opposite to P.

(ix) Q, the Green coloured house, is exactly opposite to U.

(x) P, the White coloured house, is taller than R, but shorter than S and Q.

18. What is the colour of the house diagonally opposite to the Yellow coloured house?

(a) White (b) Blue

(c) Green (d) Red

19. Which is the second tallest house?

(a) P (b) S

(c) Q (d) cannot be determined

20. What is the colour of the tallest house?

(a) Red (b) Blue (c) Green (d) Yellow

NUMERICAL TYPE QUESTIONS

1. Given two numbers x and y, define

Addition	A(x, y) = x + y
Subtraction	S (x, y) = x – y
Multiplication	M (x, y) = x, y
Division	D (x, y) = x/y

If x = 2, y = 3 and z = 5, then value of M[A(S (x, y), M (y, z)), S (A(x, z), S (z, y))] is _____

Directions Q. 2 – 3 :

These questions are based on the following data

A, B, C, D are standing in a queue. It is known that C and D are not standing adjacent to each other and that B is not in the third place.

2. If A is not in the second place and B is in the first place, then which is the place for D is _____

3. If A is not standing adjacent to B and D is the last in the queue, then the place for C is _____

Directions Q. 4 – 5 :

A boy is asked to put in a basket one mango when ordered 'One', one orange when ordered 'Two', one apple when ordered 'Three' and is asked to take out from the basket one mango and an orange when ordered 'Four'. A sequence of orders is given as
1 2 3 3 2 1 4 2 3 1 4 2 2 3 3 1 4 1 1 3 2 3 4

4. The _____total organges were in the basket at the end of the above sequence.
5. The _____total fruits will be in the basket at the end of the above order sequence.

Directions Q. 6 – 7

Each of the 11 letters A, H, I, M, O, T, U, V, W, X and Z appears same when looked at in a mirror. They are called symmetric letters. Other letters in the alphabet are asymmetric letters.

6. The_____ number of four-letter computer passwords can be formed using the symmetric letters (no repetition allowed)
7. The _____ number of three-letter computer passwords can be formed (no repetition allowed) with at least one symmetric letter.

Directions Q. 8 – 9 :

A string of three English letters is formed as per the following rules:

(a) The first letter is any vowel
(b) The second letter is m, n or p.
(c) If the second letter is m then the third letter is any vowel which is different from the first letter
(d) If the second letter is n then the third letter is e or u
(e) If the second letter is p then the third letter is the same as the first letter.

8. The _____ strings of letters can possible be formed using the above rules.
9. The _____ strings of letters can possibly be formed using the above rules such that the third letter of the string is e.

EXERCISE – II

(QUESTIONS FROM PREVIOUS GATE EXAMS)

MCQ TYPE QUESTIONS

2015

1. Mr. Vivek walks 6 meters North-East, then turns and walks 6 meters South- East, both at 60 degrees to East. He further moves 2 meters South and 4 meters West. What is the straight distance in meters between the point he started from and the point he finally reached?

(a) $2\sqrt{2}$ (b) 2
(c) $\sqrt{2}$ (d) $1/\sqrt{2}$

2. There are 16 teachers who can teach Thermodynamics (TD), 11 who can teach Electrical Sciences (ES), and 5 who can teach both TD and Engineering Mechanics (EM). There are a total of 40 teachers, 6 cannot teach any of the three subjects, i.e. EM, ES or TD. 6 can teach only ES. 4 can teach all three subjects, i.e. EM, ES and TD. 4 can teach ES and TD. How many can teach both ES and EM but not TD?

(a) 1 (b) 2
(c) 3 (d) 4

3. The given question is followed by two statements: select the most appropriate option that solves the question

Capacity of a solution tank A is 70% of the capacity of tank B. How many gallons of solution are in tank A and tank B?

Statements:

I. Tank A is 80% full and tank B is 40% full
II. Tank A if full contains 14,000 gallons of solution

(a) Statement I alone is sufficient
(b) Statement II alone is sufficient
(c) Either statement I or II alone is sufficient
(d) Both the statements I and II together are sufficient

4. Operators $\square$, $\diamond$ and $\rightarrow$ are defined by

$$a \square b = \frac{a-b}{a+b};\ a \diamond b = \frac{a+b}{a-b};\ a \rightarrow b = ab.$$

Find the value $(66 \square 6) \rightarrow (66 \diamond 6)$

(a) -2 (b) -1
(c) 1 (d) 2

5. Humpty Dumpty sits on a wall every day while having lunch. The wall sometimes breaks. A person sitting on the wall falls if the wall breaks.

Which one of the statements below is logically valid and can be inferred from the above sentences?

(a) Humpty Dumpty always falls while having lunch
(b) Humpty Dumpty does not fall sometimes while having lunch

(c) Humpty Dumpty never falls during dinner
(d) When Humpty Dumpty does not sit on the wall, the wall does not break

6. If ROAD is written as URDG, then SWAN should be written as :

(a) VXDQ (b) VZDQ
(c) VZDP (d) UXDQ

7. Select the pair that best expresses a relationship similar to that expressed in the pair :

Children : Pediatrician

(a) Adult : Orthopaedist
(b) Females : Gynaecologist
(c) Kidney : Nephrologist
(d) Skin : Dermatologist

8. The head of a newly formed government desires to appoint five of the six selected members P,Q,R,S,T and U to portfolios of Home, Power, Defense, Telecom, and Finance. U does not want any portfolio if S gets one of the five. R wants either Home or Finance or no portfolio. Q says that if S gets either Power or Telecom, then she must get the other one. T insists on a portfolio if P gets one.

Which is the valid distribution of portfolios?

(a) P-Home, Q-Power, R-Defense, S-Telecom, T-Finance
(b) R-Home, S-Power, P-Defense, Q-Telecom, T-Finance
(c) P-Home, Q-Power, T-Defense, S-Telecom, U-Finance
(d) Q-Home, U-Power, T-Defense, R-Telecom, P-Finance

9. Most experts feel that in spite of possessing all the technical skills required to be a batsman of the highest order, he is unlikely to be so due to lack of requisite temperament. He was guilty of throwing away his wicket several times after working hard to lay a strong foundation. His critics pointed out that until he addressed this problem success at the highest level will continue to elude him.

Which of the statement (s) below is/are logically valid and can be inferred from the above passage?

(i) He was already a successful batsman at the highest level
(ii) He has to improve his temperament in order to become a great batsman
(iii) He failed to make many of his good starts count
(iv) Improving his technical skills will guarantee success

(a) (iii) and (iv)
(b) (ii) and (iii)
(c) (i), (ii) and (iii)
(d) (ii) only

10. Alexander turned his attention towards India, since he had conquered Persia.

Which one of the statements below is logically valid and can inferred from the above sentence?

(a) Alexander would not have turned his attention towards India had he not conquered Persia.
(b) Alexander was not ready to rest on his laurels, and wanted to march to India
(c) Alexander was completely in control of his army and could command it to move towards India
(d) Since Alexander's kingdom extended to Indian borders after the conquest of Persia, he was keen to move further

11. Tanya is older than Enc.
Cliff is older than Tanya.
Eric is older than Cliff.
If the first two statements are true, then the third statement is

(a) True (b) False
(c) Uncertain (d) Data insufficient

12. Given below are two statements followed by two conclusions. Assuming these statements to be true, decide which one logically follows:

Statements:

I. No manager is a leader.
II. All leaders are executive.

Conclusions:

I. No manager is a executive.
II. All executive is a manager.

(a) Only conclusion I follows.
(b) Only conclusion II follows.
(c) Neither conclusion I nor II follows.
(d) Both conclusion I and II follow.

13. Find the missing sequence in the letter series below :

A, CD, GHI, _?___, UVWXY

(a) LMN (b) MNO
(c) MNOP (d) NOPQ

14. Ms. X will be in Bagdogra from 1/5/14 to 20/5/14 and from 22/5/14 to 31/5/14. On the morning of 21/5/14, she will reach Kochi via Mumbai.

Which one of the statements below is logically valid and can be inferred from the above sentences?

(a) Ms. X will be in Kochi for only one day only in May
(b) Ms. X will be in Kochi for only one day in May
(c) Ms. X will be only in Kochi for one day in May
(d) Only Ms. X will be in Kochi for one day in May

15. Given below are two statements followed by two conclusions. Assuming these statements to be true, decide which one logically follows.

Statement:

I. All film stars are playback singers.

II. All film directors are film stars.

Conclusions:

I. All film directors are playback singers.

II. Some film stars are film directors.

(a) Only conclusion I follows
(b) Only conclusion II follows
(c) Neither conclusion I nor II follows
(d) Both conclusions I and II follow

16. Lamenting the gradual sidelining of the arts in school curricula, a group of prominent artists wrote to the Chief Minister last year, asking him to allocate more funds to support arts education in schools. However, no such increase has been announced in this year's Budget. The artists expressed their deep anguish at their request not being approved, but many of them remain optimistic about funding in the future.

Which of the statement(s) below is/are logically valid and can be inferred from the above statements?

i. The artists expected funding for the arts to increase this year.

ii. The Chief Minister was receptive to the idea of increasing funding for the arts.

iii. The Chief Minister is a prominent artist.

iv. Schools are giving less importance to arts education nowadays.

(a) *iii* and *iv*
(b) *i* and *iv*
(c) *i*, *ii* and *iv*
(d) *i* and *iii*

17. Four branches of a company are located at M, N, O, and P. M is north of N at a distance of 4 km; P is south of O at a distance of 2 km; N is southeast of O by 1 km. What is the distance between M and P in km?

(a) 5.34 (b) 6.74
(c) 28.5 (d) 45.49

18. The given statement is followed by some courses of action. Assuming the statement to be true, decide the correct option.

Statement:

There has been a significant drop in the water level in the lakes supplying water to the city.

Course of action:

I. The water supply authority should impose a partial cut in supply to tackle the situation.

II. The government should appeal to all the residents through mass media for minimal use of water.

III. The government should ban the water supply in lower areas.

(a) Statements I and II follow.
(b) Statements I and III follow.
(c) Statements II and III follow.
(d) All statements follow.

2014

19. In a group of four children, Som is younger to Riaz. Shiv is elder to Ansu. Ansu is youngest in the group. Which of the following statements is/are required to find the eldest child in the group?

Statements

1. Shiv is younger to Riaz.
2. Shiv is elder to Som.

(a) Statement 1 by itself determines the eldest child.
(b) Statement 2 by itself determines the eldest child.
(c) Statements 1 and 2 are both required to determine the eldest child.
(d) Statements 1 and 2 are not sufficient to determine the eldest child.

20. X is 1 km northeast of Y. Y is 1 km southeast of Z. W is 1 km west of Z. P is 1 km south of W. Q is 1 km east of P. What is the distance between X and Q in km?

(a) 1 (b) $\sqrt{2}$
(c) $\sqrt{3}$ (d) 2

21. Rajan was not happy that Sajan decided to do the project on his own. On observing his unhappiness, Sajan explained to Rajan that he preferred to work independently.

Which one of the statements below is logically valid and can be inferred from the above sentences?

(a) Rajan has decided to work only in a group.
(b) Rajan and Sajan were formed into a group against their wishes.
(c) Sajan had decided to give in to Rajan's request to work with him.
(d) Rajan had believed that Sajan and he would be working together.

22. Find the odd one in the following group: ALRVX, EPVZB, ITZDF, OYEIK

(a) ALRVX (b) EPVZB
(c) ITZDF (d) OYEIK

23. Anuj, Bhola, Chandan, Dilip, Eswar and Faisal live on different floors in a six-storeyed building (the ground floor is numbered 1, the floor above it 2, and so on). Anuj lives on an even-numbered floor. Bhola does not live on an odd numbered floor. Chandan does not live on any of the floors below Faisal's floor. Dilip does not live on floor number 2. Eswar does not live on a floor immediately above or immediately below Bhola. Faisal lives three floors above Dilip. Which of the following floor-person combinations is correct?

	Anuj	*Bhola*	*Chandan*	*Dilip*	*Eswar*	*Faisal*
(a)	6	2	5	1	3	4
(b)	2	6	5	1	3	4
(c)	4	2	6	3	1	5
(d)	2	4	6	1	3	5

24. Find the next term in the sequence: 13M, 17Q, 19S, _____

(a) 21W (b) 21V
(c) 23W (d) 23V

25. If 'KCLFTSB' stands for 'best of luck' and 'SHSWDG' stands for 'good wishes', which of the following indicates 'ace the exam'?

(a) MCHTX
(b) MXHTC
(c) XMHCT
(d) XMHTC

26. "India is a country of rich heritage and cultural diversity."

Which one of the following facts best supports the claim made in the above sentence?

(a) India is a union of 28 states and 7 union territories.
(b) India has a population of over 1.1 billion.
(c) India is home to 22 official languages and thousands of dialects.
(d) The Indian cricket team draws players from over ten states.

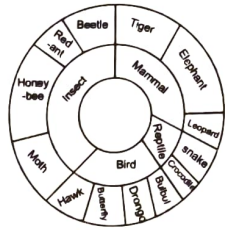
27. In which of the following options will the expression P < M be definitely true?

(a) M < R > P > S (b) M > S < P < F
(c) Q < M < F = P (d) P = A < R < M

28. Find the next term in the sequence:
7G, 11K, 13M, _____

(a) 15Q (b) 17Q
(c) 15P (d) 17P

29. The multi-level hierarchical pie chart shows the population of animals in a reserve forest. The correct conclusions from this information are:



(*i*) Butterflies are birds
(*ii*) There are more tigers in this forest than red ants
(*iii*) All reptiles in this forest are either snakes or crocodiles
(*iv*) Elephants are the largest mammals in this forest

(a) (i) and (*ii*) only
(b) (*i*), (*ii*), (*iii*) and (*iv*)
(c) (*i*), (*iii*) and (*iv*) only
(d) (*i*), (*ii*) and (*iii*) only

30. Find the odd one in the following group

Q,W,Z,B B,H,K,M W,C,G,J M,S,V,X

(a) Q, W, Z, B (b) B, H, K, M
(c) W, C, G, J (d) M, S, V, X

31. Lights of four colors (red, blue, green, yellow) are hung on a ladder. On every step of the ladder there are two lights. If one of the lights is red, the other light on that step will always be blue. If one of the lights on a step is green, the other light on that step will always be yellow. Which of the following statements is not necessarily correct?

(a) The number of red lights is equal to the number of blue lights
(b) The number of green lights is equal to the number of yellow lights
(c) The sum of the red and green lights is equal to the sum of the yellow and blue lights
(d) The sum of the red and blue lights is equal to the sum of the green and yellow lights

32. Read the statements:

All women are entrepreneurs.

Some women are doctors.

Which of the following conclusions can be logically inferred from the above statements?

(a) All women are doctors
(b) All doctors are entrepreneurs
(c) All entrepreneurs are women
(d) Some entrepreneurs are doctors

33. Find the odd one from the following group:

(a) W,E,K,O
(b) I,Q,W,A
(c) F,N,T,X.
(d) N,V,B,D

34. For submitting tax returns, all resident males with annual income below ₹ 10 lakh should fill up Form P and all resident females with income below ₹ 8 lakh should fill up Form Q. All people with incomes above ₹ 10 lakh should fill up Form R, except non residents with income above ₹ 15 lakhs, who should fill up Form S. All others should fill Form T. An example of a person who should fill Form T is

(a) a resident male with annual income ₹ 9 lakh
(b) a resident female with annual income ₹ 9 lakh
(c) a non-resident male with annual income ₹ 16 lakh
(d) a non-resident female with annual income ₹ 16 lakh

35. Which number does not belong in the series below?

2, 5, 10, 17, 26, 37, 50, 64

(a) 17 (b) 37
(c) 64 (d) 26

36. A dance programme is scheduled for 10.00 a.m. Some students are participating in the programme and they need to come an hour earlier than the start of the event. These students should be accompanied by a parent. Other students and parents should come in time for the programme. The instruction you think that is appropriate for this is

(a) Students should come at 9.00 a.m. and parents should come at 10.00 a.m.
(b) Participating students should come at 9.00 a.m. accompanied by a parent, and other parents and students should come by 10.00 a.m.
(c) Students who are not participating should come by 10.00 a.m. and they should not bring their parents. Participating students should come at 9.00 a.m.
(d) Participating students should come before 9.00 a.m. Parents who accompany them should come at 9.00 a.m. All others should come at 10.00 a.m.

37. The number of people diagnosed with dengue fever (contracted from the bite of a mosquito) in north India is twice the number diagnosed last year. Municipal authorities have concluded that measures to control the mosquito population have failed in this region.

Which one of the following statements, if true, does not contradict this conclusion?

(a) A high proportion of the affected population has returned from neighbouring countries where dengue is prevalent
(b) More cases of dengue are now reported because of an increase in the Municipal Office's administrative efficiency
(c) Many more cases of dengue are being diagnosed this year since the introduction of a new and effective diagnostic test
(d) The number of people with malarial fever (also contracted from mosquito bites) has increased this year

38. Geneticists say that they are very close to confirming the genetic roots of psychiatric illnesses such as depression and schizophrenia, and consequently, that doctors will be able to eradicate these diseases through early identification and gene therapy.

On which of the following assumptions does the statement above rely?

(a) Strategies are now available for eliminating psychiatric illnesses
(b) Certain psychiatric illnesses have a genetic basis
(c) All human diseases can be traced back to genes and how they are expressed
(d) In the future, genetics will become the only relevant field for identifying psychiatric illnesses

2013

39. Select the pair that best expresses a relationship similar to that expressed in the pair:

Medicine: Health

(a) Science : Experiment
(b) Wealth : Peace
(c) Education : Knowledge
(d) Money : Happiness

40. Abhishek is elder to Savar.

Savar is younger to Anshul.

Which of the given conclusions is logically valid and is inferred from the above statements?

(a) Abhishek is elder to Anshul
(b) Anshul is elder to Abhishek
(c) Abhishek and Anshul are of the same age
(d) No conclusion follows

41. Complete the sentence:

Universalism is to particularism as diffuseness is to ____

(a) specificity (b) neutrality
(c) generality (d) adaptation

42. After several defeats in wars, Robert Bruce went in exile and wanted to commit suicide. Just before committing suicide, he came across a spider attempting tirelessly to have its net. Time and again, the spider failed but that did not deter it to retrain from making attempts. Such attempts by the spider made Bruce curious. Thus, Bruce started observing the near-impossible goal of the spider to have the net. Ultimately, the spider succeeded in having its net despite several failures. Such act of the spider encouraged Bruce not to commit suicide. And then, Bruce went back again and won many a battle, and the rest is history.

Which one of the following assertions is best supported by the above information?

(a) Failure is the pillar of success.
(b) Honesty is the best policy.
(c) Life begins and ends with adventures.
(d) No adversity justifies giving up hope.

43. Statement: You can always give me a ring whenever you need.

Which one of the following is the best inference from the above statement?

(a) Because I have a nice caller tune.
(b) Because I have a better telephone facility.
(c) Because a friend in need is a friend indeed.
(d) Because you need not pay towards the telephone bills when you give me a ring.

44. Statement: There were different streams of freedom movements in colonial India carried out by the moderates, liberals, radicals, socialists, and so on.

Which one of the following is the best inference from the above statement?

(a) The emergence of nationalism in colonial India led to our Independence.
(b) Nationalism in India emerged in the context of colonialism.
(c) Nationalism in India is homogeneous.
(d) Nationalism in India is heterogeneous.

45. Select the pair that best expresses a relationship similar to that expressed in the pair :

water : pipe : :

(a) cart : road
(b) electricity : wire
(c) sea : beach
(d) music: instrument

46. All professors are researchers

Some scientists are professors

Which of the given conclusions is logically valid and is inferred from the above arguments :

(a) All scientists are researchers
(b) All professors are scientists
(c) Some researchers are scientists
(d) No conclusion follows

2012

47. Wanted Temporary, Part-time persons for the post of Field Interviewer to conduct personal interviews to collect and collate economic data. Requirements: High School-pass, must be available for Day, Evening and Saturday work. Transportation paid, expenses reimbursed.

Which one of the following is the best inference from the above advertisement?

(a) Gender-discriminatory
(b) Xenophobic
(c) Not designed to make the post attractive
(d) Not gender-discriminatory

48. Given the sequence of terms, AD CG FK JP, the next term is

(a) OV (b) OW
(c) PV (d) PW

2011

49. A transporter receives the same number of orders each day. Currently, he has some pending orders (backlog) to be shipped. If he uses 7 trucks, then at the end of the 4th day he can clear all the orders. Alternatively, if he uses only 3 trucks, then all the orders are cleared at the end of the 10th day. What is the minimum number of trucks required so that there will be no pending order at the end of the 5th day?

(a) 4 (b) 5
(c) 6 (d) 7

50. Few school curricula include a unit on how to deal with bereavement and grief, and yet all students at some point in their lives suffer from losses through death and parting.

Based on the above passage which topic would not be included in a unit on bereavement?

(a) How to write a letter of condolence
(b) What emotional stages are passed through in the healing process
(c) What the leading causes of death are
(d) How to give support to a grieving friend

51. The question below consists of a pair of related words followed by four pairs of words. Select the pair that best expresses the relation in the original pair :

Gladiator : Arena

(*a*) dancer : stage
(*b*) commuter : train
(*c*) teacher : classroom
(*d*) lawyer : courtroom

52. The horse has played a little known but very important role in the field of medicine. Horses were injected with toxins of diseases until their blood built up immunities. Then a serum was made from their blood. Serums to fight with diphtheria and tetanus were developed this way.

It can be inferred from the passage, that horses were

(*a*) given immunity of diseases
(*b*) generally quite immune to diseases
(*c*) given medicines to fight toxins
(*d*) given diphtheria and tetanus serums

2010

53. The question below consists of a pair of related words followed by four pairs of words. Select the pair that best expresses the relation in the original pair.

Unemployed : Worker

(*a*) fallow : land (*b*) unaware : sleeper
(*c*) wit : jester (*d*) renovated : house

54. Hari (H), Gita (G), Irfan (I) and Saira (S) are siblings (i.e. brothers and sisters). All were born on 1st January. The age difference between any two successive siblings (that is born one after another) is less than 3 years. Given the following facts:

I. Hari's age + Gita age > Irfan's age + Saira's age
II. The age difference between Gita and Saira is 1 year. However, Gita is not the oldest and Saira is not the youngest.
III. There are no twins.

In what order were they born (oldest first)?

(*a*) HSIG (*b*) SGHI
(*c*) IGSH (*d*) IHSG

55. Modern warfare has changed from large scale clashes of armies to suppression of civilian populations. Chemical agents that do their work silently appear to be suited to such warfare; and regretfully, there exist people in military establishments who think that chemical agents are useful tools for their cause.

Which of the following statements best sums up the meaning of the above passage:

(*a*) Modern warfare has resulted in civil strife
(*b*) Chemical agents are useful in modern warfare
(*c*) Use of chemical agents in warfare would be undesirable
(*d*) People in military establishments like to use chemical agents in war

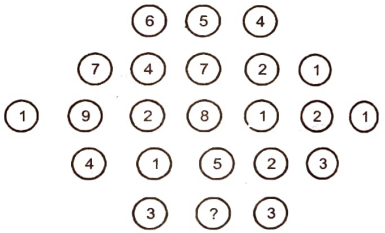
56. If 137 + 276 = 435 how much is 731 + 672?

(*a*) 534
(*b*) 1403
(*c*) 1623
(*d*) 1513

NUMERICAL TYPE QUESTIONS

2015

1. Fill in the missing value



2014

2. The next term in the series 81, 54, 36, 24,... is _____

3. Fill in the missing number in the series.

2 3 6 15 __ 157.5 630

4. What is the next number in the series?

12 35 81 173 357 ____

ANSWERS

EXERCISE – I

MCQ Type Questions

1. (*b*)	2. (*b*)	3. (*b*)	4. (*c*)	5. (*b*)	6. (*b*)	7. (*b*)	8. (*d*)	9. (*a*)	10. (*a*)
11. (*b*)	12. (*d*)	13. (*d*)	14. (*a*)	15. (*b*)	16. (*c*)	17. (*b*)	18. (*d*)	19. (*d*)	20. (*b*)

Numerical Type Questions

1. 70 2. 105 3. 2 4. 2 5. 11 6. 7920 7. 12870 8. 35 9. 10

EXERCISE – II

MCQ Type Questions

1. (*a*)	2. (*a*)	3. (*d*)	4. (*c*)	5. (*b*)	6. (*b*)	7. (*b*)	8. (*b*)	9. (*b*)	10. (*a*)
11. (*b*)	12. (*c*)	13. (*c*)	14. (*b*)	15. (*d*)	16. (*b*)	17. (*a*)	18. (*a*)	19. (*a*)	20. (*c*)
21. (*d*)	22. (*d*)	23. (*b*)	24. (*c*)	25. (*b*)	26. (*c*)	27. (*d*)	28. (*b*)	29. (*d*)	30. (*c*)
31. (*d*)	32. (*d*)	33. (*d*)	34. (*b*)	35. (*c*)	36. (*b*)	37. (*d*)	38. (*b*)	39. (*c*)	40. (*d*)
41. (*a*)	42. (*d*)	43. (*c*)	44. (*d*)	45. (*b*)	46. (*c*)	47. (*d*)	48. (*a*)	49. (*c*)	50. (*c*)
51. (*d*)	52. (*b*)	53. (*d*)	54. (*b*)	55. (*c*)	56. (*c*)				

Numerical Type Questions

1. (3) 2. (16) 3. (45) 4. (725 to 725)

EXPLANATIONS

EXERCISE – I

MCQ TYPE QUESTIONS

1. S[M(D(A(a, b)2, D(A(a, b)2)), M(D(S(a,b)2)) D(s(a,b)2)]

$$= S\left[\left|M\frac{a+b}{2}, \frac{a+b}{2}\right| \left|M\frac{a-b}{2}; \frac{a-b}{2}\right|\right]$$

$$= \left(\frac{a+b}{2}\right)^2 - \left(\frac{a-b}{2}\right)^2 = ab$$

6 – 7.

A → Lawyer → Male
B → Housewife → Female
C → Accountant → Male
D → Housewife → Female
E → Architect → Male
F → Lecturer → Female
Married Couple : AD & CF

8–9. The seating arrangement is shown below:

Left Right

7	6	5	4	3	2	1
D/C	E/F	F/E	B	C/D	G/A	A/G

10 – 13.

Level of Dissimilarity is equal to Maximum difference in ranks alloted to any two countries on any of the five traits.

For example, dissimilarity between India and China.

Trait	India'sRank	China's Rank	Difference
C	1	2	1
P	2	3	1
N	3	1	2
V	4	5	1
D	5	4	1

From the above table, maximum difference is 2.

So dissimilarity level between India and China is 2.

10. Calculating the level of dissimilarity for India Vs remaining countries.

China : For the parameter N (3 – 1) = 2
Japan : For the parameter D (5 – 1) = 4
Malaysia : For the parameter C (4 – 1) = 3
Thailand : For the parameter V (4 – 1) = 3

Hence least level of dissimilarity is 2 which is for China.

11. From above calculations, highest level of dissimilarity is 4 which is for Japan.

12. *The level of dissimilarity between*

China and Japan is for the parameter D
$(4 - 1) = 3$

India and China is for the parameter N
$(3 - 1) = 2$

Malaysia and Japan is for the parameter V
$(4 - 1) = 3$

Thailand and Japan is for the parameter D
$(5 - 1) = 4$

13. *The level of dissimilarity between*

Malaysia and China is for the parameter V
$(5 - 1) = 4$

China and Thailand is for the parameter V
$(5 - 1) = 4$

Thailand and Japan is for the parameter D
$(5 - 1) = 4$

Japan and Malaysia is for the parameter V
$(4 - 1) = 3$

14 to 17 :

The given set of questions is based on distribution and sequencing.

Firstly, let us put the names of women and men,

Women : Shanthi, Sridevi, Joya, Anita

Men : Sunil, Anil, Raj, Raman

It is given that(2 kids family) ← came before (No kid family)

Also Shanti (0 kids) → came before Sridevi

As Sunil and his wife came with their only kid, hence Sunil cannot be the husband of Shanthi.

As Anil and Raj are fathers, hence neither of them is the husband of Shanthi. Hence Raman is the husband of Shanthi.

Also, Joya came before Shanthi and met Anita on the venue, this gives the following order of their reaching the venue. [Anita, Joya, Shanthi, Sridevi]

As Sunil reached last, Sunil is the husband of Sridevi. This means that Raj is Joya's husband and Anil is Anita's husband.

Following table gives the desired details

Order	Women	Men	No. of Kids
1	Anita	Anil	at least 1
2	Joya	Raj	2
3	Shanthi	Raman	0
4	Sridevi	Sunil	1

14. Shanthi arrived third.

15. Out of the given choices. Sunil and Sridevi is the correct pair.

16. The daughters of Anita and Sridevi go to the same school. Anita's husband is Anil and Sridevi's husband is Sunil. Hence, Sunil and Anil is the correct answer.

17. Raj's family consists of two kids for certain (Anil has at least one kid, hence could be having exactly one kid also).

18 to 20 :

From statements (vii), (viii) and (ix), we get

P U S
Orange
R Q
Yellow Green

Then, T is opposite S and the colour of S is red.

From (x), the colour of house P is white, hence the colour of house T is blue.

P	U	S
White	Orange	Red
R	Q	T
Yellow	Green	Blue

Now, from (v) and (x) , we get

$T > S, Q > P > R$

From (vi), U is the shortest.

i.e., the order of houses in the descending order of their heights is T, S/Q, Q/S, P, R, U.

18. R is yellow coloured house and the house diagonally opposite R is S. And the colour of house S is Red.

19. The second tallest house is either S or Q.

20. The tallest house is T and its colour is blue.

NUMERICAL TYPE QUESTIONS

1. M[A(S(x, y), M(y, z)), S(A(x, z), S (z, y))]
= M[A((2 – 3), (3, × 5)), S((2 + 5), (5 – 3))]
= M[A(--1, 15), S(7, 2)]
= M[14, 5] = 70.

2. Since B is in 1, and C and D are not adjacent, the only places possible for them are 2 and 4.

3. Since B is not in the third place and A is not adjacent to B, and D is last in the queue. The position of A will be beside D and B will be at the other corner. C will be between B and A. Hence the positions will be BCAD. Place of C is 2, hence choice (b)

The sitting position is as follows.

White	Green	Red	Yellow
Maharashtra	U.P.	West Bengal	A.P.

From the above table,

Ms. Andhra Pradesh ⟶ Yellow

4. (1) (2) 3 3 2 1 4 (2) 3 (1) 4 (2) (2) 3 3 (1) 4 1 (1) (3) (2) 3 4

Total no. oranges were in basket = 2

5. Total fruits in the basket = 11

6. Total number of four letter password formed
$= {}^{11}C_4 \times \angle 4 = 11 \times 10 \times 9 \times 8 = 7920$

7. Required number of passwords
$= {}^{26}C_3 \times \angle 3 - {}^{15}C_3 \times \angle 3$
$= 26 \times 25 \times 24 - 15 \times 14 \times 13 = 12870$

8.

1st Letter	2nd letter	3rd letter

There are 5 vowels, a, e, i, o,, u,.

hence 1st letter can be any of the 5 vowels ;

∴ 5 different methods. ...(i)

The second letter can be any of the m, n, or p.

But each has its own rule for the third letter.

Hence, each case to be considered separately.

1 Vowel		m

In the above case; third letter is one of the remaining 4 vowels

Hence there are 5 × 4 = 20 ways of forming such a string. ...(ii)

If the second letter is n, the third is either e or u, *i.e.* 2 different ways.

∴ total of 5 × 2 ways = 10 ways ...(iii)

When 2nd position is filled by P, the third letter is same as the first letter. As the 1st letter is already selected, third can be filled only one way.

Hence, there are 5× 1 = 5 ways .....(iv)

From equations (ii), (iii) and (iv),

Total number of strings that can be formed
= 20 + 10 + 5 = 35

9. Third letter find is 'e'.

Once again considering each case of the second position being filled by m, n or p, we have

1st position	n	e
	2nd position	3rd position

The 3rd position shall be different form vowel in first position, *i.e.* vowel for 1st position can be selected out of the vowels a, i, o, u only.

Hence 4 different ways .....(i)

When 2nd possible is filled by n, occupying 2nd position, does not create any restriction on the vowel is position 1.

e	n	
3rd	2nd	1st

Hence, 1st position can be occupied by any of the 5 vowels.

Hence, 5 different ways .....(ii)

When 2nd position is filled by p,

e	p	
3rd	2nd	1st

In this case, letters in 3rd position out 1st position shall be the same.

Hence, 1st position shall be filled by 'e'.

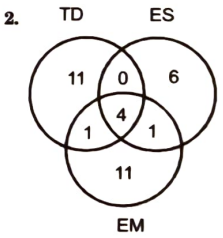
i.e. only one way of filling .....(iii)

Total number of possible strings = 4 + 5 + 1 = 10

EXERCISE – II

MCQ TYPE QUESTIONS

2.



3. Statement-I can be used to solve the question if capacity of both tanks is already known

Statement-II can be used if it is known what quantity of each tank is full/empty.

Therefore, by using both statements

Let capacity of tank B is x

$$\frac{70}{100}x = 14000$$

$$\Rightarrow \quad x = 20000 \text{ gallons}$$

Solution in tank A = $\frac{80}{100} \times 14000 = 11200$ gallons

Solution in tank B = $\frac{40}{100} \times 20000 = 8000$ gallons

$\therefore$ Total solution = 11200 + 8000 = 19200 gallons

4. $66 \square 6 = \frac{66-6}{66+6} = \frac{60}{72} = \frac{5}{6}$

$66 \diamond 6 = \frac{66+6}{66-6} = \frac{72}{60} = \frac{6}{5}$

$(66 \square 6) \rightarrow (66 \diamond 6) = \frac{5}{6} \times \frac{6}{5} = 1$

6. R + 3 = U, O + 3 = R,
A + 3 = D, D + 3 = G;
S + 3 = V, W + 3 = Z,
A + 3 = D, N + 3 = Q

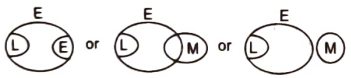
7. Community of people : Doctor

8. Since U does not want any portfolio, (*c*) and (*d*) are ruled out. R wants Home, or Finance or No portfolio, (*a*) is not valid.

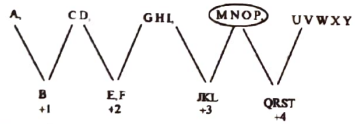
Hence option (*b*) is correct.

12. S + 1: S + 2 :

Therefore concluding diagram can be



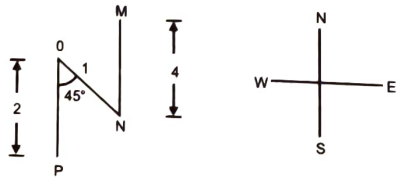
13.



14. Second sentence says that Ms. X reaches Kochi on 21/05/2014. Also she has to be in Bagdogora on 22/05/2014.

$\therefore$ She stays in Kochi for only one day in may.

17.



19. Riaz > Som

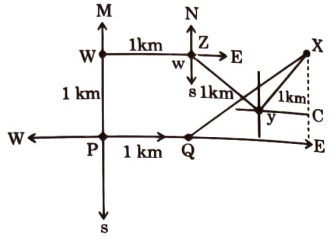
Shiv > Ansu ...(*i*)

Ansu is youngest in the group ...(*ii*)

$\therefore$ From equations (*i*) and (*ii*)

$\therefore$ Riaz > Som > Shiv > Ansu

20.



Here XE = 1 km

$$QE = \frac{2}{\sqrt{2}} = \sqrt{2}$$

$\therefore$ using Pythagoras theorem,

$$XQ = \sqrt{(QE)^2 + (XE)^2}$$

$$= \sqrt{2+1} = \sqrt{3} \text{ km}$$

$\therefore$ Statement I itself determines the eldest child which statement 2 is incorrect as Ansu is youngest in the group is not satisfied by statement 2

Hence option (*a*) is correct

22. ALRVX → only one vowel

EPVZB → only one vowel

ITZDF → only one vowel

OYEIK → three vowels

23. (*a*) Anuj: Even numbered floor (2, 4, 6)

(*b*) Bhola: Even numbered floor (2, 4, 6)

(*c*) Chandan lives on the floor above that of Faisal.

(*d*) Dilip: not on 2nd floor.

(*e*) Eswar: does not live immediately above or immediately below Bhola

From the options its clear, that only option (*b*) satisfies condition (*e*).

So, correct Ans is (*b*).

25. KCLFTSB: BST-Best, F-Of, LCK-Luck (Reverse order)

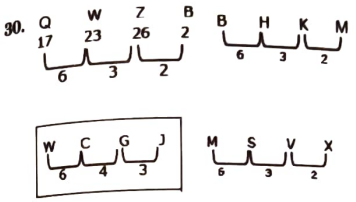
SHSWDG: GD-Good, WSHS-Wishes (Reverse order)

Similarly "ace the Exam'- C-Ace, T-The, XM-Exam

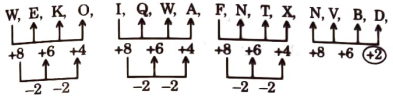
26. Diversity is shown in terms of difference language.

29. It is not mentioned that elephant is the largest animal.

30.



33.

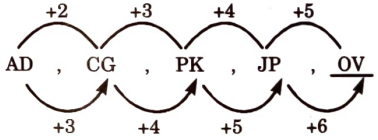


Hence the odd one from the following group is N, V, B, D.

34. By reading the instructions, it is clear that person who fills form T is a resident female with annual income of ₹ 9 Lakh.

37. Among given option, option (*d*) *i.e.* the number of people with malarial fever has increased this year do not contradic the conclusion.

48.



49. Let each truck carry 100 units.

$$2800 = 4n + e$$

where n = normal

$$3000 = 10n + e$$

where e = excess/pending

$$\therefore \quad n = \frac{100}{3}, \quad e = \frac{8000}{3}$$

$$5 \text{ days} \Rightarrow 500x = \frac{5100}{3} + \frac{8000}{3}$$

$$\Rightarrow \quad 500x = \frac{8500}{3} = 17$$

$$\Rightarrow \quad x > 5$$

Minimum possible = 6

50. The given passage clearly deals with how to deal with bereavement and grief and so after the tragedy occurs and not about precautions. Therefore, irrespective of the causes of death, a school student rarely gets into details of causes—which is beyond the scope of the context. Rest all are important in dealing with grief.

51. Given relationship is worker: workplace. A gladiator is

(*i*) a person, usually a professional combatant trained to entertain the public by engaging in mortal combat with another person or a wild.

(*ii*) A person engaged in a controversy or debate, especially in public.

52. From the passage it cannot be inferred that horses are given immunity as in option (*a*), since the aim is to develop medicine and in turn immunize humans. Option (*b*) is correct since it is given that horses develop immunity after some time. Refer "until their blood built up immunities". Even option(*c*) is invalid since medicine is not built till immunity is developed in the horses. Option (*d*) is incorrect since specific examples are cited to illustrate and this cannot capture the essence.

54. H + G > I + S ...(1)

and G – S = 1 ...(2)

G is not oldest, S is not youngest

$\therefore$ H + 1 > I

Irfan older than Hari

Gita older than Sarita

From given option SGHI

55. Use of chemical agents in warfare would be undesirable

56.

$$\begin{array}{llllll} 137- & \overline{001} & \overline{011} & \overline{111} & \\ +276- & 010 & 111 & 110 & \\ \hline & \underline{100} & 011 & 101 & = 435 \end{array}$$

Octal number system

$$\begin{array}{llllll} \therefore \quad 731- & 111 & 011 & 001 & \\ 672- & 110 & 111 & 110 & \\ \hline & \underline{100} & 010 & 011 & \end{array}$$

Hence 1623

NUMERICAL TYPE QUESTIONS

1. Middle number is the average of number on both sides

$\therefore$ Average of 3 and 3 is $\frac{3+3}{2} = \frac{6}{2} = 3$

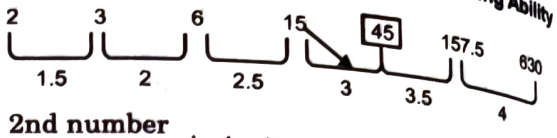
2. $81 - 54 = 27; 27 \times \frac{2}{3} = 18$

$54 - 36 = 18; 18 \times \frac{2}{3} = 12$

$36 - 24 = 12; 12 \times \frac{2}{3} = 8$

$\therefore$ $24 - 8 = 16$

3.



$\frac{\text{2nd number}}{\text{1st number}}$ is in increasing order as shown above

4. The given series is

12, 35, 81, 173, 357, ...

The given series follows the following pattern

$12 \times 2 + 11 = 35$

$35 \times 2 + 11 = 81$

$81 \times 2 + 11 = 173$

$173 \times 2 + 11 = 357$

$357 \times 2 + 11 = 725$